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Germany

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OECD ECONOMIC SURVEYS

2000-2001

Germany

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BASIC STATISTICS OF GERMANY, 2000

THE LAND

Area (thousand sq. km)	357.0
Agricultural area (thousand sq. km)	195.4
Forests (thousand sq. km)	104.3

Major cities, June 1999 (thousand inhabitants)	
Berlin	3 393
Hamburg	1 702
Munich	1 194
Cologne	963
Frankfurt	645
Essen	601
Dortmund	590
Stuttgart	581
Düsseldorf	569
Bremen	542

THE PEOPLE

Population (thousands)	82 127
Number of inhabitants per sq. km	230
Net natural increase in population (thousands), 1999	–77
Net migration (thousands), 1999	205

Labour force (thousands)	40 614
Employment (thousands)	38 534
of which:	
Agriculture, forestry, fishing and hunting	978
Industry (including construction)	11 313
Private services	26 244

THE PRODUCTION

GDP (billion of DM)	3 976
GDP per capita (US$)	22 812
Gross fixed investment	
Per cent of GDP	21.4
Per capita (US$)	4 881

Origin of GDP (per cent)	
Agriculture, forestry, fishing and hunting	1.2
Industry (including construction)	30.1
Services	68.8

THE GOVERNMENT

Public consumption (per cent of GDP)	18.9
General government current revenue (per cent of GDP)	44.4
Public debt (ratio to general government current revenue), 1999	136.3

Composition of Parliament:	Seats
Social Democratic Party (SPD)	298
Christian Democratic Party (CDU)	198
Christian Social Union (CSU)	47
Greens	47
Free Democratic Party (FDP)	43
PSD	36

Last general election: September 1998
Next general election: 2002

FOREIGN TRADE

Exports of goods and services (per cent of GDP)	33.4
Main exports (per cent of total merchandise exports):	
Products of agriculture, forestry and fishing	0.9
Basic materials and semi-finished goods	21.4
Manufactured foods and tobacco	3.6
Other consumer goods	9.7
Investment goods	58.2
End products	6.2
Total	100.0

Imports of goods and services (per cent of GDP)	33.0
Main imports (per cent of total merchandise imports):	
Food	6.9
Raw materials and semi-finished goods	14.8
Finished goods	68.3
of which: Primary products	9.9
End products	58.4
Other exports	10.0
Total	100.0

THE CURRENCY

Irrevocable conversion rate	1.95583

Currency units of euro per US$, average of daily figures:	
Year 2000	1.0851
March 2001	1.1010

N*ote*: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Germany by the Economic and Development Review Committee on 7 May 2001.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 6 April 2001.

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The previous Survey of Germany was issued in November 1999.

Assessment and recommendations

An economy in recovery but still facing some major challenges

The German economy grew by 3 per cent in 2000, the strongest rate since 1992. While the present deceleration in world trade will have a dampening impact this year and next, domestic sources of growth remain relatively robust so that only a moderate slowing of GDP is projected. This is largely related to the fact that the policy environment in Germany has improved, notably due to the major income and business tax reform that comes into force in 2001. The main macroeconomic factors influencing the recovery and the Secretariat's projections are reviewed in the first part of this *Survey* (Chapter I). However, major challenges to fiscal and structural policies remain and these are reviewed in the following sections (Chapters II and III). Although the size of the deficit reduction over the medium term announced by the government is smaller than in several other OECD countries, meeting the targets will require strict spending control given the substantial loss of revenues implied by the tax reform and several fiscal risks that are apparent for 2001 and the years thereafter. The slowing of the economic expansion to be expected this year and next reinforces the need to continue with structural reform in product and labour markets – within a stability- and growth-oriented macroeconomic policy setting – so as to raise the growth potential of the German economy above its present level, estimated at around 2 per cent by the OECD. Finally, somewhat more than ten years after German reunification, the *Survey* takes stock of the economic development of the new states and derives policy recommendations to give new impulses for the convergence with the west, which should rely more on market forces (Chapter IV).

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The economy recovered in 2000 driven by buoyant exports and investment…

The German economy has rebounded from a growth pause that ended in mid-1999 largely due to strongly accelerating exports. While this recovery was largely driven by the rapid expansion in world trade, German exporters also gained market share. This reflected a marked improvement in the country's competitiveness due to the depreciation of the euro in 2000 and the year before, as well as roughly constant unit labour costs, in contrast to an increase in the euro zone as a whole. With external demand expanding rapidly, investment in machinery and equipment grew buoyantly as well, and the resulting rise in imports offset some of the growth contribution from strong exports. Improvements in Germany's policy framework – notably the prospects for the major income and business tax reform to be implemented in 2001 – probably also contributed to the positive investment climate. But construction slid back into recession, further downsizing in eastern Germany being the primary reason. Employment growth picked up considerably in 2000, but a large part of this increase appears to be attributable to jobs with very few hours, having been taken on by new entrants into the labour market. The unemployment rate dropped from 8.3 per cent in 1999 to 7.8 per cent in 2000, while consumer price inflation edged up by 1.3 percentage points, mainly due to the increase of energy prices.

… but with world trade slowing GDP growth should decelerate, the decline being limited by favourable domestic conditions

With world trade growth expected to slow this year, the net contribution to growth of the external sector will decline, though exports should pick up again in the course of 2002, reflecting the projected recovery in world trade. Final domestic demand is likely to strengthen somewhat this year and the next. Private consumption is projected to accelerate in 2001, based on the income tax reductions and improved labour market conditions. Investment (equipment) should remain strong, benefiting from tax cuts (despite a tightening of depreciation rules), from high capacity utilisation in manufacturing and from the favourable investment climate associated with wage moderation embodied in collective agreements that are largely fixed until spring 2002. Construction, however, will remain weak. All in all, GDP growth is projected to decelerate to 2¼ per cent in 2001 and to strengthen to 2½ per cent in 2002. The unemployment rate should decline from 7.8 per cent in 2000 to 6¾ per cent in 2002, and headline inflation should slow because of the

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reversal of the oil price effects. Risks appear balanced so long as the global slowdown is not greater than projected: continuing declines in unemployment may fuel increased wage pressures, which could in turn dampen investment. On the other hand, tax reductions might induce a higher consumption of private households than assumed in the projections.

Monetary conditions are roughly neutral with respect to economic activity

Monetary conditions were accommodative in 1999 but became more neutral thereafter. From the end of 1999 to October 2000 the European Central Bank (ECB) undertook successive interest rate increases, totalling 2¼ percentage points, in response to rising inflation risks associated with the economic recovery in the euro zone and especially with strongly accelerating oil and import prices. Short-term market rates increased accordingly, though long-term rates remained virtually unchanged over this period, indicating stable inflationary expectations. More recently, market rates have declined. In the process, real long-term rates have fallen to their lowest level since 1995. While the ECB has not made any adjustments to interest rates since last October, a cut in short-term rates underlies the OECD projections. Monetary conditions should therefore not act as a brake for GDP growth.

The general government balance continued to improve…

General government balances continued to improve over the last two years, reflecting larger-than-expected income tax receipts, and spending restraint. In 2000 the general government balance swung into a surplus of 1.5 per cent of GDP, owing to one-off receipts from the auctioning of UMTS mobile phone licenses. Extra revenues from the sales of the licences amounted to 2.5 per cent of GDP. Net of these receipts, the deficit improved by 0.4 per cent of GDP, less than in 1999, while the structural (cyclically-adjusted) balance remained roughly unchanged.

… but consolidation will pause in 2001…

General government finances in 2001 will be heavily affected by the revenues foregone associated with the 2001 step of the income and business tax reform. These come on top of the tax relief stemming from the earlier steps of the reform project. While declining unemployment-related transfers and moderate wage settlements in the public sector will bring some relief to the budget on the spending

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side, further consolidation gains appear unlikely. In the Secretariat's assessment, the general government deficit will be some ¾ per cent of GDP higher than in 2000, net of the UMTS receipts, and roughly in line with the deficit target laid down in Germany's Stability Programme for the EU (1½ per cent). Nevertheless, with the debt redemption associated with the UMTS receipts being distributed over 2000 and 2001, debt will decline further this year, and should move from 60 per cent of GDP in 2000 to some 58 per cent in 2002 (Maastricht definition).

… and strict spending control will be needed for fiscal consolidation to progress thereafter

The government has reinforced its commitment to continue fiscal consolidation aiming at balancing the budget on the general government level by 2004, and on the federal level two years later. While targeted deficit reductions are slow in comparison to several other OECD countries, they appear acceptable in view of the substantial revenue losses associated with the phased income and business tax reforms. But tight spending control is all the more necessary to ensure that the targets are met. Indeed, there are risks that, despite the savings programme presented in summer 1999, consolidation might proceed on a less stringent path than envisaged. While the pension reform is progressing, it appears that savings might be less than initially planned. Social spending is being extended in some fields, and the long-term care insurance is now producing deficits after the surpluses observed in the first years after its introduction. There is also a risk of spending over-runs in the health funds. Already implemented or forthcoming income tax reductions and increases in family benefits might involve larger revenue losses than projected. Furthermore, medium-term growth might turn out to be lower than the 2½ per cent officially projected. The necessary spending restraint will need to be combined with further structural reforms to identify economically important priorities for savings, notably with respect to subsidies, to achieve a more effective control over social spending and to implement measures to improve public sector efficiency.

A comprehensive income tax reform is being implemented

The main income and business tax reform, whose key elements have already been highlighted in the previous *Survey*, passed parliament in 2000 and became effective in January 2001. They imply large tax reductions totalling more than 1 per cent of GDP in this year; some of the measures

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are being phased in over a five-year horizon. Statutory income and corporate tax rates have been reduced substantially, and the tax base for profits has been broadened. The reform will place Germany in the middle range of income and business tax rates observed in the OECD, down from the group of countries with the highest rates. Overall, the reform will improve the allocation of investment and the attractiveness of Germany as a business location, and it will also foster work incentives.

The pension reform is being implemented and needs to be carefully monitored

Limiting the fiscal impact of ageing on the public pay-as-you-go pension system has been a vital part of the government's fiscal consolidation programme introduced in September 2000. Meanwhile a draft reform law has passed the first chamber of parliament, and new legislation is planned to become effective in spring 2001. The reform will introduce a funded non-mandatory component into the pension system with associated contributions being subsidised on the basis of individual choice either by cash transfers or tax allowances up to a certain limit. Correspondingly, there will be phased reductions in the level of pay-as-you-go pensions relative to average wages. Both features are important steps in improving the fiscal viability of the overall system, and reform along these lines has been proposed in earlier *Surveys* on Germany. However, given the long time horizon involved, it is not clear that the measures will suffice to stabilise contribution rates at targeted levels, and further action to consolidate the pension system may become necessary. Indeed the new legislation commits the government to take action if it appears likely that the targeted contribution rates will not be attained. Moreover, making part of the funded layer mandatory rather than giving tax preferences to contributions would be preferable on fiscal grounds and is worth considering. Defining the range of admissible funded pension instruments has also been an issue. In this respect, regulation should allow for maximum returns on assets while at the same time securing the long-term solvency of the (defined-contribution) schemes. This might be best achieved by putting more emphasis on the prudent regulation of funds' reserve requirements associated with certain investment instruments, rather than restricting the range of admissible investment instruments itself.

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Rulings by the Constitutional Court concerning the inter-governmental revenues equalisation system should be taken as an opportunity for thorough reform…

In order to equalise revenues per capita across federal states Germany's fiscal equalisation system transfers tax revenues – after primary tax sharing – from the wealthier to the poorer states, and federal transfers are also directed towards states with below average per capita revenues. The Federal Constitutional Court has demanded clarification of the principles on which such transfers should be based. The government should take this opportunity to remedy identified shortcomings that hamper the system's allocative efficiency. At present, the marginal effective tax retention facing states which generate extra tax revenues is confiscatory. This may be expected to lead to inefficient decisions with respect to attempting to increase the states' own tax revenues, and empirical evidence suggests that this is indeed the case. One system, which would meet present ambitious social and political objectives to improve economic performance, would be to make greater use of block grants to poorer states over a multi-annual period and to reduce annual transfer rates based on financial capacity. Furthermore, to the extent income redistribution in the present system is biased against states with a low population density corrections are necessary.

… while implications for federal fiscal relations of the new tax and pension legislation reinforce the need to increase the revenue autonomy of local governments

But reform should not be confined to the inter-governmental fiscal equalisation system. As has been outlined in the 1998 *Survey*, there has been a secular trend over many years reducing the connection between autonomous tax revenues and spending responsibilities of states and communities. Also, there is a risk that social spending obligations of the communities will increase as replacement ratios in the pay-as-you-go pension system are reduced over the next years insofar as these reductions give rise to increased claims for locally-provided social assistance transfers. These developments reinforce the need, highlighted in the 1998 *Survey*, to increase the revenue raising autonomy of local governments.

Labour market agreements have continued moving in the direction of more flexibility, and this needs to be supported by policy measures

As has been outlined in the previous *Survey*, an important element in the government's economic and social policy approach has been to establish round table talks between the social partners and the government for the purpose of developing a joint strategy for employment and growth (Alliance for Jobs). In important respects these talks appear to have produced positive results. The Alliance had reached broad understandings that wage policies

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should be longer-term oriented and primarily be geared towards increasing employment. In 2000 relatively moderate wage agreements, mostly extending over two years, validate this agreement. It is remarkable that these settlements came about in a period of rapidly rising oil and energy prices. The wage negotiations also confirmed the trend towards more flexibility on the company level as opening clauses often were renewed. Nevertheless, there has been a tendency in recent years to undercut collective agreements, and non-compliance is particularly pronounced in eastern Germany. Overall, labour market institutions are adjusting, but not yet sufficiently to cope with the substantial labour market imbalances still persisting in Germany. The authorities should support the process of introducing greater flexibility into the wage bargaining system. New legislation gives employees the legal right to transform full-time employment into part-time employment, except under certain conditions. While this measure might meet demands for more part-time employment it runs the risk of increasing labour costs. It is welcome that restrictions for employing older people on fixed-term contracts have been relaxed. However, the renewability of fixed-term contracts has been restrained further and this reduces working time flexibility.

Labour taxation needs to be lowered further

Steps have been taken to increase work incentives, but further effort is needed. Legislation allowing older unemployed to receive unemployment benefits without obligation to search for a job expired in 2000 and has not been renewed. This is a welcome move towards better labour utilisation of older persons. Lowering labour taxation and non-wage labour costs remains an issue, despite the progress associated with the phased reduction in income tax rates that are part of the income tax reform. Marginal effective tax rates are still particularly high for recipients of social assistance due to high benefit withdrawal rates, as outlined in the 1996 *Survey*. Experiments with reducing effective labour taxation by providing in-work benefits are currently under way. But reform is also necessary to better integrate the provision of social assistance benefits into the workplacement system.

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Regulatory reform of network industries has proven fruitful, but further policy action is required

Advances have been made in liberalising product markets although there are many regulatory challenges that still need to be tackled. Considerable progress has been made in liberalising network industries. In telecommunications, electricity supply and postal services, Germany has made major progress and opened markets more rapidly than EC Directives required. While the new regulation of gas supply also carries further the opening of markets than in many other European countries, the pace of reform needs to be stepped up. In telecommunications regulatory reform has proved to be a major success which has been reflected in sizeable price cuts, increased product variety, and improved quality of services. Price cuts have also been observed in the electricity sector, but little progress has been made yet in reforming the gas sector, and the effectiveness of regulatory reform in the railway system with respect to competition has also been only modest. Any attempts to delay the further development of competition in network industries – either by incumbents or other forces – need to be resisted and further policy initiatives may be necessary to strengthen competition in these markets. Policy action in these fields is warranted by the potentially very substantial welfare improvements for consumers and the higher productivity growth in the economy overall. Major outstanding issues are:

- In *telecommunications*, competition is most underdeveloped with respect to local calls – as is the case in most other countries with a liberalised market – requiring measures to foster access to the local loop. More generally, tight sector-specific regulation of the *Deutsche Telekom* needs to be maintained as long as the incumbent dominates the telecommunication infrastructure.

- Regarding the *Post Office*, the present charges for mail delivery, which seem to be high by international comparison, have been extended until the end of 2002. The prolongation implies higher costs for consumers, and a reduced competitiveness for enterprises. Hence, the monopoly rights need to be terminated. Moreover, the Post Office needs to be privatised to exclude future conflicts of interest of the owner.

- In the *railway sector* the powers of the regulator should be strengthened. As an additional measure

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the provision of transport services should be unbundled from the operation of the grid in order to support non-discriminatory access to the net.

– With vertically integrated conglomerates dominating *electricity and gas markets*, non-discriminatory access to networks is still under threat. Further unbundling of generation, distribution and transmission in electricity and gas markets is therefore necessary to foster competition. Also, further consideration needs to be given to establish an independent regulatory authority for both electricity and gas.

State ownership of banks needs to be phased out and shop opening hours need to be liberalised further

Important regulatory issues also arise with respect to non-network sectors:

– Community or state governments typically act as a guaranteeing authority for *public-sector banks*, effectively lowering the refinancing costs – especially for Länder-owned banks – in comparison with private-sector banks. There are no convincing reasons, however, why banks, especially those operating in commercial markets, should be government owned at all, given that meritorious services could also be commissioned with private banks. Hence, the state-owned banks should be privatised.

– In *retailing*, comparatively restrictive regulations on rebates and bonuses will be abolished without replacement. This will put German retailers on a more equal footing with international competitors in e-commerce, and should strengthen retail competition more generally. In addition, the regulations on shop opening hours should be liberalised further, and several state governments have demanded such a move, but no new legislative initiative has yet emerged.

Progress in the new states is mixed: while living standards have improved remarkably, market-driven convergence has not

From the outset, financial support policies for the new states were meant to serve two purposes.

– First, recognising that viable competitive economic structures cannot be established overnight, financial support was designed to compensate for various disadvantages facing firms and households in the new states. These derived from the fundamental distortions imposed on all economic activities by the ill-defined central planning system of the German Democratic Republic.

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– Secondly, financial support was intended to pave the way to establishing new competitive economic structures that, after a transition period, would generate self-sustainable growth in the eastern economy.

Regarding the first goal, policies have been successful. Spurred by both high levels of investment aid and income support for households, eastern Germany experienced buoyant GDP growth in the first half of the 1990s, mainly originating from the construction sector. Wages increased rapidly and so did transfer income, with western Germany's comprehensive social security system having been fully extended to the east already at the beginning of the 1990s. By the end of the decade net household income in eastern Germany is estimated to have been some 85 per cent of the western level in real terms. Living standards have approached western levels in other respects as well. Heavily polluted industrial sites have been cleaned up and emission levels have been reduced to the western norm, the housing stock has largely been renewed and new infrastructure has been installed. The second goal, however, of reaching self-sustainable economic growth that would lead to convergence with the west, has not yet been attained. Growth decelerated in the second half of the 1990s to low rates that even fell short of those in the west. In line with this development, real GDP per capita in the new states stagnated at around 60 per cent of the level in the old states. These developments occurred against the background of persistently high levels of west-to-east transfers amounting annually to some 4½ per cent of western German GDP over the 1990s on a net basis, excluding tax concessions and certain take-overs of eastern debt by federal institutions. In 1999 such transfers amounted to one-third of the GDP of the new states. Mainly financed by these transfers, total absorption in the new states has exceeded domestic production by about 50 per cent.

Structural change occurring over the past decade should have established eastern Germany on a high-growth path…

It is nonetheless true that a number of factors characterising the structural change that has occurred over the past ten years should have been favourable for putting the eastern German economy on a high-growth path. The infrastructure capital stock has been substantially renewed and extended. Similarly, the capital stock of eastern German business increased rapidly as well, supported by massive investment aid extended by the Federal Government and

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the new states. In manufacturing, the average capital intensity (gross fixed capital stock by employed person) has been estimated to have reached 92 per cent of the western level in 1999, though there are large differences within the sector with the capital intensity in final goods production on average falling substantially short of the level in the west.

… but productivity remained subdued, and unit labour costs have failed to come down to competitive levels, while the unification challenges turned out to be large and persistent

In part, the failure to achieve a self-sustaining strong growth path must be ascribed to the size and persistence of the initial transition challenge. Indeed, the exposure to competition of the eastern German economy revealed that its structural deficiencies were too severe to be remedied quickly by restructuring or wage moderation. Consequently a large part of eastern German industry had to be closed down. The re-industrialisation thereafter had to start from scratch and its performance is to some extent masked by the simultaneous downsizing of construction. Indeed, the manufacturing sector grew strongly and high-growth branches within the sector experienced marked improvements in their export performance. However, two largely policy-related characteristics of the eastern German economy also account for the fact that the actual growth performance over the last decade has remained subdued and below its potential.

- The productivity gap with the west is significantly larger than what the gap in capital intensity would suggest.
- Unit labour costs on average in the eastern German economy have not come down to competitive levels. Although declines were substantial – in particular in manufacturing – unit labour costs are still significantly above the level in western Germany.

Investment subsidisation after unification has distorted the economic structure to become too capital intensive while productivity gains remained modest

The massive subsidisation of enterprises after unification introduced various distortions. The evidence suggests that persistent capital subsidisation has biased resource allocation in favour of capital intensive industries and construction. In the process, to some extent investment was pushed into low-productivity sectors. Furthermore, in a number of sectors the capital intensity of production was driven to high levels relative to the west while associated gains in relative productivity were only modest. In manufacturing this phenomenon is largely concentrated in basic goods industries and mainly reflects action in the first years after unification. In view of these adverse developments the Federal Government

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phased out special depreciation tax allowances in 1998 and restructured support schemes in favour of small and medium-sized enterprises and capacity widening investment.

Rapidly rising labour costs, underpinned by massive transfers, depressed the creation of new employment opportunities

Regarding the second characteristic – unit labour costs above competitive levels – this is mainly attributable to massive wage increases that ensued immediately after the currency union with western Germany. The currency union implied *ceteris paribus* a substantial hike in labour costs in the new states whose neutralisation would have required, at the minimum, a considerable delay of further nominal wage increases. Instead, the social partners agreed on a predetermined wage path targeting rapid convergence with western wage levels, and this led to massive wage increases in the first years after unification, and convergence towards western wage levels has continued, though on a considerably slower path. These wage hikes meant that much of the eastern industry was rendered uncompetitive. Despite the significant labour shake-out after reunification, and persistently weak employment creation in the primary labour market thereafter, productivity growth has been insufficient to close the competitiveness gap. This result would not have been sustainable without massive transfers from the west. Industrial subsidies effectively helped safeguard the viability of enterprises despite the rise in labour costs, and unemployment-related benefits provided income support to displaced workers. Simultaneously the drop in employment was cushioned by work provision and training schemes, while early retirement schemes reduced labour force participation. In the absence of the formulation and execution of a clear exit strategy from the granting of financial support for inherited difficulties, the compensation policies instituted to tide over the new states during the transition are in danger of perpetuating a transfer dependency of the east on the west and forestalling the onset of self-sustaining growth.

Another part of the productivity gap can be explained by specific factors affecting eastern Germany

While the eastern German labour force is generally considered to be highly qualified, and migration to western Germany does not appear to have led to a systematic erosion of skill availability in the east, there does appear to be a lack of managerial and marketing skills that might have hampered the competitiveness of eastern German products and lowered the effective productivity levels. Various other factors such as a

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slightly lower rate of capacity utilisation, differences in the size distribution of enterprises relative to the west, a lower population density, limited access to new international markets and the crisis in old (eastern and FSU) markets and furthermore a lack of networking between establishments may also explain another part of the productivity gap to the west.

Fiscal and structural policies need to rely more on market forces to bring about convergence with the old Länder

In the future, fiscal and structural policies need to rely much more on market forces to bring about economic convergence with the west than they have done in the past. In fact, *Surveys on Germany* published in the early 1990s warned that massive and sustained levels of financial support could risk severe distortions in eastern Germany's economic structure. Similarly, setting wages above market clearing levels was judged to jeopardise convergence with the west because of adverse effects on investment and job creation. At the present juncture, the OECD would emphasise the following:

- Government spending needs to be both streamlined and further redirected in favour of infrastructure bottlenecks so as to widen the scope for market forces to operate.
- The efficiency of the inter-governmental transfer system needs to be strengthened.
- Financial aid to enterprises needs to be scaled down further and restructured. Similarly employment provision and training schemes need to be reduced and restructured as well.
- High unemployment in the east reinforces the need for comprehensive labour market reform in Germany overall.

Infrastructure development needs to continue and fiscal authorities need to restructure spending patterns so as to enhance economic growth

Infrastructure development needs to continue, but with more selectivity than up to now. Barriers to growth originating from insufficient infrastructure endowment need to be identified on a project basis via cost-benefit analysis. Rough east-to-west comparisons of the value of the infrastructure capital stock installed cannot replace such in-depth project evaluation. Transfers from the west will continue to be needed to support such infrastructure investment. But restructuring of the budgets of the states and the communities in eastern Germany itself is also required. Over most of

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the past decade governments in the new states have devoted a declining share of their revenues to financing investment – while indebtedness increased – and their infrastructure spending appears to have been biased in favour of areas that are not of primary importance for fostering the economic integration of the east. Redressing these imbalances requires measures in the following areas:

- While government employment has been reduced substantially over the past decade, there is still overstaffing that needs to be further reduced. The functional distribution of government employment appears to be significantly biased towards fields such as central administration, social services and culture, and this should be adjusted. Moreover, except for an east-west differentiation that is being reduced, the present remuneration system for the public sector in Germany overall does not allow for wage differentiation across regions and activities, and this should be changed so as to underpin the needed reallocations.
- Public sector services need to be priced more closely to the market.
- Greater efforts should be made to engage non-governmental bodies and associations in the provision of social and cultural services that have been supplied till now directly by the government.

Subsidies need to be reduced and restructured

The original rationale for special industrial subsidies given to the east was to temporarily compensate enterprises for comparative disadvantages in economic framework conditions relative to the west. With improving conditions in various fields this rationale is vanishing. Consequently, to avoid further distortions to the economic structure associated with subsidisation, aid needs to be further scaled down and restructured:

- The recent trend to apply the same set of rules for subsidisation for Germany overall should be enforced and continued. To the extent that special support has been fixed until 2004, this policy should be decided and announced soon to trigger the necessary adjustments on the side of enterprises. With a unified subsidy regime for Germany overall, the new states

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would still continue to obtain the larger share of aid per capita since they remain still structurally weaker than the old ones.

– The share of financial aid to enterprises for training the management and the workforce should be increased at the expense of investment subsidies.

– The heterogeneity of support programmes extended by both the federal and Länder governments is likely to increase the distortionary effects of subsidisation and favours "subsidy competition" between the states. Efforts should be made to co-ordinate support programmes between the governments involved.

– Also, all programmes should be continuously evaluated and associated with a pre-announced time path specifying their phasing out. These conditions, designed to minimise rent seeking, should not be restricted to the new states.

Labour market conditions in the east reinforce the need for flexible work arrangements and comprehensive reforms

High wages relative to productivity, low unemployment and a low degree of net east-to-west migration are mutually incompatible goals if high levels of transfers are not to be perpetuated. Wage growth would have to undercut the evolution of labour productivity for several years in a row to firmly establish eastern unemployment on a downward path towards western levels. Also, since high wage policies obstruct employment creation they are not an effective means to prevent migration to the west. Even so, commuting and migration to western Germany can be beneficial to the extent this improves employment prospects of the eastern German labour force, preserves and develops human capital in the primary labour market, and fosters east-west convergence of economic activity more generally. The various facets of labour market reform that have been recommended in the OECD *Jobs Strategy for Germany* overall are particularly urgent when seen from the perspective of the eastern states. This holds especially for the need to allow the wage distribution to reflect differences in skills and economic activity. While the labour market after unification was in a difficult condition and keeping unemployed in work programmes under such exceptional circumstances may have had a value in itself, it cannot be ignored that work

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provision and training schemes (ALMPs) applied on a large scale have not proven to be effective. Rather than in bringing the unemployed back into regular employment ALMPs have effectively been applied as an expensive instrument for social policy. They are also a fruitless means for keeping young people in the new states and should be reduced considerably, to levels where the programmes and subsequent placement are better manageable than at present. Programme evaluation by independent bodies needs to be institutionalised to widen the scope for corrective action to improve the effectiveness of programmes as placement instruments.

Summary

Although economic activity is projected to slow due to external factors, Germany should achieve solid growth over the next two years. This reflects both a higher degree of competitiveness and an improvement in framework conditions. However, many important challenges remain. A major factor contributing to this relatively favourable outlook is the income and business tax reform that becomes effective this year. Moderate wage settlements, often agreed for a two-year period, have also been important. Tight spending control is necessary for medium-term fiscal consolidation targets to be met. Within a stability- and growth-oriented macroeconomic policy setting further structural reforms are needed to increase the capacity of the economy to generate jobs and thereby strengthen future growth prospects and to underpin the long-term viability of the pension and health systems. Increasing the flexibility of labour markets needs to rank high on the policy agenda, and labour taxation should be reduced further in order to increase work incentives and labour demand. While good progress has been made with respect to competition in network industries, the potentially large gains to be expected from fostering competition in these sectors and product markets more generally warrant reinforcing regulatory reform. Living standards in the new states have improved remarkably since 1990. While much remains to be done to eliminate the inherited deficiencies, framework conditions have been established that would allow eastern Germany to embark on a high-growth path, and the strong recovery of final goods manufacturing in the eastern Länder is signalling a promising devel-

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opment. However, in order for this recovery to broaden and bring about a sustained process of economic convergence fiscal and structural policies will need to have a sharper focus on improving efficiency and rely much more on market forces than they did in the past. This will require policy adjustments in several fields, including, further downsizing and restructuring of aid given to industry, revisions to spending priorities by the new states, and a more focused approach to active labour market policies. Pushing ahead with such policy reforms should both ensure the fuller integration of the new states and improve Germany's overall economic performance.

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I. Recent trends and prospects

Overview

The German economy experienced a broadly-based recovery within the past two years, although activity moderated in the second half of 2000. Economic activity began to gather pace in the second half of 1999 as export growth accelerated on the back of a recovery in foreign demand for German produced goods and an improvement in external competitiveness. Subsequently, domestic private demand was bolstered by high investment growth in machinery and equipment, which responded to increasing capacity utilisation, and a pick-up in private consumption following a rise in real disposable income. As a result, the pace of economic activity accelerated and reached a peak during the first half of 2000. Thereafter activity decelerated in the second half of the year, reflecting higher oil and import prices depressing real disposable incomes and profits. Net exports declined because of accelerating imports. Strong employment gains, particularly in part-time employment, accompanied by a substantial inflow into the labour force, has led to a decline in the unemployment rate. Employment growth was primarily taking place in the service sector, notably with respect to business-near services. Labour demand in the manufacturing sector has to some extent been curbed by the cost increasing impact of higher oil and import prices. As the current wage agreements point to continued subdued wage inflation, no second round effects of the higher oil and import prices should materialise and headline inflation should decline as the negative terms-of-trade effects peter out. Economic growth is projected to remain slightly above potential with the stimulating effects associated with the income tax reform and a fall in oil and decelerating import prices sustaining domestic demand, while export growth should develop in line with a projected softening of export market growth in 2001 and a recovery the following year. Broadly-defined monetary conditions – reflecting the recent fall in long-term interest rates and a projected cut in the policy rates – should not act as a brake for economic activity.

The economic recovery gathered pace in 1999

The economic recovery gathered momentum during the second half of 1999 as first export and then private domestic demand growth accelerated (Figure 1 and Table 1), which together with an increase in capacity utilisation led

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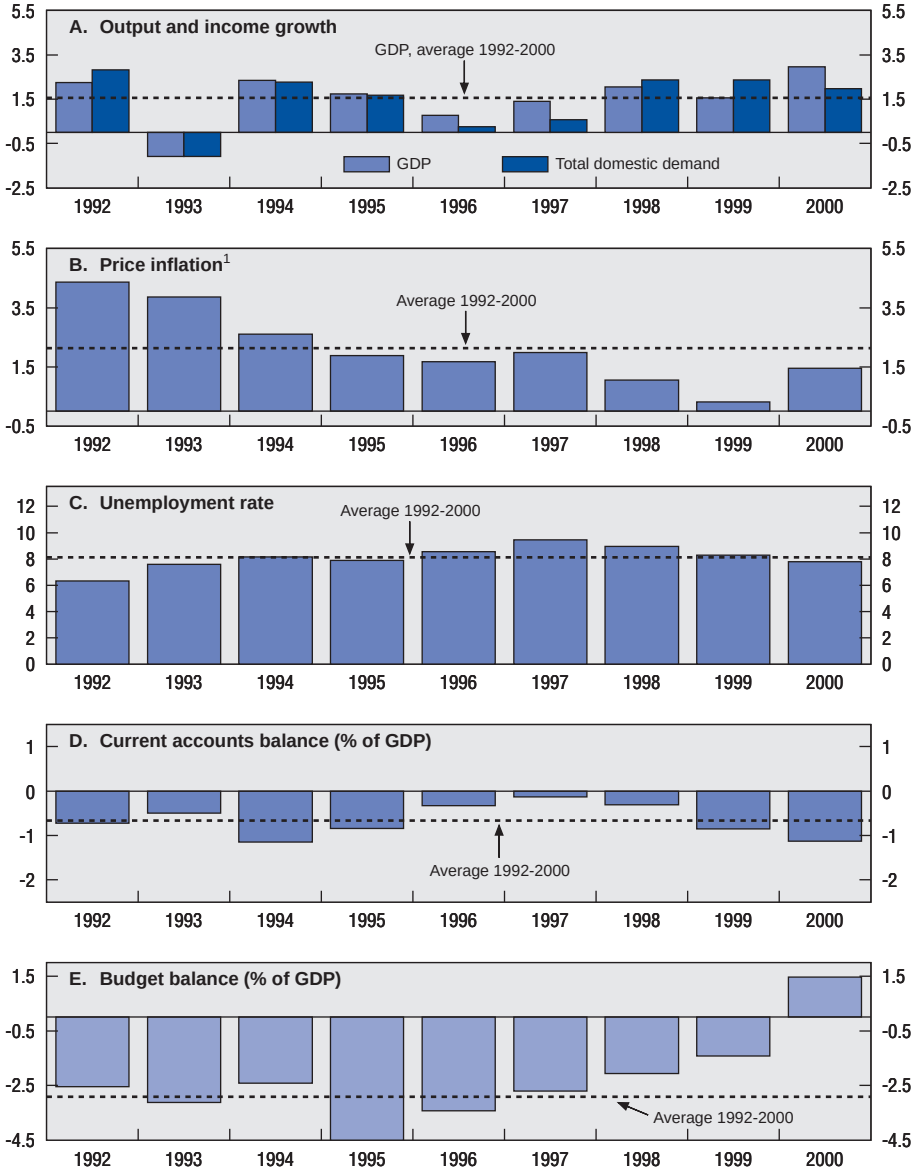
Figure 1. **Macroeconomic performance**
Per cent



1. Growth in private consumption deflator.
Source: OECD.

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Table 1. **Demand and output**
Percentage change from previous period, 1995 prices

	1994	1995	1996	1997	1998	1999	2000	2001[1]	2002[1]
Private consumption	1.0	2.0	1.0	0.7	2.0	2.6	1.6	2.2	2.2
Government consumption	2.4	1.5	1.8	–0.9	0.5	–0.1	1.4	0.5	0.5
Total fixed investment	4.0	–0.7	–0.8	0.6	3.0	3.3	2.4	2.1	2.9
Construction	6.9	–1.8	–2.8	–1.5	–1.0	0.5	–2.5	–1.2	0.1
Machinery and equipment	–1.0	1.5	2.8	4.0	9.4	7.4	9.0	6.1	6.1
Stockbuilding[2]	0.3	0.3	–0.5	0.2	0.4	0.2	0.2	0.2	0.0
Total domestic demand	**2.3**	**1.7**	**0.3**	**0.6**	**2.4**	**2.4**	**2.0**	**2.0**	**2.0**
Net exports[2]	0.1	0.1	0.5	0.8	–0.3	–0.8	1.0	0.2	0.5
Exports of goods and services	7.6	5.7	5.1	11.3	7.0	5.1	13.2	8.7	7.4
Imports of goods and services	7.4	5.6	3.1	8.4	8.6	8.1	10.2	8.4	6.4
GDP at market prices	**2.3**	**1.7**	**0.8**	**1.4**	**2.1**	**1.6**	**3.0**	**2.2**	**2.4**
Memorandum items:									
Total employment[3]	–0.2	0.2	–0.3	–0.2	0.9	1.1	1.5	0.9	0.8
Household savings rate	11.6	11.2	10.8	10.4	10.2	9.9	9.8	10.2	9.9
Relative ULC in manufacturing	1.1	7.7	–3.0	–4.6	2.2	–0.3	–6.6	. .	. .
Export market growth in manufacturing	11.5	10.3	6.4	10.2	8.8	6.6	13.0	8.0	8.0
GDP deflator	2.5	2.0	1.0	0.8	1.1	0.9	–0.4	1.1	1.4
Private consumption deflator	2.6	1.9	1.7	2.0	1.1	0.3	1.4	1.8	1.5
Unemployment (in millions)[3]	3.3	3.2	3.5	3.9	3.7	3.4	3.2	3.1	2.9
Unemployment (per cent of labour force)[3]	8.2	7.9	8.6	9.5	8.9	8.3	7.8	7.3	6.8
Current account (per cent of GDP)	–1.1	–0.8	–0.3	–0.1	–0.2	–0.9	–1.1	–1.3	–0.8

1. OECD projections.
2. Contributions to GDP growth.
3. According to the labour force survey.
Source: OECD.

to a pick-up in investment in machinery and equipment to double digit growth rates in the first half of 2000 before slowing in the second half of 2000. Private consumption growth firmed in the first half of 2000, before weakening in the second half as real disposable incomes were negatively affected by higher oil and import prices. Accelerating imports, following the strengthening of domestic demand, reduced net exports. Convergence between the new and old states suffered a setback as the fast expansion of the new states' output in manufacturing and business-near services was offset by a further contraction in the construction sector. Overall real GDP accelerated from an annualised growth rate of just above 1½ per cent in the first half of 1999 to some 4 per cent in the first half of 2000, before the economic expansion slowed towards around 2 per cent.

As export growth accelerated…

Total export growth accelerated in mid-1999, reaching an annual rate of around 13 per cent in the second part of 1999, and maintained that momentum

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throughout 2000. This strong recovery occurred against the background of a marked pick-up in export market growth for goods from an annual rate of around 4 per cent in the first half of 1999 to about 13 per cent in the second semester of 1999 and 12½ per cent in 2000, as the expansion in North America maintained its strength and the European recovery came under way (Table 2). The resulting gain in Germany's market share for manufactured goods was partly the result of German exports having a high content of investment goods and benefiting from strong demand for capital goods among its trading partners and partly related to the marked improvement in external competitiveness following the depreciation of the euro.[1] Competitiveness also improved *vis-à-vis* the other EMU countries as

Table 2. **Trade by region**

	1996	1997	1998	1999	99 I	99 II	00 I	99 I	99 II	00 I
	Percentage change from previous year							Share		
Exports to[1]										
EU	3.1	8.7	9.6	6.4	2.9	10.0	16.0	58.6	56.3	57.9
of which: Euro area	3.4	7.1	10.2	7.1	3.7	10.6	15.7	45.3	43.1	44.6
EFTA	–3.1	6.3	10.0	0.6	0.3	0.8	11.4	5.2	5.2	5.0
Central and Eastern Europe	17.1	28.8	11.4	–5.6	–13.9	2.9	20.8	9.5	10.1	9.8
of which: Russia	. .	43.4	–13.4	–31.8	–54.1	1.3	37.9	0.9	1.2	1.0
North America[2]	9.7	30.3	16.1	11.0	4.2	17.6	23.9	10.3	11.0	10.9
Japan	12.5	–2.7	–11.0	11.1	4.4	17.7	26.6	2.0	2.1	2.1
Asia[3]	5.3	8.3	–25.1	1.3	–12.7	17.0	28.4	3.5	3.8	3.8
Central and South America	4.7	28.3	20.4	0.9	–1.9	3.3	10.4	2.7	3.1	2.5
Other	4.2	12.6	2.3	1.0	–1.9	3.7	14.5	8.2	8.5	8.1
Total	4.9	12.6	7.7	4.5	–0.1	9.1	17.4	100.0	100.0	100.0
Memorandum item:										
Share of machinery and transport equipment in total exports	48.9	50.2	52.0	52.0	52.4	51.6	51.0			
Imports from[1]										
EU	3.5	9.1	6.5	3.9	0.1	7.8	14.7	55.0	53.0	53.0
of which: Euro area	2.8	8.5	7.4	3.6	0.2	7.1	13.1	44.2	42.1	42.0
EFTA	4.6	7.0	3.2	2.9	–3.8	9.6	16.5	4.9	5.3	4.8
Central and Eastern Europe	6.7	20.6	13.3	14.0	13.9	14.1	24.8	10.7	10.8	11.2
of which: Russia	13.4	12.1	–13.1	26.4	23.1	29.7	36.6	2.1	2.0	2.4
North America[2]	7.4	19.0	13.8	7.5	2.4	10.2	24.8	8.9	8.8	9.3
Japan	–5.1	11.7	5.7	3.3	–3.5	10.6	25.0	4.8	4.8	5.1
Asia[3]	–4.9	13.6	1.0	5.8	–2.8	14.2	33.8	4.7	5.1	5.3
Central and South America	0.8	16.6	6.9	5.6	3.9	7.3	12.0	2.1	2.0	2.0
Other	8.4	15.6	2.9	15.5	5.6	25.3	31.7	9.0	10.1	9.9
Total	3.6	11.8	6.9	6.0	1.5	10.7	19.0	100.0	100.0	100.0

1. F.o.b., including adjustment.
2. United States and Canada.
3. Dynamic Asian economies.

Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik*, OECD, *Foreign Trade Statistics Series* A.

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the corresponding German relative unit labour costs improved, contributing to a gain in market shares within the European Union.

... and domestic demand improved...

The business climate improved markedly in 1999 and remained at a high level in the first half of 2000 as export growth accelerated, and incoming orders – particularly from abroad – continued to increase (Figure 2). This stimulated investment in machinery and equipment, a process that was further underpinned by capacity utilisation in manufacturing reaching post-reunification highs during 2000. Moreover, the tax reform's tightening of depreciation rules from 2001 onwards might have brought forward some business investment into 2000.

Non-financial corporations' profits deteriorated slightly in 1999, but remained at the second highest level in the 1990s.[2] In 2000, the 5 per cent deterioration in the terms of trade (national account statistics) points to a further squeeze on profit margins at an aggregated level, as producers have been unable to fully pass on higher costs associated with higher import and oil prices to final prices, as is partly reflected in a 0.4 per cent decline in the price deflator for GDP in 2000. Some sectors have experienced a profit squeeze, such as the transport sector as the result of higher input costs, and the construction sector arising from declining demand. On the other hand, industrial companies have reported higher profits,[3] in particular those which are larger and more export oriented.

Construction investment hardly grew in 1999 as further declining activity in the new states partly offset the mildly positive development in the old ones. Construction activity contracted sharply in 2000. This is mainly the result of the ongoing adjustment process in the new states. Residential construction turned out to be particularly weak, and this might have been related to a reduction in subsidies for new owner-occupied housing. The fiscal consolidation process also has led to a substantial lowering of orders placed by the public sector.

Household real disposable income benefited in 1999 from very low consumer price increases, and real private consumption grew by more than 2½ per cent. Consumer confidence was volatile in both 1999 and 2000, reflecting on the one hand increasing energy prices, and on the other hand further income tax cuts to come. While private consumption grew at an annual rate of 2 per cent in the first half of 2000, it decelerated subsequently to an annual rate of around ¾ per cent in response to the pickup in inflation and a softening of the general economic outlook.

... growth peaked in the first half of 2000

Export growth peaked in the first half of 2000 and foreign orders continued to come in strongly before decelerating at the beginning of 2001 (Figure 2). At the same time, the domestic impetus to growth weakened as incoming domestic

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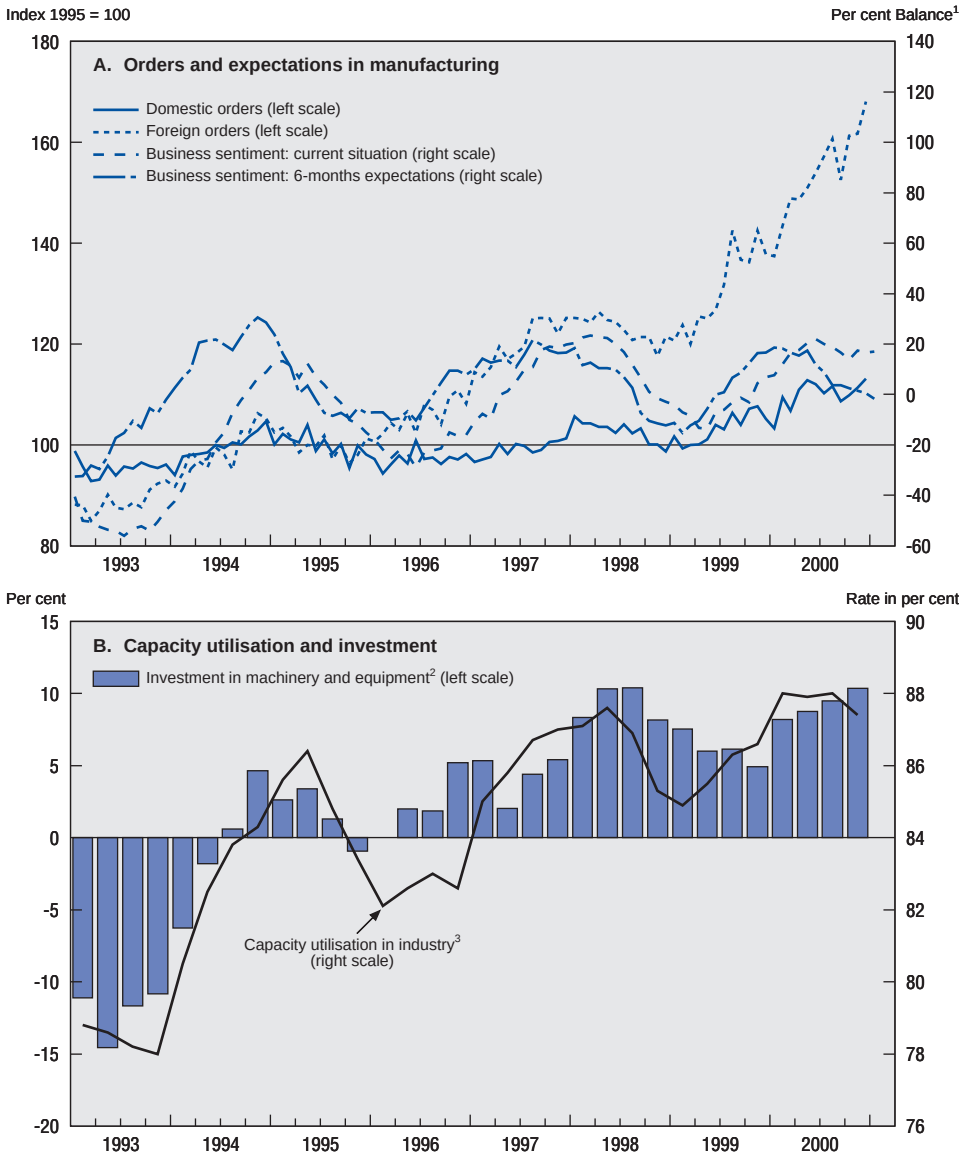
Figure 2. **The business climate**



1. Balance of positive-negative responses. A level of zero corresponds to normality.
2. At 1995 prices. Percentage change over previous period.
3. Refers to western Germany.

Source: Ifo Institut für Wirtschaftsforschung; Deutsche Bundesbank and OECD, *Main Economic Indicators*.

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Table 3. **The current account of the balance of payments**
DM billion

	1993	1994	1995	1996	1997	1998	1999	2000
Goods	68.1	82.7	93.2	106.3	125.1	140.2	133.8	120.7
Exports (f.o.b.)[1]	632.7	696.4	749.9	786.5	885.9	953.7	996.5	1 165.8
Imports (f.o.b.)[1]	564.6	613.7	656.7	680.2	760.8	813.6	862.7	1 045.1
Services	–55.9	–66.4	–67.3	–68.4	–74.7	–83.3	–96.0	–107.9
of which: Tourism	–44.3	–51.6	–51.4	–53.0	–54.1	–56.5	–58.3	–61.8
Factor income	27.4	4.9	0.2	1.4	–2.5	–11.6	–23.4	–16.9
of which: Investment income	26.6	4.5	1.6	3.2	–0.8	–9.9	–21.6	–15.0
Transfers	–55.7	–59.9	–55.7	–51.3	–52.7	–53.3	–50.3	–53.3
of which:								
Net contribution to the EU	–27.3	–31.7	–30.0	–27.6	–28.5	–30.4	–27.1	–30.7
Other public transfers	–14.1	–12.9	–10.3	–7.7	–8.3	–6.9	–6.5	–7.4
Current account	–16.2	–38.8	–29.7	–12.0	–4.8	–8.0	–35.9	–57.3
(As a per cent of GDP)	–0.5	–1.1	–0.8	–0.3	–0.1	–0.2	–0.9	–1.4

1. Special trade, according to the foreign trade statistics. Including supplementary trade items; imports exclude freight and maritime transport insurance costs, which are included under services.
Source: Deutsche Bundesbank, *Zahlungsbilanzstatistik*.

orders in manufacturing softened in the second half of 2000. As import growth accelerated in the second half of 2000 the net contribution to growth of net exports diminished, reinforcing the deceleration of economic activity. Looking at the composition of orders, it is remarkable that domestic orders for consumption goods have fallen, while foreign orders have grown strongly until the end of 2000, reflecting the larger reliance on exports to sustain growth in Germany as compared with its main trading partners' relative greater reliance on private consumption.

The worsening terms of trade and the acceleration of imports in 2000 further reduced the trade surplus (Table 3). This deterioration can partly be explained by a larger deficit on the energy balance, which rose by some DM 14 billion from the first half of 1999 to the first half of 2000 as energy prices increased. As the service balance continued to deteriorate as well, the current account deficit widened to nearly 1½ per cent of GDP, despite an improvement of the factor income balance.

Strong employment growth has allowed unemployment to come down…

Between the second quarter 1999 and the last quarter 2000, employment creation in terms of employed persons has been rather strong, increasing by nearly 2½ per cent (more than 900 000 jobs, seasonally adjusted). Measured in terms of hours worked the employment gains are less impressive with an increase of less than ½ per cent, although the decline in effective working days in 2000 overstates the difference between the two concepts (Figure 3, panel A).[4] Nevertheless, if average hours worked had remained unchanged, employment creation in terms of persons would

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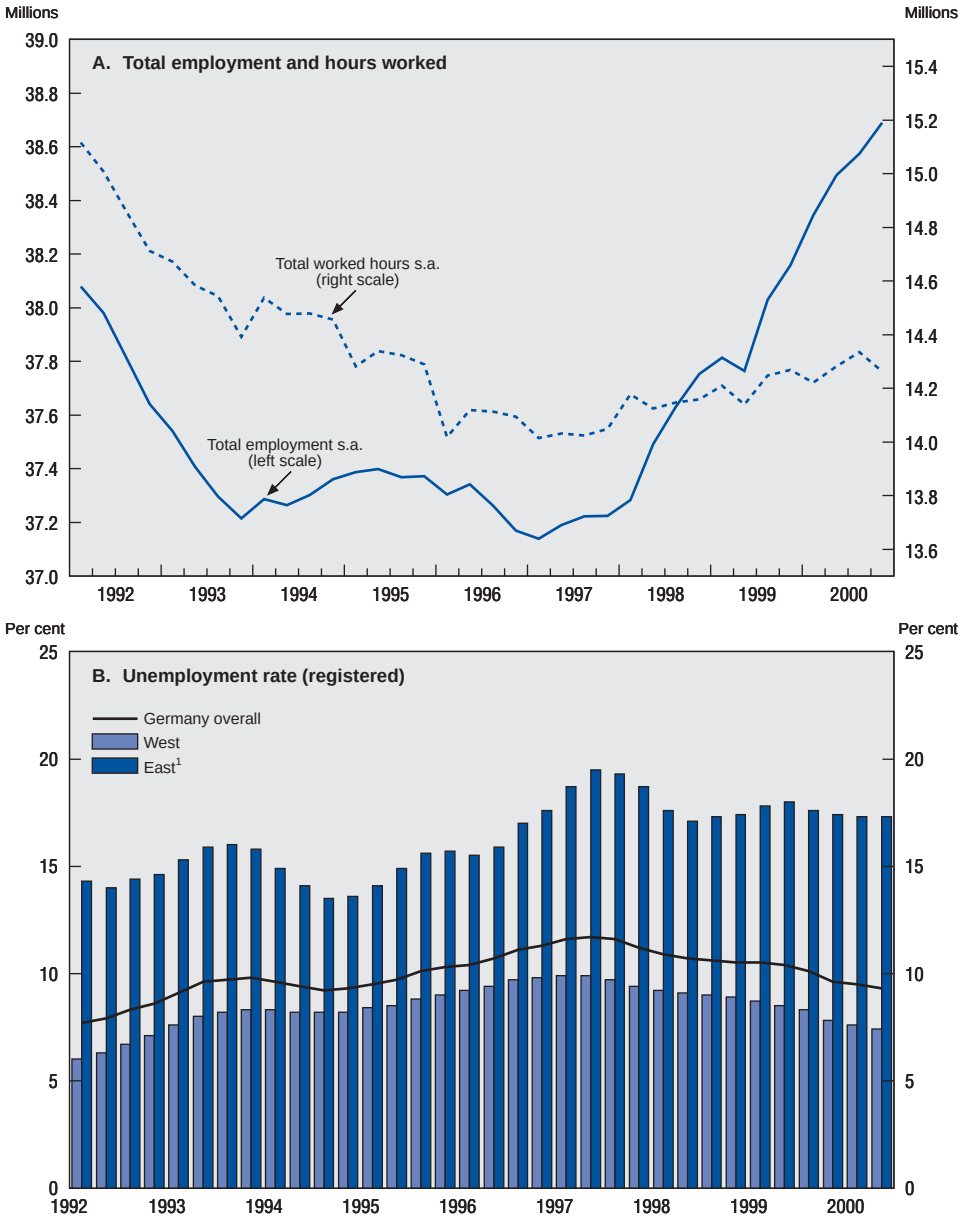
Figure 3. **Employment and unemployment**



1. Data not available before Q3 1992.
Source: Deutsche Bundesbank; Federal Labour Office and OECD.

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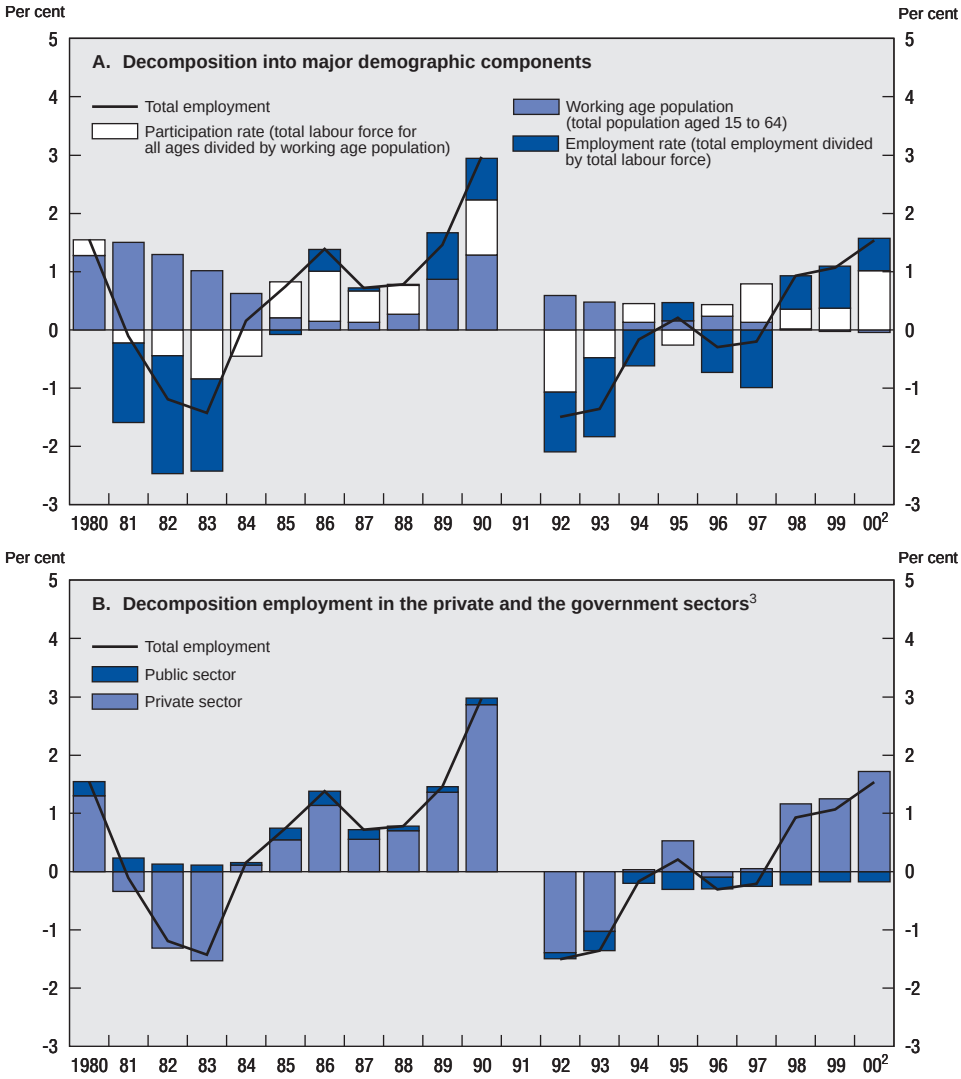
Figure 4. **Decomposition of change in total employment**[1]
Percentage change



1. Prior 1991 figures relate to western Germany and are on the old national accounts basis.
2. Estimates except for total employment and labour force.
3. The decomposition of employment growth has been approximated by a weighted sum of employment growth in the private and the government sectors, with the weights being the share of each type of employment in the total recalculated each year.

Source: Federal Statistical Office and OECD.

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have been substantially lower. Thus, the expansion of part-time employment explains a large part of the increase in employed persons. An important factor behind this development was the imposition of employees' social security contributions in April 1999 on "small hours" jobs when such employment is being held as a secondary job.[5] Thus, it appears that a substitution in the composition in "small hours" jobs has taken place away from secondary employment to primary employment. As earnings in "small hours" jobs are restricted to a level that falls short of average unemployment benefits, the expansion of secondary "small hours" jobs has mainly been filled by pulling in substantial numbers of marginal workers into the labour force.

Employment growth in the private sector was strong in 2000, although it decelerated at the end of the year, following the slowing of the economic expansion (Figure 4). The construction sector continued its shedding of labour. Productivity growth picked up in 2000. However, the growth discrepancy between hours worked and persons employed also led to substantial differences in productivity

Box 1. Revisions to the German labour market statistics

The employment statistics from the Federal Statistical Office have been revised upwards in two steps over the past couple of years. The first step, as reported in the 1999 *Survey*, was to adapt to the International Labour Office (ILO) definition and increased employment by about 2½ million jobs, mainly through the inclusion of jobs involving less than 15 hours per week – the so-called "small hours" jobs. The second step, enacted in August 2000, led to a further upward adjustment in the number of "small hours" jobs, bringing total employment to 37½ million in 1998 as compared with previously reported 34 million, an increase of 10½ per cent. The revisions also reflected previous under-estimations as compared with the annual micro census and new reporting requirements. Concurrently, average hours worked were revised downwards, implying that total hours worked only increased by 5 per cent. With no revisions to output levels, implied productivity levels were lowered correspondingly. The unemployment data were also adjusted to comply with the ILO definition to include all unemployed of 15 years or more who have actively been looking for a job over the past four weeks and are able to take up employment within two weeks. In 1999, the survey based unemployment level was 3.4 million as compared with registered unemployment of nearly 4.1 million, or in terms of unemployment rates 8.3 per cent and 10.5 per cent, respectively. The registered unemployment data set is larger than the survey based concept. It includes all job seekers that have registered with the labour office, including holders of jobs involving less than 15 hours per week, and who are looking for employment of at least 15 hours per week and for a duration of more than seven days. The Federal Labour Office adjusts employment data to avoid double counting in the calculation of the registered unemployment rate.

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growth derived from the two concepts with hourly productivity growth being twice as high as the per employee productivity increases.

The favourable employment trends resulted in both a rather strong labour supply response and a fall in unemployment. According to the national account statistics, the labour force increased by around ¾ million between the second quarter 1999 and the last quarter 2000 and at the same time unemployment declined by some 250 000, resulting in a ¾ percentage point fall in the unemployment rate to just above 7½ per cent (seasonally adjusted; Figure 4). In comparison, the registered unemployment rate fell by more than a full percentage point, reflecting different definitions (Box 1). Employment provision measures have been scaled back roughly in line with improved labour market conditions (Table 4). The regional development

Table 4. **The German labour market**
Thousands of persons

	Germany		Western Germany		New Länder (eastern Germany)	
	1999	2000	1999	2000	1999	2000
	Annual average					
Labour force	41 370	41 773				
of which:						
Unemployed[1]	3 428	3 247				
Employed (domestic)	37 942	38 526	31 977[2]	32 658[2]	5 965	5 874
Memorandum item:						
Registered unemployed[3]	4 099	3 889	2 756	2 529	1 344	1 359
Labour market programmes						
of which:						
Short-time workers[4]	50	47	37	34	13	13
Subsidised jobs[5]	430	317	82	70	348	247
Training courses[6]	402	394	248	242	155	153
Early retirement programmes[7]	809	791	511	516	271	266
	Unemployment rate (per cent)					
Registered unemployment	10.5	9.6	8.8	7.8	17.6	17.4
Registered unemployment and participants in ALMPs as a percentage of the labour force	15.5	14.0	12.0	10.9	29.4	27.5

1. Source: national accounts, ILO definitions.
2. Including Berlin.
3. Source: Federal Labour Office (*Bundesanstalt für Arbeit*).
4. Programmes subsidised by the Federal Labour Office; number of persons calculated as equivalent to unemployment.
5. Public employment programmes subsidised by the Federal Labour Office.
6. Full-time participants in further education programmes, retraining courses, and German language courses subsidised by the Federal Labour Office.
7. Programmes subsidised by the Federal Government, by the Federal Labour Office and by the Public Pension Systems. Excludes invalidity pensions.

Source: Institute for Labour Market Research, Federal Labour Office and OECD.

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in unemployment in 2000 has been rather uneven with a more marked lowering of unemployment in the old Länder as compared with the new Länder, where the male unemployment rate even increased. A similar discrepancy has emerged in the development of youth and long-term unemployment, which has been lowered considerably in the old Länder, in contrast to increases in the new states. On the other hand, unemployment among older workers (+55 years) has declined in the east and the west.

... without the emergence of wage pressures

Despite the recovery of the labour market, wage increases have receded. Collectively-bargained hourly wages increased by some 3 per cent in 1999, but reductions in non-wage labour costs (notably lowered pension contribution rates) and faster growth in – relatively low paid – part-time employment led to lower increases in per-capita overall compensation. Thus, compensation per employee increased by just above 1 per cent and compensation per hour worked by just above 1½ per cent. In 2000 growth in collectively-bargained wages came down to 2 per cent, and the continued negative wage drift held increases in compensation per employee at 1¼ per cent. In addition, the wage negotiations led to mostly two-year wage contracts with average collectively bargained wage increases of around 2¼ per cent for 2001. Wage increases were lower in the contracting construction sector (see Chapter III below). Public sector wage increases have also remained moderate, and are fixed until the autumn 2002.

Higher productivity growth and a weaker euro have further improved external competitiveness

Unit labour costs have remained roughly stable in 2000 as higher productivity levels have offset the modest increases in compensation. On the other hand, the marked depreciation of the real effective exchange rate that occurred in 1999 and 2000 has improved international competitiveness. Measured in a common currency, there has been a substantial improvement in Germany's relative unit labour costs *vis-à-vis* its trading partners over this period (Figure 5).

Nonetheless inflation accelerated somewhat following the boost in oil prices

The inflation environment has deteriorated, but can still be described as favourable overall. The increases in the consumer price index (CPI) accelerated from around ½ per cent in mid-1999 to 2½ per cent in the autumn of 2000, mostly as the result of double digit increases in the energy component of the index. Following a slight dip in late 2000, consumer price inflation returned to 2½ per cent in early 2001 partly on account of higher energy taxes. Core inflation, measured as the consumer price index excluding energy and food, accelerated less, rising from below ½ per cent in mid-1999 to around 1½ per cent in early 2001 (Figure 6).

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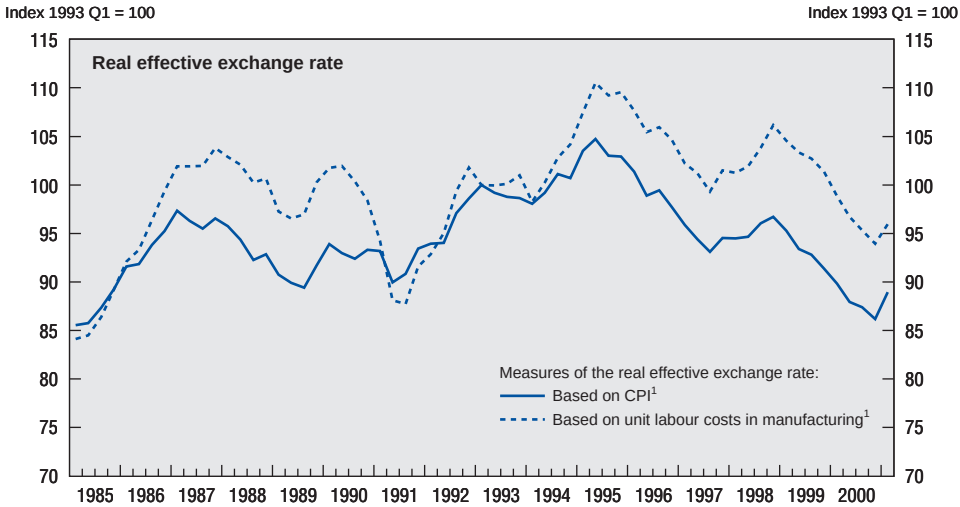
Figure 5. **International competitiveness**



1. Weighted external value of the DM against currencies of 41 industrial countries.
Source: OECD.

Higher competition in network industries – in particular the telecommunication sector – helped to contain inflation.

Increasing import and oil prices have had a direct effect as well as an indirect effect on inflation through an increase in producer price inflation. This is particularly so for the manufacturing sector, where producer prices increased at around twice the rate of the consumer price index (Figure 5). In addition, expectations of rising inflation have increased in 1999, but were partly reversed in 2000. With import prices in winter 2000 being more than 8½ per cent higher than a year ago, the inflation outlook appeared sensitive to additional negative external supply shocks, although this risk faded with the euro recovery and oil price declines in end-2000.

Despite some increases in policy rates, monetary conditions remain conducive to growth in Germany

Monetary policy tightened over the review period. The European Central Bank undertook a series of gradual interest rate increases between November 1999 and October 2000 in response to increasing inflation pressure emerging from the economic recovery in the euro zone as well as to prevent additional inflation pressures to emerge from the higher oil and import prices. The combined increases amount to 225 basis points for all policy rates, leaving their structure unchanged (Figure 7,

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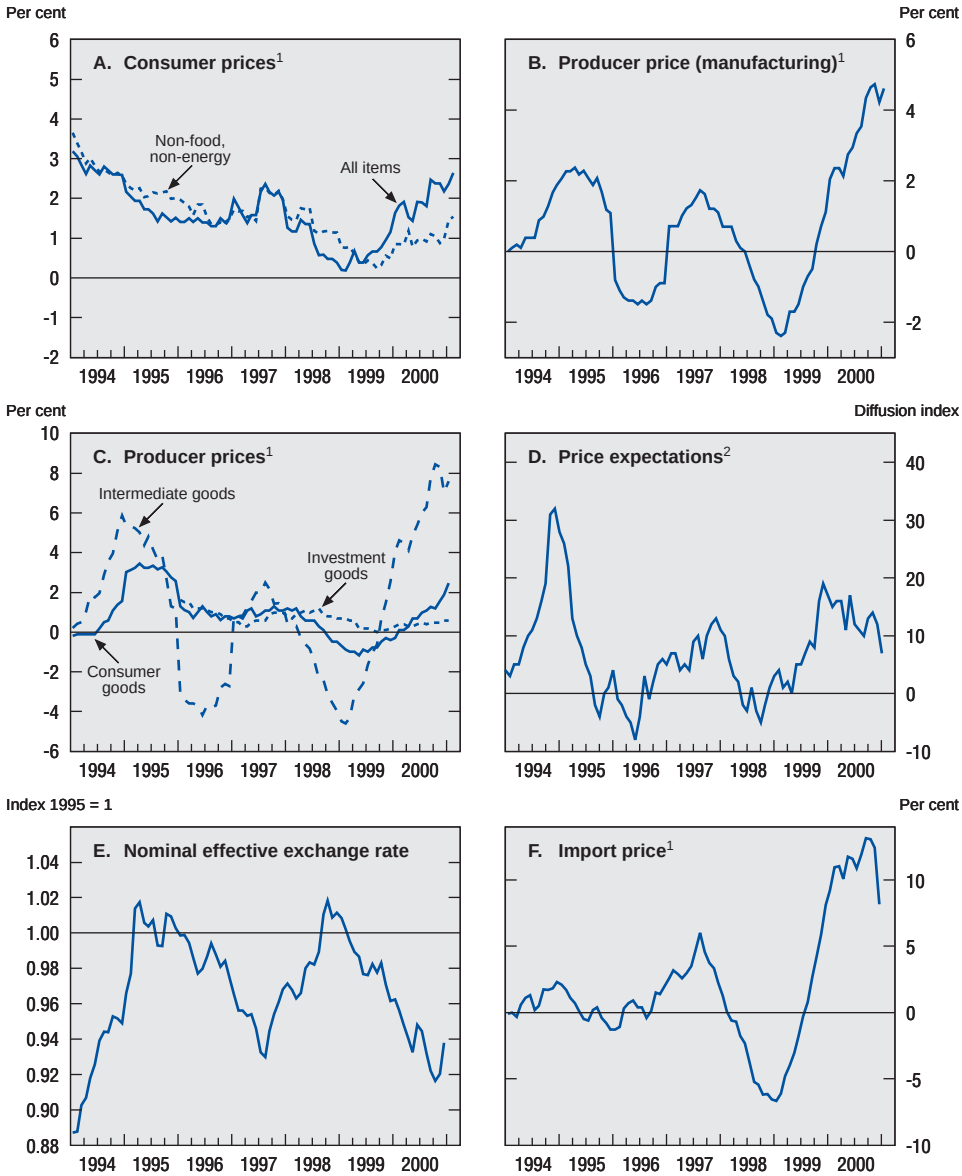
Figure 6. The inflation environment



1. Percentage change over 12 months.
2. Balance of higher and lower expectations. Seasonally adjusted.

Source: Ifo Institut für Wirtschaftsforschung; Deutsche Bundesbank and OECD, *Main Economic Indicators.*

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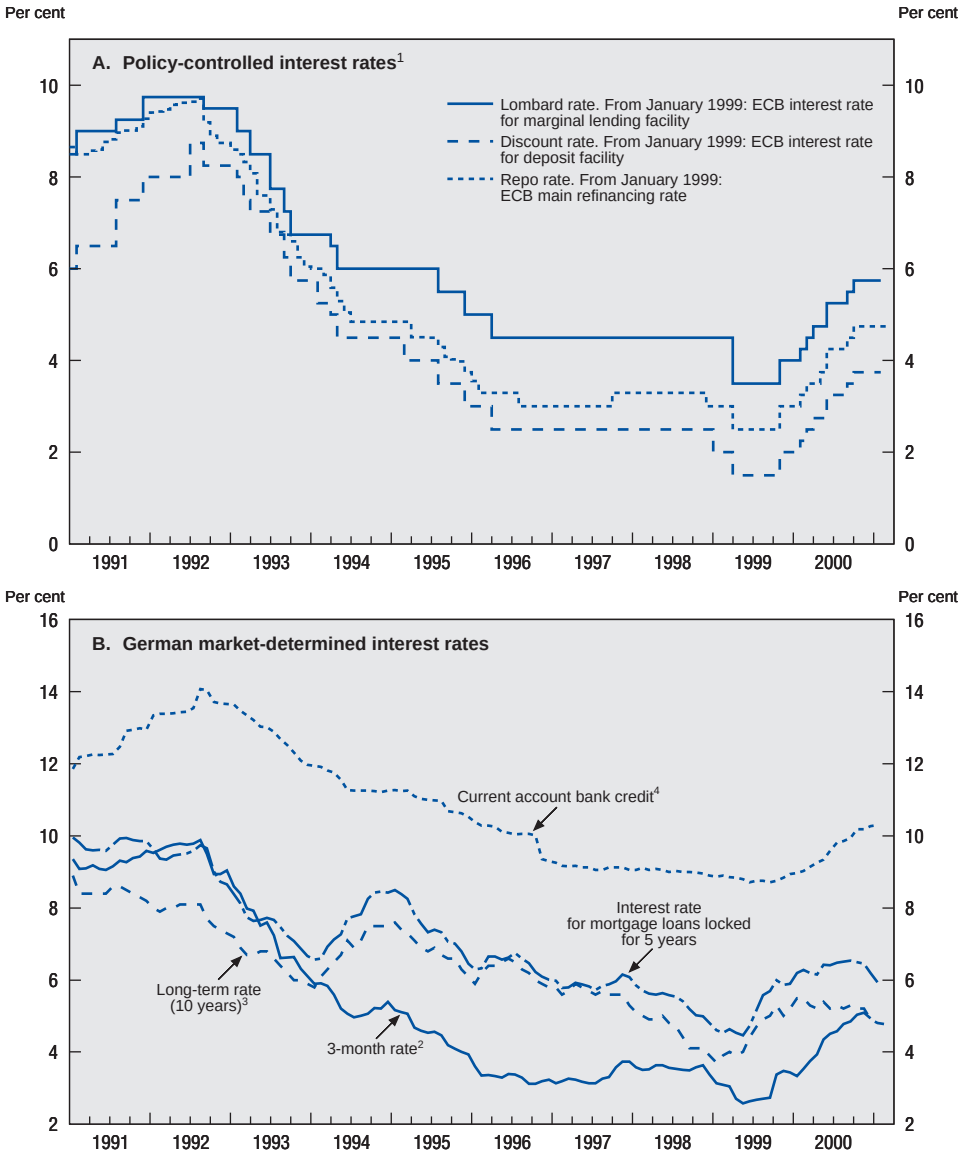
Figure 7. **Interest-rate developments**



1. End of period.
2. Prior to January 1999: FIBOR, thereafter: Euribor.
3. Yield on government bonds outstanding.
4. Credit of less than DM 1 million.

Source: Deutsche Bundesbank.

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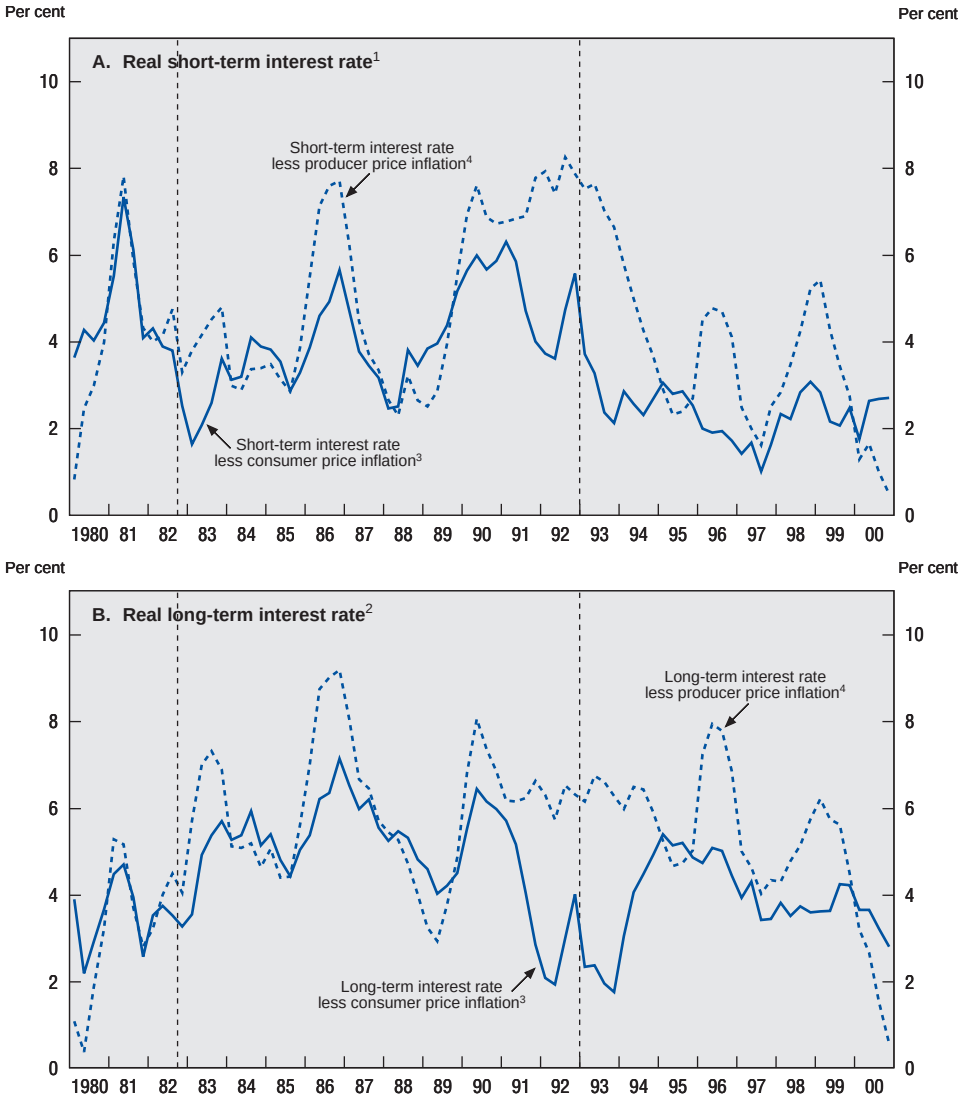
Figure 8. **Real interest rates**
Quarterly average in per cent



The vertical lines indicate the throughs.

1. 3-month interbank rate. Prior to January 1999: FIBOR, thereafter: Euribor.
2. Until 1990 7-15 year German public sector bonds, from 1990 yields on listed federal securities with residual maturity of 9 to 10 years.
3. As measured by the 4-quarter percentage change in the consumer price index.
4. As measured by the 4-quarter percentage change in the producer price index.

Source: Deutsche Bundesbank and OECD, *Main Economic Indicators.*

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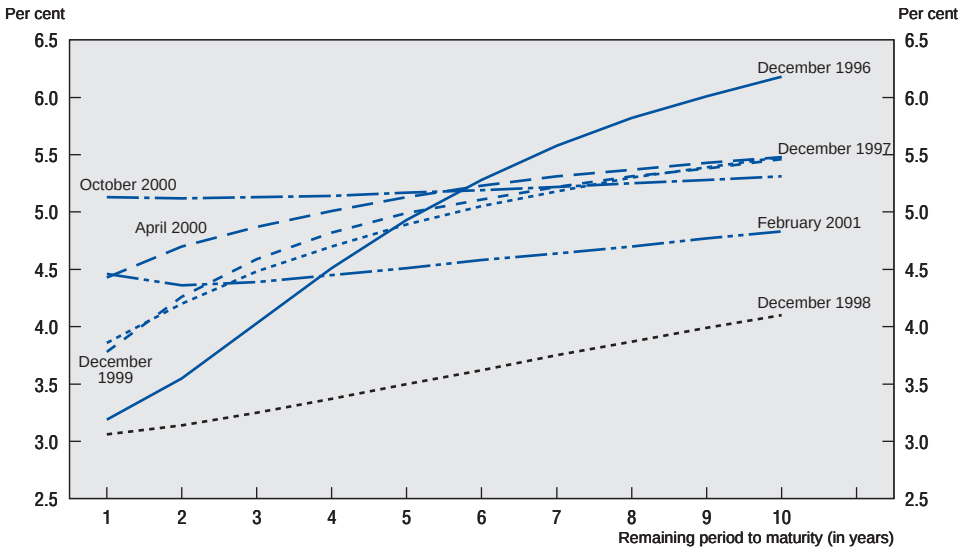
panel A). Long-term interest rates, as measured by 10-year government bonds, remained virtually unchanged at around 5¼ per cent during the period of monetary tightening and fell to 4¾ per cent in the beginning of 2001 (Figure 7, panel B), indicating subdued inflation expectations in view of the monetary policy action. Moreover, the interest rate differential *vis-à-vis* similar US government bonds has changed little over the period.

Rates on instruments with shorter maturity increased in reaction to the higher policy rates (Figure 8). But real long-term interest rates have come down to levels below those in the second half of the 1990s. On the other hand, the declines in the German stock market in 2000 indicate that equity financing has become less favourable.

The yield curve has flattened since mid-1999 and became almost horizontal in autumn 2000 before it shifted downwards in end-2000, leaving rates on instruments with maturities longer than three years below their levels a year earlier. Overall, this should have a stabilising impact on economic activity, despite higher short-term interest rates, as longer-term interest rates are relatively more important in German credit markets (Figure 9). In addition, the real effective exchange rate has depreciated over the past year (despite the appreciation of the

Figure 9. **Yield curves**



Source: Deutsche Bundesbank.

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euro in end-2000). A cut in short-term interest rates also underlies the present OECD projections. Thus, broadly defined monetary conditions in Germany should not act as a brake for GDP growth.

Short-term prospects and risks

Growth should decelerate this year, but recover in 2002

With world trade growth expected to slow this year, the net contribution to growth of the external sector seems likely to decline. In 2002 export growth should strengthen again, reflecting the projected recovery in world trade. Robust domestic demand is likely to support growth both this year and next. Private consumption is projected to accelerate significantly in 2001, based on rising real disposable incomes induced by the income tax reductions and falling energy and decelerating import prices. Declining unemployment will also support consumption. Investment in machinery and equipment is projected to continue growing strongly benefiting from tax cuts (despite a tightening of depreciation rules), continued high capacity utilisation in manufacturing, lower interest rates and the favourable investment climate associated with wage moderation embodied in collective agreements that are largely fixed until spring 2002. Construction, however, is projected to remain in recession this year, and is unlikely to contribute to growth in 2002. This reflects the ongoing downsizing of the construction sector in the new states, high capacities in residential construction and weak demand from the public sector. All in all, GDP growth is projected to decelerate to 2¼ per cent in 2001, before recovering moderately to 2½ per cent in 2002. Employment will continue to grow although at a slower path. This – together with a slowing in labour force participation – will reduce the unemployment rate from 7.8 per cent in 2000 to 6¾ per cent next year. With energy prices declining, inflation is likely to decrease in the course of 2001.

Risks to these projections appear balanced so long as global economic activity does not weaken more than expected: continuing declines in unemployment might increase wage pressures, which could in turn dampen investment and growth. On the other hand, tax reductions might eventually induce a stronger private consumption than assumed in the projections, and generate stronger growth.

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II. Fiscal policy

The government introduced a multi-year savings package in June 1999 in order to set the deficit on a declining path in accordance with the medium-term consolidation plans. The need for such a package had become necessary in the light of a number of new demands perceived to weigh heavily on general government budgets, notably a scheduled re-design of the family support system and significant revenues foregone associated with ambitious tax reform plans. The first part of this chapter reviews recent budgetary developments. and assesses the scope for continued fiscal consolidation in the context of slowing, but still robust growth, and against the major income and business tax reform to be implemented. Future spending restraint appears necessary, and this should be linked to structural reform in several fields. Finally, the chapter highlights major structural policy issues which will shape the stance of fiscal policy in the medium-term, namely tax reform, pension reform, health reform and the reform of the inter-governmental fiscal equalisation system.

Progress in fiscal consolidation

The 1999 *and* 2000 *budgets*

The general government deficit (Maastricht Treaty definition) fell to 1.4 per cent of GDP in 1999, the reduction totalling 0.7 percentage points (Table 5). All territorial authorities benefited from higher-than-projected tax revenues. Higher corporate and assessed income tax receipts largely reflected recovering profits and the lagged effects from phasing out special depreciation allowances for the new states. Indirect tax receipts also rose strongly, but most of this increase is attributable to the introduction of the energy tax in 1999 and spill-over effects from the 1998 increase in VAT by 1 percentage point. Receipts from the energy tax have been channelled into the pension system, matched by a lowering of pension contribution rates. Consolidation occurred from the spending side as well as with Federal Government consumption and interest outlays lower than budgeted (Table 6).[6] In addition, further downsizing of government employment and relatively low wage increases relieved the budgets of all levels of government, in particular those of the states and the communities.

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Table 5. **Public sector financial balances**
DM billion

	1995[1]	1996	1997	1998	1999	2000[2]
General government						
(Maastricht basis)	–116.8	–122.5	–99.3	–77.8	–55.0	58.3
Excluding UMTS proceeds)						(–41.1)
Per cent of GDP	–3.3	–3.4	–2.7	–2.1	–1.4	+1.5
(Excluding UMTS proceeds)						(–1.0)
of which:						
Territorial[3] (authorities)	–109.4	–121.9	–95.2	–54.4	–54.9	–65½
Federal government	–50.5	–78.5	–63.4	–56.6	–51.3	–46½
State government west[4]	–28.6	–32.3	–24.9	–17.6	–11.7	–12½
Local government west[4]	–12.4	–6.1	–4.7	5.7	5.0	2
State government east[4]	–16.4	–15.0	–13.8	–8.9	–7.2	–5
Local government east[4]	–1.9	–2.6	–1.4	–0.8	–0.4	0
Federal funds[5]	0.5	12.6	13.1	23.8	10.6	–3½
National accounts adjustments[6]	7.4	12.5	–6.0	–28.6	–10.9	+19½
Social security	–14.9	–13.1	1.9	5.1	10.8	+5.1
Cyclically-adjusted, per cent of potential GDP	–2.7	–2.4	–1.6	–1.1	–0.4	–0.5

1. 1995 figures do not include (as a capital transfer item) the take-over by the federal government of debts of the Treuhand (DM 204.6 billion) and of the East German Housing Fund (DM 30 billion). Inclusion of these items would imply a general government deficit in 1995 of the order of 9 to 10 per cent of GDP.
2. Preliminary data and Government estimates.
3. Until 1997 including hospitals. The sum of the individual items differs for some years from the summary entry for the territorial authorities due to discrepancies in statistical reporting.
4. West including Berlin-West; east including Berlin-East.
5. Economic Recovery Programme Fund, Financial Equalisation Fund, Compensation Fund, Railway Fund, Coal Fund, German Unity Fund, Inherited Debt Fund.
6. Including lending operations, privatisation receipts, timing adjustments, and, until 1995, profits paid by the Bundesbank into a debt sinking fund.

Source: Ministry of Finance, Finanzbericht and submission; Deutsche Bundesbank; *Monthly report*; and OECD.

The communities also further reduced their investment spending, in particular in the new Länder. The surplus of the social security system roughly doubled in 1999, but this was largely due to discretionary measures affecting the pension system requested to replenish its statutory reserves. In contrast, the long-term care insurance system produced a deficit, for the first time since its introduction in 1995. Gross debt remained roughly constant at around 61 per cent of GDP (Maastricht Treaty definition), helped by privatisation receipts totalling 0.6 per cent of GDP.

In 2000 the federal deficit declined by 0.1 per cent of GDP, net of the receipts from the UMTS mobile phone licence sales. Spending was slightly lower than foreseen in the budget, which was relieved by lower interest outlays resulting largely from debt restructuring. Unemployment-related outlays declined due to

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Table 6. **The Federal Budget**
DM billion

	1998 Outcome	1999		2000		2001 Budget
		Budget	Outcome	Budget	Outcome	
Expenditure	456.9	485.7	482.8	478.8	478.0	477.0
(Per cent of GDP)	(12.1)	(12.5)	(12.5)	(12.0)	(12.0)	(11.5)
of which:						
Consumption	90.7	94.7	92.7	93.8	92.8	85.9
Interest payment	56.2	81.5	80.4	78.5	76.6	77.0
Investment	13.5	14.1	13.9	13.9	13.2	14.0
Transaction and lending	298.5	296.5	295.8	293.4	295.5	301.2
of which:						
To other administrations	78.8	48.6	53.0	48.8	51.5	45.7
Others	217.7	247.9	242.9	244.5	244.1	255.6
Revenues	400.3	432.1	431.5	429.2	431.3	433.2
(Per cent of GDP)	(10.6)	(11.1)	(11.1)	(10.8)		(10.5)
of which:						
Taxes	341.5	371.7	376.3	387.5	388.8	384.6
Others	58.8	60.4	55.2	41.7	42.7	48.5
Financial balance	–56.6	–53.6	–51.3	–49.6	–46.7	–43.9
(Per cent of GDP)	(–1.5)	(–1.4)	(–1.3)	(–1.2)	(–1.2)	(–1.1)

Source: Ministry of Finance and OECD.

unemployment falling in the course of the labour market recovery and tightened eligibility conditions for unemployment assistance benefits. On the other hand, transfers to the pension system increased further reflecting both rising receipts from the eco tax affecting the pension financing and larger budgetary contributions for child-raising periods of the insured. The budget was also subject to unforeseen spending such as compensation for forced labour during the Second World War and compensation paid to households for higher heating expenses. While revenues were budgeted to decline because of low perceived income tax receipts (partly due to the 2000 tax reductions) as well as declining privatisation proceeds, they turned out to roughly maintain the 1999 level, mainly because of higher-than-expected income tax receipts. Revenues stemming from the selling of shares in the Telekom and the Post Office were transferred from the Federal Government to the Post Funds, following an earlier arrangement concerning the funding of the Post Office's pension liabilities.

The states' deficit increased in the west, but declined in the east. While the states benefited from higher-than-expected tax revenues, spending in the new states rose less than in the west, due to larger reductions in personnel and contin-

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ued cuts in investment. For similar reasons, surpluses shrunk for the communities in the west while in the new states they broke even. The surplus in the social security funds deteriorated in 2000 as the surplus in the pension insurance declined strongly in comparison to 1999 when the reserves had to be replenished to the statutory minimum level. The long-term care insurance and the health funds have been more or less in balance. Revenues of both the pension and health care system have benefited from the extension of the obligation to pay social security contributions to "small hours" jobs.

With the extra revenues from the sales of the universal mobile phone licences, of the order of 2.5 per cent of GDP, being counted as affecting the budget balance, the general government moved to a surplus of 1.5 per cent of GDP (see Box 2). But net of these receipts the deficit would have improved by only 0.4 percentage points and the structural (cyclically-adjusted) balance remained roughly unchanged. One-third of the proceeds from the auctioning of the mobile phone licences are being used for debt redemption in 2000 and two-thirds in 2001, thus generating savings on interest paid in these and the following years. Privatisation receipts totalled 0.3 per cent of GDP. Overall, the gross debt-to-GDP ratio declined by 1.1 per cent.

Box 2. The fiscal impact of the sale of UMTS licenses

The auctioning in 2000 of the third generation mobile-phone licenses (UMTS), valid for 20 years, yielded proceeds of nearly DM 100 billion, equivalent to 2½ per cent of nominal GDP. Following guidelines from the Eurostat, the proceeds enter the public finances in 2000 as one-off income from the sale of a non-tangible asset, implying a surplus of general government balances in the order of 1½ per cent of GDP. Reflecting the one-off nature of the transaction, there will be no correspondent improvement in the structural balance. The government has decided to utilise the proceeds to repay government debt, with one third allocated in 2000 and the rest the following year. Annual interest savings implied by the debt redemption total some 0.1 per cent of GDP. Until 2004 the government plans to spend about four-fifths of these interest savings on projects considered to be of high importance for maintaining and further developing the growth potential of the economy: infrastructure of the railways system (mainly transfers to the German Railways), education and R&D. No spending decisions have been made beyond this time horizon. Alternatively, the interest savings could be used for further debt redemption. A full assessment of the budgetary value of the receipts would also take into consideration that the outlays of the auctions in 2000 *ceteris paribus* imply tax revenues foregone in the future. To what extent these will materialise depends on future induced profits associated with the licence sales.

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The 2001 *budget*

General government finances in 2001 will be heavily affected by the phasing in of the major 2001 step of the income and business tax reform. According to the official tax estimates, this will imply revenues foregone on a cash basis of DM 45 billion – more than 1 per cent of GDP – on top of the tax relief stemming from the earlier steps of the tax reform in 1999 and 2000 (see Table 10 below).[7] The federal budget for 2001 envisages a reduction of the deficit by DM 6 billion (more than 0.1 per cent of GDP) relative to the 2000 budget plan (and net of the receipts from the UMTS licence sales in 2000) (Table 5). While tax receipts are set to decline, other revenues, notably from privatisation, are scheduled to increase

Table 7. **Appropriation account for general government**
Including social security

	1998	1999	2000	2001	2002
	DM billion				
Current receipts[1]	**1 656.4**	**1 724.5**	**1 765.3**	**1 766.5**	**1 832.4**
of which:					
Direct taxes	434.5	463.9	496.6	453.0	481.9
Social security contributions	727.7	733.6	741.9	757.3	774.8
Indirect taxes	437.5	472.1	478.8	498.3	521.8
Sales and property income	56.7	54.9	48.0	57.9	53.9
Current disbursements[1]	**1 696.3**	**1 737.9**	**1 760.5**	**1 785.3**	**1 827.4**
of which:					
Government consumption	722.4	738.0	750.3	761.5	772.4
Property income payable	136.1	137.3	131.7	131.0	136.0
Subsidies	69.4	67.6	67.3	66.7	66.4
Social security outlays to households	712.9	731.9	743.3	759.0	780.1
Other current transfers	53.7	63.1	67.9	67.2	72.5
Saving, net	**–39.9**	**–13.4**	**4.8**	**–18.8**	**5.0**
Net capital outlays	**37.9**	**41.6**	**–53.5**	**51.2**	**55.5**
Gross investment	68.6	71.4	70.9	71.5	72.2
Net capital transfers received	–33.3	–34.4	58.8	–45.5	–49.0
Consumption of fixed capital	64.0	64.2	65.6	65.8	65.8
Net lending	**–77.8**	**–55.0**	**58.3**	**–70.0**	**–50.5**
	Per cent of nominal GDP				
Memorandum items:					
Net lending, Maastricht basis	–2.1	–1.4	1.5	–1.7	–1.2
Cyclically-adjusted net lending (potential GDP)	–1.1	–0.4	–0.5	–1.3	–1.0
Financial liabilities	63.0	60.6	59.7	58.0	57.7

1. The grouping shown here is not that utilised by the Federal Statistical Office. The data is subject to revision.
Source: Federal Statistical Office for raw data and OECD.

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so as to contain the overall decline in revenues.[8] Federal expenditures are budgeted to decrease slightly in nominal terms and government consumption is set to decline as well. Interest savings generated by the debt redemption on account of the UMTS license sales are being spent mainly on rail and road infrastructure projects and partly for other purposes, notably research and education. Transfers overall are set to decline, with major savings being expected from the projected reduction in unemployment. But transfers to the pension system will continue to rise, partly linked to further legislated increases in the eco tax (see below). Other social transfers are also scheduled to expand. This is true for housing benefits but also concerns extended tax allowances for commuting costs (Table 5). Other risks include compensating transfers on account of the BSE crisis.

The tax reductions will also significantly affect the budgets of the states and communities, which receive a share of 42.5 and 15 per cent of the income tax receipts, respectively (prior to further inter-governmental equalisation transfers). It is likely that the states and especially the communities will respond by further cutting their investment outlays. Spending for personnel, which is the biggest item of government consumption, in particular for the states and communities, will be relieved by relatively modest wage agreements for the public sector, and all levels of government are planning to continue downsizing their staff. Nevertheless, fiscal balances of the states and communities are likely to deteriorate, and for the general government as a whole the Secretariat projects that the deficit will increase in 2001 to 1.7 per cent of GDP, ¾ percentage point higher than the deficit in 2000 net of the receipts from the UMTS sales. This will be roughly in line with the deficit target laid down in Germany's Stability Programme (1½ per cent of GDP). Nevertheless, with the public debt redemption associated with the UMTS receipts being distributed over 2000 and 2001, debt will decline further this year. Consolidation will resume in 2002, with the structural deficit projected to improve slightly. Debt is projected to fall to some 58 per cent by 2002 (Maastricht definition).

Medium-term objectives

The Federal Government has reinforced its commitment to continue fiscal consolidation both in the German Stability Programme to the EU Commission and in the medium term Finance Plan presented to the Parliament (*Deutscher Bundestag*). These programmes foresee a balanced budget on the general government level by 2004. According to the official projections the federal budget will still be in a deficit in 2004, but the government aims at balancing it two years later. Further fiscal targets until 2005 concern reductions in the debt, tax and spending to GDP ratios (Table 8).

While tax receipts in 2000 came in higher than expected the implementation of the income and business tax reform in 2001 in combination with further tax relief to be phased in successively until 2005 will imply a significant drop in the

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Table 8. **Medium-term fiscal objectives of the general government**
Per cent of GDP

	1999	2000	2001	2002	2003	2004
Expenditure without UMTS	48.6	48.0	46.5	45½	44½	44
Expenditure with UMTS (treated as a negative capital transfer)		45½				
Tax and social charges	43.1	43.0	41	41	40½	40½
Budget deficit	–1.4	–1.2[1]	–1½	–1	–½	0
Gross debt	61.1	60.0	58.0	57½	56½	54½

1. Excluding UMTS proceeds.
Source: Ministry of Finance.

tax-to-GDP ratio in the years to come. Moreover, as was outlined in last year's *Survey*, the government introduced in mid-1999 a comprehensive programme specifying measures of spending restraint, although some of these effectively shift the savings burden from the federal to other layers of the general government. Restraining the growth of pension outlays and cutting expenses for personnel are major elements of this programme, and these have already been incorporated partly into the 2000 budget. However, there are risks that additional restraint measures will be needed to achieve the consolidation targets. As highlighted above, social transfers are set to expand. While the pension reform is progressing, there is a risk that implied savings might be less than envisaged (see the section on pensions further below). Moreover, following a ruling of the Constitutional Court, unemployment insurance benefits will be related to a more broadly defined wage base effectively increasing outlays. Since 1999 the pay-as-you-go long-term care insurance produces deficits and, in view of its benefits having been extended, future deficits appear to be likely. With the health care system still under investigation for further reform, there is also a risk of future spending over-runs in the health funds and associated increases in social security contributions. Moreover, fiscal risks are also associated with the revenue side. To some degree the tax reductions are likely to self-finance themselves, given the positive incentive effects associated with the reform. But larger-than-expected revenue losses associated with already implemented or forthcoming income tax reductions and increases in family benefits cannot be excluded, given the large degree of uncertainty surrounding such assessments. This is particularly true in view of the fact that relative to original plans tax reductions have been accelerated and additional tax relief has been legislated for 2005 (see below).

Caution is also warranted with respect to the growth assumption underlying the medium-term consolidation plans. It stipulates annual real GDP growth of 2.5 per cent on average until the middle of the decade. However, there is a risk that this rate might not be attained. Over the last decade annual real growth

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totalled 1.6 per cent on average, and estimates by the Secretariat suggest that potential output might at present be growing at a rate of around 2 per cent. Various factors suggest that the growth of potential has increased recently and may rise further over the next years to come. Labour markets appear to be more flexible today than they were only a few years ago owing to more flexible collective bargaining arrangements, product market competition has increased in the network industries, and the income and business tax reforms can also be expected to raise potential output. Hence, potential output growth over the next half a decade might well be significantly higher than 2 per cent. On the other hand, the shrinking of the working-age population that will occur in Germany over the coming years and decades would substantially suppress potential output growth if it were not balanced by significant increases in labour force participation, and it is not clear that those will indeed materialise (see the sections on pensions further below). Moreover, with respect to cyclical movements in economic activity, the short-term projections by the Secretariat suggest that real growth will decline over the next two years, to 2¼ per cent this year. While a slight recovery is projected for 2002, it is not clear yet in what direction the cycle will move thereafter. On balance, it appears that assuming average real growth of 2½ per cent into the second half of the decade is surrounded by considerable risks and may be on the optimistic side. Failure to achieve such growth, however, would put the government's medium-term consolidation targets at risk.

Hence tight control of spending is imperative if consolidation targets are to be met. As has been suggested in the 1999 S*urvey* spending restraint needs to be combined with further structural reform to identify economically important priorities for savings. Subsidies and tax expenditures have declined somewhat over the last years, but are still estimated to total around 8 per cent of GDP, being prime candidates for substantial cuts.[9] More effective control of social spending and measures to improve public sector efficiency are also required. Major issues in this respect are addressed further below.

Tax reform

Income and business taxation

Personal income and business tax reform in Germany is being phased in since 1999, based on various reform laws. In summer 2000 the centrepiece of the reform, the "Tax Reduction Law" (S*teuersenkungsgesetz*) has passed both houses of parliament. This has effectively terminated a controversial debate on income tax reform in Germany that extended over the second half of the 1990s. The major features of the reform law were already discussed in the last S*urvey*. However during the mediation process between both houses of parliament some important elements were modified or added, and these are considered in turn. Box 3

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Box 3. The major components of tax reform

Income and business tax reform measures are being phased in as from 1999, and on present legislation changes in income tax parameters will continue until 2005. The measures have often been legislated in separate laws and consecutive steps. There is also a phased introduction of ecological taxation (electricity and fuel). The box summarises important tax measures legislated since 1999.

I. Personal income taxes

- The bottom statutory tax rates are being reduced from 25.9 per cent in 1998 to: 23.9 per cent (January 1999); 22.9 per cent (January 2000); 19.9 per cent (January 2001); 17 per cent (January 2003) and 15 per cent (January 2005).
- The top statutory tax rate is being decreased from 53 per cent in 1998 to 51 per cent (January 2000); 48.5 per cent (January 2001); 47.0 per cent (January 2003) and 42 per cent (January 2005). Until 2000 entrepreneurial income is subject to lower statutory top income tax rates, but this will be abolished in 2001 (see below under "partnerships").
- The basic tax-free allowance (*Grundfreibetrag*) is being increased in annual steps (except in 2002 and 2004) from DM 12 300 in 1998 to DM 15 000 in 2005. Benefits and tax credits for dependent children are being raised in 2000 and 2002 (the latter increase not yet legislated).

II. Business taxation

***Incorporated companies*:**

- Reduction of the tax rate to 25 per cent for both retained profits (from 40 per cent) and distributed profits (from 30 per cent) – in 2001.
- At present the corporate tax system is fully integrated with the personal income tax: domestic shareholders receive a tax credit for the tax on distributed profits paid by a company so that profits are effectively taxed at the personal income tax rate. Because the tax credit is not available for non-residents who are not liable to pay income tax in Germany, this system has been criticised for treating residents and non-residents in an unequal way. In 2002 the imputation will be abolished and companies and shareholders will be treated as entirely separate entities. Profits are taxed at 25 per cent on the company level. As regards distributed profits, 50 per cent will enter the shareholder's personal income tax base and will be taxed at their personal tax rate.
- Capital gains tax for equity participation of corporations in other domestic incorporated companies will be abolished – in 2002.

***Non-incorporated partnerships*:**

- The top statutory tax rate is lowered from 47 per cent in 1998 to 45 per cent (January 1999); and 43 per cent (January 2000). From 2001 on the preferential statutory tax rates for entrepreneurial income will be abolished.

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Box 3. **The major components of tax reform** (*cont.*)

- Deductibility of the local business tax (1.8 times the basic tax) from the income tax liability – in 2001.
- Increase of the tax-free allowance for selling the firm from DM 60 000 to DM 100 000 – in 2002. For entrepreneurs aged 55 or older only half of the personal income tax rate will be applied (once in a lifetime) – reintroduction in 2001.

***Financing measures comprise* inter alia:**

- A number of measures have been implemented in 1999 to broaden the tax base of business income including changes in the ways inventories are valued, discounting of reserves against future liabilities, and in the way tax free reserves are accumulated in the insurance and energy sectors. In the last two sectors tax-free reserves will need to be unwound in coming years.
- Lowering the threshold for full taxation of capital gains from selling stakes in incorporated companies to a share of 1 per cent or more in the company's capital, from 10 per cent or more (except for the participation of corporations in other corporations, which are exempted from capital gains tax, see above) – in 2002.
- Reduction of the maximum declining-balance depreciation rate for investment in machinery and equipment to 20 per cent (from 30 per cent); reduction of the linear depreciation rate for buildings to 3 per cent (from 4 per cent) – in 2001.
- Extension of tax depreciation schedules to assessed economic life times – in 2001.

III. Ecological tax reform – legislated

New taxes on energy consumption, implemented in April 1999 (DM 8½ billion revenues in 1999) are used to lower pension contribution rates. The taxes have been introduced in April 1999 and are being increased in annual steps between 2000 and 2003 (DM 22.3 billion revenues in 2001).

summarises the major changes in income and business taxation that are being phased in between 1999 and the final stage of reform in 2005.

As regards the personal income tax, reductions in taxes, which originally were scheduled for 2002, have been brought forward into 2001. Moreover, statutory income tax rates scheduled for the final stage of reform will be reduced to below the levels originally planned. Between 2000 and 2005 the statutory entrance tax rate will decline by 7.9 percentage points to 15.0 per cent, and the top statutory rate by 9 percentage points to 42.0 per cent (Figure 10).[10] Overall, this will

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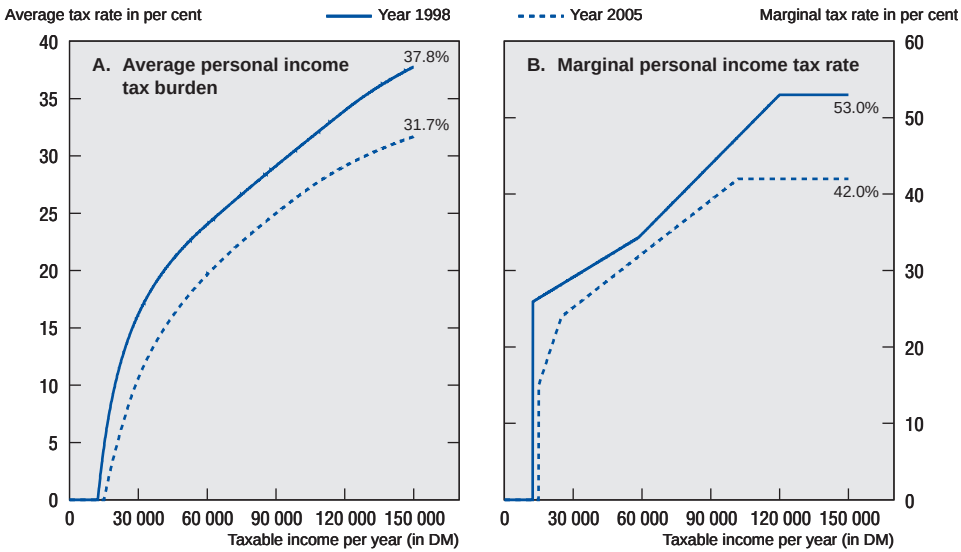
Figure 10. **Statutory income tax rates before and after reform**



Source: Federal Ministry of Finance.

place Germany in the middle range of statutory income tax rates observed in the OECD, down from the group of countries with the highest rates to which it belonged until recently (Table 9).

On the business side, the law proposal foresaw an option for non-incorporated business to be assessed as if they were incorporated. This provision aimed to compensate for the increasing (with the level of profits) spread between the top personal income tax rate on the one hand – relevant for retained and distributed earnings of partnerships – and the flat 25 per cent final corporate tax rate relevant for incorporated business. However, such an option proved to be technically complex and would have been associated with a considerable degree of uncertainty on the side of unincorporated business. It was thus dropped in the parliamentary mediation process. As a substitute, partnerships can now deduct from their income tax liability a standardised amount of their local business tax liability. While this provision will effectively imply a more equal taxation of profits of partnerships and corporations, there are drawbacks associated with this approach.[11]

Another new feature is the abolition of capital gains taxes for equity participations held by incorporated business from 2002 on. Allegedly, up to now capital gains taxation has prevented significant resales of equity participations that

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Table 9. **Statutory tax rates within the OECD**

Per cent

	1986	1991	1999
United States	46.0	34.0	35.0
Japan	43.0	38.0	34.5
Germany[1]	**56.0-36.0**	**50.0-36.0**	**40.0-30.0**
Applicable from 2001 onwards			25.0
France	45.0	34.0-42.0	33.3
Italy	36.0	36.0	37.0
United Kingdom	35.0	34.0	31.0
Canada	36.0	28.0	28.0
Australia	49.0	39.0	36.0
Austria[2]	50.0	30.0	34.0
Belgium	45.0	39.0	39.0
Denmark	50.0	38.0	32.0
Finland	33.0	23.0	28.0
Greece[3]	49.0	46.0	35.0-40.0
Iceland	51.0	45.0	30.0
Ireland	50.0	43.0	28.0
Luxembourg	40.0	33.0	30.0
Mexico[4]	39.2	35.0	35.0
Netherlands	42.0	35.0	35.0
New Zealand	45.0	33.0	33.0
Norway	28.0	27.0	28.0
Portugal	42.0-47.0	36.0	34.0
Spain	35.0	35.0	35.0
Sweden	52.0	30.0	28.0
Switzerland[5]	4.0-10.0	4.0-10.0	7.8
Turkey	46.0	49.0	25.0
Average basic rate[6]	43.5	36.3	32.9
Range (maximum to minimum)[6]	28.0	27.0	15.0
Standard deviation[6]	7.0	6.8	4.0

Note: This table represents the rates of central government corporate income tax. Austria, Canada, Finland, Germany, Italy, Japan, Norway, Portugal, Switzerland and the United States also have sub-central corporate taxes. Rates rounded to nearest percentage point. Many countries also have special rates for firms with lower profits and for particular sectors.

1. First figure = retained; second figure = distributed earnings. In 1995, a "solidarity" surcharge was introduced on the corporate income tax. It is not included in the figure shown in the table. The local business tax is also not included.
2. 1986 data are national data.
3. The last column applies to 1997 instead of 1999.
4. Mexican corporate tax data for the first year considered are from national sources and concern 1988 instead of 1986.
5. The figure for 1998 is a net rate. The federal tax is deductible from its own base.
6. Excludes Switzerland, takes into account the rates on retained earnings for Germany (respectively 50 per cent in 1991 and 45 per cent in 1998), 34 per cent for France in 1991 (rate applicable from the second part of the year onwards), 40 per cent for Greece in 1997 (rate applying to non-listed corporate profits), and 42 per cent for Portugal in 1986.

Source: Owens and Whitehouse (1996). The OECD Tax Data Base (2000).

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Table 10. **Financial effects of the 2001 income and business tax reform**[1]
DM billion
– = tax relief; + = tax burden

	Revenues expected in year (financial base)					
	2001	2002	2003	2004	2005	2006
Business taxation						
Tax relief	–23.3	–34.8	–37.1	–35.6	–35.6	–35.8
Financing measures[2]	+6.3	+14.3	+21.6	+23.7	+24.3	+21.4
Net tax relief	–17.0	–20.5	–15.5	–11.8	–11.2	–14.4
Income tax schedule	–28.4	+1.3	–13.8	–13.2	–51.5	–52.8
Net tax relief, total	–45.4	–19.2	–29.3	–25.1	–62.8	–67.2

1. Cash basis.
2. Including the adaptation of depreciation tables to economic life spans.
Source: Ministry of Finance and OECD.

have been acquired in the past by banks, other financial intermediaries and business in general. One might therefore expect a major reshaping of the ownership structure of German business after 2002, and in general this should improve the efficiency of capital allocations. For the same reason, inwards foreign direct investment should also expand. The tax exemption is not valid for partnerships, however, implying a non-neutrality with respect to the legal form of the enterprise. Further action to rectify this inertia should therefore be considered (Table 10).

Recent energy tax measures are distortionary

With oil prices increasing strongly in 2000, demands were mounting to give up further increases in energy tax rates that were to be phased in according to current legislation in three steps between 2001 and 2003. While the current legislation remained unchanged and eco taxes will be raised as scheduled, the government responded to the demands for income compensation by introducing transfers and tax allowances related to heating oil and fuel consumption. One-off grants for the purpose of relieving heating expenses in the winter period 2000/2001 are given to low-income families. This support is proportional to the size of the recipient's housing space. Moreover, the distance-related tax allowances for commuting expenses to the workplace, contained in the income tax code, are being increased in size and extended to cover other means of transportation in addition to private cars.

While "ecological taxation" is sensible for fostering sustainable development, the present system of energy taxation is not efficient in reducing greenhouse gas emissions (see the 1999 *Survey*). Hence, energy taxation should not be limited in scope at times when fuel prices are fluctuating upwards, but the system

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should be reformed to better conform with targets to reduce green house gas emissions. Also, the initiatives taken to compensate for income losses associated with fuel price increases are at variance with the principal policy intention to foster sustainable development. Although the grants for heating expenses are planned to be one-off only, they weaken incentives to economise on heating energy to the extent consumers are expecting income compensation on account of energy price increases to be a repetitive exercise. Extending the coverage of commuting allowances to other means of transportation reduces the taxation bias in favour of private car utilisation. However, the subsidisation of commuting expenses in general financially supports traffic generation rather than reducing it. For reasons of environmental protection the commuting allowance should therefore be abolished rather than extended, and this would also be a significant step to reducing the level of subsidies.[12]

Pension reform

Improving fiscal sustainability of the public pay-as-you-go pension system has been a vital part in the government's consolidation programme (*Zukunftsprogramm*) introduced in June 2000. The existing defined-benefit system is already under stress, and the rapid rise in the share of old people in the population expected over the next three decades will ensure that Germany is among a handful of countries within the OECD with very large increases in public pension spending. According to recent population projections by the Federal Statistical Office, the old-age dependency ratio – defined as the ratio of persons aged 65 and older to those aged between 20 and 64 – will rise from 25 per cent in 1999 to 56 per cent in 2040 (Figure 11).[13] Thus, without major reform Germany's public pension system will require substantial increases in pension contribution rates, with corresponding effects on labour costs.

Previous legislation of the preceding government, designed to reduce pension replacement ratios, has not been implemented by the new administration, and the pension adjustment in 2000 was based on consumer price inflation rather than net wages (see the last *Survey*).[14] In January 2001 the first chamber of parliament passed a new reform law, which, if finally passed by both chambers of parliament, is to become effective in spring 2001. Its key features are the introduction of a voluntary funded system, subsidised by the government through grants and tax concession, and a gradual reduction in replacement rates of the pay-as-you-go system. The reform aims at stabilising the contribution rate for the pay-as-you-go system at a level below 20 per cent of the wages until 2020 and not more than 22 per cent until 2030 (from 19.1 per cent in 2001). Replacement ratios are planned to be reduced such that "reference pensions" of new retirees drop from presently almost 70 per cent of the net wage base to between 67 and 68 per cent of the wage base in 2030 while overall replacement ratios – including annuities

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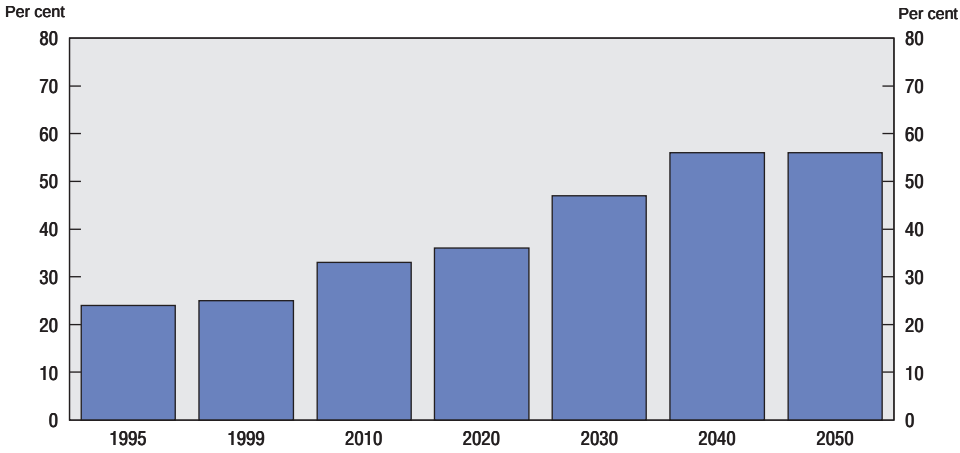
Figure 11. **Projected evolution of the old age dependency ratio**[1]
Per cent



1. Number of persons aged 65 and over as a percentage of number of persons aged between 20 and 65 years. The projections assume a net immigration into Germany of 100 000 persons annually.
Source: Federal Statistical Office.

from the funded part – are projected to rise.[15] While details are still being negotiated in the parliamentary process, the main features of the reform plan are:

– Voluntary contributions to funded pension or annuity plans of all persons obliged to contribute to the statutory pay-as-you-go system are promoted by the government through cash transfers or deductions from the tax base within certain limits, whose level rises with the number of children. The mode of support most favourable for the person will be applied. The maximum amount of financially supported savings increases in three equal steps in 2004, 2006 and 2008. This amount will rise from 1 per cent of the income threshold that caps pension contributions in 2002 to 4 per cent in 2008. The contributions to the pension plans will not be taxed, and the returns will remain untaxed in the accumulation phase but will be taxed in the pay-off phase. Pension plans are either company based or consist of individual schemes. Collective contracts between the social partners can specify the type of pension plans to be chosen. Subsidised pension plans need to guarantee non-decreasing annuities totalling at least the size of the contributions. They are also subject to other conditions that aim at raising the security standard of the investment.

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– Pensions will increase slower in the future than they would in accordance with the old adjustment procedure that was valid until 1999. While in the past pension adjustments were indexed to the evolution of net wages the new adjustment formula will index pensions for new and old retirees to gross wages net of changes in statutory pension contribution rates only. Hence, changes in general taxation (as well as in health and unemployment social security contributions) will leave pensions unaffected. This implies, in particular, that pensions will not rise as a result of the income tax reductions phased in between 2001 and 2005. Moreover, the wage determining the level of pensions will be reduced in 2002, 2004, 2006 and 2008 by the maximum amount deductible from the income tax base. From 2011 onwards pension adjustment will be further dampened by an annual discount from the reference wage path.

– Periods of child raising will be higher valued in determining pension rights. Also, married couples obtain the option of splitting between them their pension claims, which can lead to higher benefits for survivors.

Reducing the pay-as-you-go part of pensions and supplementing it with a funded layer is an important step to improving the fiscal stance of the overall system in view of the significant "population ageing" the German economy is facing, and has been recommended repeatedly in previous *Surveys*. As has been argued there, such a mixed system has the advantage of diversifying the risks associated with a pay-as-you-go system on the one hand (demographic, political risk) and a funded system on the other hand (capital market risk). Also, the measures envisaged are consistent with the goal of reducing non-wage labour costs – at least to the extent that contributions to funded pension plans would effectively be co-financed out of general tax revenues, and more so if the workers' own contributions are perceived by them as saving rather than tax. Restricting the catalogue of subsidised pension plans to those that promise non-decreasing annuities over the life span after the contribution phase helps contain the risk of moral hazard that would arise if contributors were able to cash in accumulated pension savings while claiming social assistance benefits. Indeed, for persons aged 65 and over and disabled persons, the pension reform also introduces a special type of means-tested transfers, which differ from the common social assistance benefits in that the income and wealth of the recipient's children are not subject to the means test. Defining the range of admissible pension funds has been an issue in the policy debate. Portability of pension plans needs to be required if labour market flexibility is to be secured. Moreover, regulation should allow for maximum returns on assets while at the same time securing the long-term solvability of the funds. This might be best achieved by putting more emphasis on prudent regulation of funds' reserve requirements associated with certain investment instruments rather than restricting the range of admissible investment instruments itself.

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Given the long time horizon involved, it is not clear, however, that the measures envisaged will suffice to stabilise contribution rates to the pay-as-you-go system at the targeted rates. There is a risk that the future evolution of employment and labour force participation might turn out less favourable than officially assumed.[16] The reduction in replacement rates appears mild in comparison to the projected dramatic increase in the old-age dependency ratio over the next 30 years.[17] Hence, further action to consolidate the pension system may become necessary and indeed the new legislation commits the government to take action if it appears likely that the targeted contribution rates will not be attained. This could be combined with widening the scope for voluntary pension arrangements (see the 1999 S*urvey*). Such reform would be aided by explicitly linking future pensions to life expectancy, and increasing discounts for short contribution periods, thereby reducing incentives for early retirement. Making part of the funded layer mandatory, rather than giving tax preferences to contributions, would be preferable on fiscal grounds and is worth considering. Unifying taxation rules for different types of pension schemes would be important as well. Moreover, as has been argued in previous S*urveys*, eligibility for invalidity pensions should be set exclusively on health grounds making them independent from labour market conditions.

Health reform

New health reform measures came into effect in January 2000 (G*esundheitsreform* 2000). To contain costs in the health insurance system, spending by hospitals and doctors are subject to budgets. A positive list of reimbursable pharmaceuticals is being developed to support spending restraint. It is scheduled to become effective in 2002. The provision of preventive health care services has been widened. Moreover, the health funds obtained greater powers to negotiate directly with groups of doctors and clinics to supply a package of health services without the need to seek permission of the doctors' association. Contracts between health funds and health providers are allowed to establish a (voluntary) gatekeeper function of general practitioners, via bonuses given to patients who first visit their doctor before going to a specialist. Regarding the reimbursement of hospital services, from 2003 onwards a more performance oriented price structure for will be introduced. It is based on uniform case-fees – "diagnosis related groups" – that will replace the present mixed system of per diems, uniform case fees and procedural fees.[18]

A major goal of the government's health care policies is keeping contribution rates stable. This goal has been reached for 2000 and contribution rates have even slightly declined. The strategy of containing health costs largely rests on the effectiveness of sectoral budgets. Past experience suggests, however, that while spending caps might be effective in containing costs in the short run, they are no

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appropriate means to increase the efficiency of providing health services in a more medium-term perspective (*Germany Surveys* 1997 *and* 1999). Some of the measures implemented, notably the extended scope for contractual relations between health funds and health care providers, the widening of the scope of general practitioners to act as gatekeepers, and the scheduled revision of the remuneration system for hospitals, are steps in the right direction which are likely to produce efficiency gains. But they do not go far enough to build a health care system, which is consistent and sustainable over the medium term. Health funds compete for members, but cannot compete on the basis of services or costs. They should be given wider powers to deal directly with health care suppliers. Investment financing of the hospitals on the one hand and the financing of hospital maintenance and other costs on the other hand will remain divided between the states and the health insurers preserving incentives for the states to create overcapacities. Overall, more fundamental reform is necessary, and reform strategies have been highlighted in the 1997 *Survey on Germany*.

Reform of the inter-governmental fiscal equalisation system

Policy action to reform the system of inter-governmental fiscal equalisation transfers has become necessary for 2002, due to a ruling of the Federal Constitutional Court dating back to the end of 1999. In order to secure roughly equal living conditions across the federation, Germany's fiscal equalisation system transfers tax revenues – after primary tax sharing – from the wealthier to the poorer states, and federal transfers are also directed towards states with below average per capita revenues. The Federal Constitutional Court has demanded to clarify the principles on which such transfers should be based. In a further legislative step the legislator is then required to revise the system in accordance with these principles. For the fiscal equalisation system to remain constitutional the first part of the legislation needs to be implemented by the beginning of 2003 and the second by January 2005. It is important that this need for action be taken as an opportunity to remedy identified shortcomings that hamper the system's allocative efficiency.

As has been argued in the 1998 *Survey*, the major shortcoming of the present fiscal equalisation system consists of disincentives in terms of the high effective outflow of additional tax revenues generated by the states that is implied by the rules of secondary tax redistribution both between the states and between the Federal Government and the states. By construction the burden affects both the wealthy and the poor Länder. For wealthy states the equalisation system transfers part of the additional revenues after tax sharing to poorer states. For the poorer states additional revenues after tax sharing reduce the eligibility for equalising transfers from the wealthy Länder and the Bund. In effect, individual states lose between 86 and 49 per cent of the additional income tax revenues they

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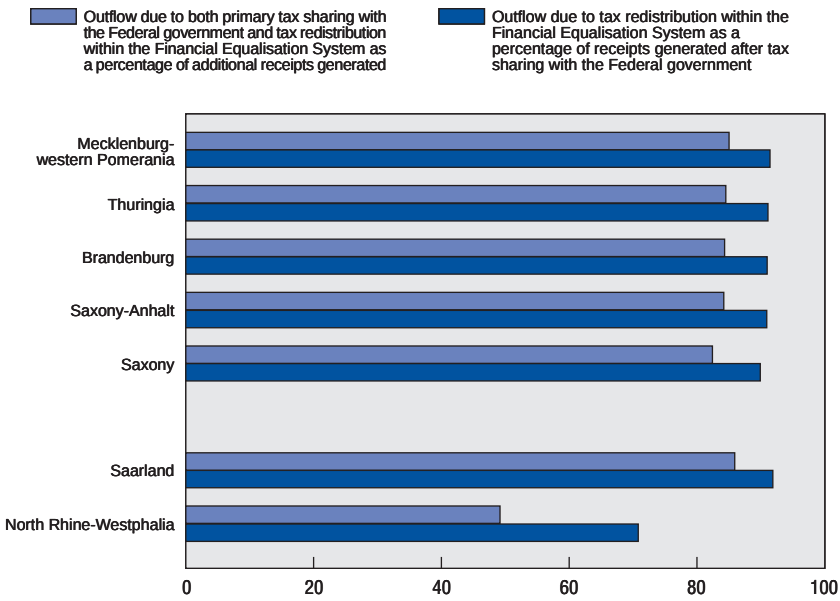
generate, measured as a percentage of revenues available after the primary tax sharing with the Federal Government (in 1999). The new states range among those states with the highest loss rates, amounting to between 82½ and 85 per cent. These rates are even higher if primary tax sharing is also included as an outflow from the Länder's tax revenues (Figure 12).

With effective taxation of additional income tax revenues at such high levels, the equalisation system in itself produces few incentives to create conditions which attract companies and to support the development of taxable activity (notwithstanding the fact that such incentives exist outside the sphere of fiscal redistribution).[19] Indeed, empirical evidence suggests that the marginal effective taxation of the states' tax revenues has a significantly negative impact on tax enforcement by the states, although there appear to be improvements in the technology of tax auditing.[20, 21] Moreover, tax auditing by the new states was criticised by the Federal Auditor-General's Office. The Court, in its 2000 annual report to the Federal Government, stated that for enterprises of all size classes the

Figure 12. **Marginal effective burden due to financial equalisation and tax sharing**[1]

Per cent



1. In 1999. It is assumed that a Land generates an increase in income tax receipts of DM 1 million.
Source: Huber, B. and K. Lichtblau (2000).

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auditing density in the new states fell short of the density in the old states by a wide margin. In the Court's assessment, in eastern Germany between one-third and more than one-half of posts required for the auditing of enterprises (*Betriebsprüfer*) are not filled, implying serious shortfalls in tax revenues and requiring immediate action by the states.[22]

Hence, introducing a more incentive compatible system is necessary. This requires reducing the link between the level of equalising transfers and the annual development of the Länders' financial capacity as measured by their tax revenues. As has been argued in the 1998 *Survey*, one such reform that is consistent with ambitious redistribution objectives could be realised by allocating to poorer states lump-sum transfers that are fixed over a multi-annual period. At a second stage, taxes could be redistributed between states conditional on their financial capacity as it prevails after stage one. Given that the financial endowment of the poor Länder would have been already raised at the lump sum stage, the transfer rate for redistributing revenues on an annual base could be reduced. After the reference period has passed, the allocation of lump-sum transfers would be reviewed and new allocations decided.[23] With the periodically revised lump-sum allocations likely to reflect the Länder's long-term economic development, this system is unlikely to fully abolish adverse incentives for developing the Länder's own revenues base. However, the reformed system would be more incentive compatible than the rules currently in operation.

Another aspect that has gained importance in the policy debate concerns the fact that in the present equalisation system income redistribution appears to be biased in favour of states with a high population density. When establishing a measure for the states' "financial needs" the equalisation system is based on the assumption that financial requirements increase with population density. Therefore "needs premia" are allocated to Germany's three city-states (Berlin, Bremen and Hamburg) as well as to states with large municipalities. The latter increase with the size of the community. The proposition that "financial needs" rise with population density has been questioned by the Constitutional Court in its ruling. The Court demanded examining whether high degrees of agglomeration or particularly low population densities within federal states could imply higher financial needs per inhabitant. In addition, the Court demanded investigating the link between taxing power and agglomeration. The ruling explicitly referenced three thinly populated states in eastern Germany. Empirical work, following the Court ruling, on the adequateness of compensation given to the city states produced mixed results. While one study found that the redistribution system over-compensates additional spending needs by the city states another study arrived at the conclusion that even higher premiums might be warranted to compensate the city states for their overall fiscal disadvantages.[24] More generally, the cost of providing public sector services on the level of state governments has been found neutral with respect to population density.[25] Other work, which considers both the state

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and community levels combined, found significant additional per-capita costs of providing a given standard of infrastructure endowment associated with sparsely populated states.[26] While these studies differ with respect to the degree the various levels of government are covered as well as the methodological approach chosen, they cast doubt on the adequateness of the extent to which the fiscal equalisation system compensates states with higher degrees of agglomeration for presumed higher per-capita costs of public sector activity. Similarly, for the purpose of establishing the area states' financial endowments prior to redistribution, the equalisation system adds in only 50 per cent of communal tax revenues. Empirical evidence suggests, however, that this rule disadvantages low-density areas because they produce smaller revenues per inhabitant than agglomerations.[27] Hence, if provision of similar standards of infrastructure across the federation is a goal, these biases in the redistribution system would have to be abolished. This is of particular relevance for the new states, where infrastructure gaps relative to the west have been ascertained (see Chapter IV below).

But reform should also address the division between the different layers of government of revenue raising and spending powers. As has been outlined in the 1998 S*urvey*, over many years there has been a secular trend reducing the connection between the competency to levy taxes and spending responsibilities of the states and communities Also, there is a risk that social spending obligations of the communities will increase as replacement ratios in the pay-as-you-go system are reduced over the next years to the extent these reductions give rise to increased claims for locally-provided social assistance transfers. These developments reinforce the need, highlighted in the 1998 S*urvey*, to increase the revenues raising power of local governments.[28]

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III. A review of progress in structural reforms

The previous *Survey* described in some detail the shift in labour market policy that had taken place after the new administration came to office. An important element in the government's structural and social policy approach has been the institution of round table talks between the social partners and the government for the purpose of developing a joint strategy for fostering employment, vocational training, and the competitiveness of the economy (Alliance for Jobs). The Alliance succeeded in reaching a broad understanding to promote employment, and in particular contributed to limiting the adverse impact on job creation of the recent oil price shock. Progress in labour market reform more generally, however, appears to be mixed. As regards product markets considerable progress has been made over the last years in liberalising network industries. In telecommunications, electricity supply and postal services Germany has opened markets more rapidly than European regulation requires. While the new regulation of gas supply also carries the opening of markets further than in many other European countries, the pace of reform has slackened. In the field of postal services, the German government has now decided to link further market opening to progress on the European level. Regulatory reform with respect to water supply is still under discussion, and not much progress has been made with respect to fostering competition in the railway system.

The first part of this Chapter reviews recent developments in labour market policies. The second part then examines the accomplishments and shortcomings of regulatory reform. The final section presents an agenda for further policy action.

Recent labour market reforms to tackle high structural unemployment

Increasing wage and labour cost flexibility

The OECD *Jobs Strategy* has identified wage and labour cost flexibility as essential elements of a successful strategy against unemployment. In the past, wage developments often had not adequately taken into account the high and increasing level of unemployment. As has been outlined in the last *Survey*, the Alliance for Jobs reached a broad understanding that wage policy should be geared primarily towards increasing employment, and to facilitate this goal they

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agreed on some informal guidelines for wage negotiations: negotiated wage increases shall not exhaust productivity progress, multi-year wage contracts shall be preferred over one-year contracts, and regional and sectoral differentiation of wages shall be given more room. The relatively moderate wage agreements that were made in spring 2000, mostly extending over two years, validated this agreement to a large extent. It is remarkable that these settlements came about in a period of rapidly rising oil and energy prices, and that unions did not demand additional compensation when prices accelerated.

Previous OECD *Surveys of Germany* described in detail the emerging wage flexibility at the company level. The wage negotiations that took place in 2000 confirmed this trend as unions and employers' associations often agreed to renew opening clauses, which had been introduced in many sectors in recent years.[29] To some extent branch wide agreements in 2000 also responded to uneven economic conditions across sectors and regions. For instance, in eastern German construction, which has been in recession for four years in a row and experienced substantial employment losses, wage increases will be well below the national average both in 2000 and in 2001. However, there has also been a trend in recent years to undercut collectively agreed wages – with enterprises either leaving the employers' association or not observing collective agreements – suggesting that labour market institutions have not adjusted sufficiently to cope with the imbalances on the labour market. In mid-1999 just 55 per cent of all companies in eastern Germany and 68 per cent in western Germany still paid wages according to contracts negotiated with unions (see Table 11).[30]

Table 11. **Coverage by collectively negotiated wage agreements**
Per cent, 1999[1]

	Enterprises	Employed persons
Eastern Germany		
Collective branch wage agreements	21 (26)	46 (44)
Firm-specific collective wage agreement	5	11
No collective wage agreement	74	43
Wage paid in accordance with collective branch		
Wage agreement but without obligation	29	22
Western Germany		
Collective branch wage agreements	44 (49)	65 (65)
Firm-specific collective wage agreement	3	8
No collective wage agreement	52	27
Wage paid in accordance with collective branch		
Wage agreement but without obligation	20	13

1. Figures for 1997 in brackets.
Source: R. Schäfer and I. Wahse (2000).

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Compulsory extensions by the Minister of Labour of collectively bargained wage agreements to non-contracting parties – designed to restrain outsider competition – declined in 1999. Overall, they applied to 1.1 per cent of collective wage agreements, down from 1.2 per cent in 1998. Most of these administrative extensions cover framework agreements that regulate general job conditions, such as working time conditions and bonus payments. But some 7 per cent also determine the level of wages – as is the case with respect to minimum wages in the construction sector. The right of the Minister of Labour to extend agreements even without the consensus of the social partners has been challenged by construction firms in the new states. However, in summer 2000 the Federal Constitutional Court ruled that such non-approved extensions do not violate the constitution.

Increasing working time flexibility and easing employment security provisions

In the eighties and the first half of the nineties global reductions of working time were launched to fight unemployment. At the end of 1999 demands for general working time reductions were raised again, following the reorientation of labour market policies in autumn 1998 (see *Germany Survey* 1999). Overall, the policy of general working time reductions did not succeed in bringing down unemployment.[31] However, as has been outlined in the last *Survey on Germany*, substantial progress has been made during recent years with respect to allowing more flexible working time arrangements at company level. This trend is continuing, and surveys indicate that in 1999 working time accounts existed for 35 per cent of the employed in western Germany and 33 per cent in eastern Germany.[32] The accounts allow for compensating periods of overtime by reduced hours worked within a certain time span extending from a short period to several years. Long-run working time accounts are in some cases denominated in Deutschemark and invested on the capital market. The proceeds can then be used later to take sabbaticals or retire prior to the statutory retirement age. In autumn 2000 the social partners agreed for the first time in a branch-wide collective contract in the steel industry – on a tariff contract that allows for working-time accounts spanning over the full working life. In a similar vein, a "benchmarking group" of the Alliance for Jobs – a group of experts analysing working time and other issues on the agenda of the Alliance – agreed, in February 2000, that future working time arrangements should neither hamper investment, nor increase costs, nor shorten operating hours, nor reinforce skill imbalances on the labour market. Instead, they should accommodate both individual preferences of workers and the needs of employers.[33] However, the Alliance for Jobs itself has not yet adopted this assessment.

The government has introduced new legislation to foster part-time employment. From January 2001 on, employees will have the legal right to transform a full-time job unilaterally into a part-time job. The management can reject the employees' request if it is able to claim (partly pre-specified) urgent needs for

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maintaining full-time employment. Furthermore, part-time employees wishing to extend working hours shall be given priority to external applicants when filling vacancies. This measure might meet employees' demands for more part-time employment. However, it runs the risks of increasing labour costs and legal uncertainty in that it might require a plurality of court rulings to establish a consensus on the definition of urgent business needs exempting companies form granting part-time jobs to full-time staff. Legal uncertainty increases transaction costs and may thus impede employment creation. However, the law authorises the social partners to identify in collective contracts such urgent business requirements. This in turn may help to reduce the legal ambiguity.

Collective contracts that establish the right for employees to claim "part-time work for older employees" (*Altersteilzeit*) have further increased. Part-time employment can be distributed over an admissible time span of up to 10 years, and often employees work full time for the first half of this span and do not work within the second part (see OECD *Germany Survey* 1999). Effectively, this scheme allows early retirement to be subsidised by the Federal Labour Office provided older workers are replaced by unemployed or by young people having completed their apprenticeship. Most wage agreements in 2000 included the introduction, prolongation or modification of part-time schemes for older employees. At the end of 2000, 530 collective contracts, covering about 15.4 million employees, included such schemes, and the Federal Labour Office subsidised 35 000 employees participating in the programme.[34]

Small hours jobs held as a secondary job are now subject to the same taxation and contribution rules as regular jobs (full-time or part-time). This has increased the overall tax and contribution burden for secondary small hours jobs while little has changed for small hours jobs held as a primary occupation (for which, typically, contributions to social security now have to be paid, replacing a previous system of lump-sum taxes).[35] There is evidence that this has resulted in a substitution from secondary small hours jobs to those held as main occupation. The composition of the workforce holding small hours jobs was changed accordingly[36] with students, pensioners and especially housewives increasing their share. The reallocation of labour indicates that the elasticity of labour supply might be high in this segment of the labour market. Accordingly, the welfare costs of differential tax treatment of different types of jobs might be substantial. It is not clear whether the new regulation contributed to net gains in standard part-time or full-time jobs.[37] Another important question will be whether small jobs ease the transition into regular employment, in particular for the unemployed. Little is known about this issue. The transition from a marginal employment to a standard part-time or full-time job is hindered by a high implicit marginal tax rate, which may in some cases be well above 100 per cent.[38]

The new regulation for small jobs has proved to be rather complex as contributions to social security funds and taxation now depend on a number of the

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employees' characteristics (see 1999 *Germany Survey*). It places substantial additional administrative obligations on employers who have to compute and pass on social security contributions and taxes. Various companies, especially in some service branches, temporarily experienced bottlenecks when employees with small hours jobs held as a secondary occupation cancelled their contracts in response to the new regulations. Over time, however, most businesses seem to have succeeded either in recruiting substitute staff, – part-time or full-time – or adapting working hours of the existing staff, by introducing working time accounts or by paying overtime premia.[39]

Temporary help agencies, which sub-contract their employees to other firms for a limited amount of time, can also be an important tool to add to the flexibility of the labour market and foster the transition into regular employment. Indeed, the evidence suggests that temporary help agencies have played a significant role in improving the efficiency of matching labour supply and demand and contributed to lowering unemployment (Table 12). Agencies substantially increased their staff in recent years: by 57 per cent between 1996 and 1999 with employment growth in 1999 alone totalling 12 per cent. Some 0.7 per cent of the employed are under contract with a temporary help agency (1999). Typically, about 60 per cent of the persons newly hired by an agency have not been in employment before, and some 20 per cent have been without a job for more than a year. In most cases employment in an agency terminates within three months, and many of the workers on temporary loan finally get a permanent job. The rapid expansion of this industry took place despite tight regulations.[40] Relaxing these restrictions might foster employment overall, although the available evidence is ambiguous in this respect as in some cases relaxing only rules for temporary

Table 12. **Employment in temporary help agencies**

	1996	1997	1998	1999
Average employment	175 798	200 541	245 780	275 838
Percentage change from previous year	. .	14.1	22.6	12.2
Share in total employment	0.4	0.5	0.6	0.7
Hirings	149 936	183 243	233 091	270 536
of which: (per cent)				
Employed immediately before being hired	38.2	36.1	37.5	38.4
Not employed before being hired	61.8	63.9	62.5	61.6
Employed within the previous 12 months	41.1	42.8	41.3	40.7
Not employed within the previous 12 months	10.9	11.9	11.9	11.3
Never employed before	9.7	9.1	9.3	9.6

Source: Neunter Bericht der Bundesregierung über Erfahrungen bei der Anwendung des Arbeitnehmerüberlassungsgesetzes – AÜG – sowie über die Auswirkungen des Gesetzes zur Bekämpfung der illegalen Beschäftigung – BillBG –, Berlin 2000 and OECD.

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workers rather than a general relaxation of employment protection legislation may lead to large composition effects in employment.[41, 42]

Fixed-term contracts have also gained in importance. The share of young persons entering the labour market via fixed-term contracts is increasing, and the duration of many contracts with reduced working time is fixed in advance (Table 13). In 1999 13.1 per cent of all jobs in eastern Germany and 7.1 per cent in Western Germany were fixed-term.[43] But this is still lower than in many other European countries (Table 14). Two types of fixed-term contracts exist in Germany.[44] The first type requires that a "reasonable motive" be given. For instance, employment on trial, seasonal employment, and temporary replacement for employees being on parental leave are covered by this clause. The second type – accounting for only a small share of the total – does not require such justification, but these contracts are subject to a maximum duration of 24 months with limited possibilities for renewals within this time span. This variant was introduced temporarily in 1985 and was later extended until the end of 2000 (see *Germany Survey* 1999).

While the unions demanded the elimination of fixed-term contracts without "reasonable" motives – arguing that less restrictive employment protection rules would not increase employment[45] – in autumn 2000 the regulation of fixed-term contracts was unified in a new law. The new legislation consolidates earlier written law with case law, thus potentially diminishing legal uncertainty. For instance, motives defining type 1 fixed-term contracts are now specified by law,

Table 13. **Utilisation of fixed-term contracts in Germany**
Percentage share of employed persons[1]

	June 1985	April 1991	April 1995	April 1999
Eastern Germany				
Overall		10.3	12.1	13.1
Below 25 years of age		13.0	17.0	27.1
Men		9.8	10.4	12.2
Men, less than 20 hours/week		. .	. .	31.9
Women		10.9	14.2	14.1
Women, less than 20 hours/week		. .	. .	19.2
Western Germany				
Overall	5.4	5.1	5.5	7.1
Below 25 years of age	10.3	9.7	13.0	20.2
Men	4.8	4.3	5.1	6.8
Men, less than 20 hours/week	31.7	36.6	29.1	21.4
Women	6.3	6.2	5.9	7.6
Women, less than 20 hours/week	13.0	11.7	8.4	7.8

1. Including public servants and persons employed in active labour market measures, without persons in vocational training. Mikrozensus data.
Source: H. Rudolph (2000).

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Table 14. **Utilisation of fixed-term contracts in selected countries, 1998**
Percentage share of employed persons

	Overall	*of which:* not in vocational training	*of which:* in vocational training
United Kingdom	7.0	6.6	0.4
Austria	7.8	3.6	4.2
Italy	8.5	5.9	2.6
Germany	**12.2**	**6.3**	**5.8**
Netherlands	12.7	12.5	0.2
France	13.9	11.8	2.0
Portugal	17.4	16.9	0.6
Spain	32.9	31.6	1.3

Source: H. Rudolph (2000). Data from Eurostat Labour Force Survey.

which was not the case before. The law also relaxes limitations on fixed-term contracts for older employees.[46] On the other hand, the new law further restraints the renewability of those fixed-term contracts that are not based on "reasonable" motives. This feature has been criticised by industry arguing that an important degree of freedom has been lost. However, the social partners are allowed to fix in collective agreements conditions under which fixed-term contracts can be extended. In general, the relaxation of employment protection legislation with respect to temporary and fixed-term contracts has positive effects on both employment and unemployment levels.[47]

Reducing the distortions arising from unemployment insurance and related benefits

As has been outlined in previous *Germany Surveys*, unemployed aged 58 or older have been eligible for receiving unemployment benefits without obligation to accept a job. This temporary provision, which has been criticised by the OECD as it effectively encourages early retirement at the expense of higher unemployment insurance contributions, expired at the end of 2000 without replacement.[48]

In spring 2000 the Federal Constitutional Court ruled it unconstitutional to base unemployment benefits on basic monthly earnings only, while the social security contribution base includes extra payments such as gratifications for Christmas and holidays. The policy response to this ruling could have been either to take bonus payments off the contribution base and leaving unemployment benefits unchanged, or to leave the contribution base unchanged and raising the base for computing unemployment benefits. The Federal Government opted for the latter approach, because of its more favourable implications in terms of maintaining the revenue base for the unemployment insurance system. Without further action bene-

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fit levels and replacement rates will increase, reducing work incentives. Moreover, contribution rates will have to rise *ceteris paribus*, thereby raising non-wage labour costs, although recent employment gains eased the contribution burden.

Achieving a better integration of social assistance transfers within the system of unemployment related benefits is still an unresolved issue. Many unemployed persons receive supplementary social assistance because unemployment insurance benefits or unemployment assistance do not cover their basic needs. Provision of social assistance is in the responsibility of the communities, and this has become a heavy burden for local budgets.

Local authorities have chosen different responses to relieve their budgets and to ease the (re-) integration of recipients of social assistance into the primary labour market. As the active labour market measures of the Federal Labour Office cover only those non-employed entitled to unemployment benefits or unemployment assistance, many municipalities have started activation programmes of their own.[49] Typically they provide temporary subsidised jobs to social assistance recipients, either in the private sector or in public local employment schemes.[50] This is also meant to test the readiness of the beneficiaries to work.[51] Some municipalities grant additional assistance for up to 12 months upon taking up employment in the primary labour market, thus reducing the strong disincentives associated with the withdrawal of social assistance benefits (see *Germany Survey* 1996). Furthermore, some local authorities monitor potential problem groups and offer advice, thus trying to prevent the formation of cores of un- or non-employment. Such initiatives do appear to have reduced the number of beneficiaries of social assistance as some unwilling to work stopped claiming benefits. At the same time, anecdotal evidence indicates that some discouraged long-term unemployed have found back into the primary labour market. Still local market programmes have to be framed carefully to prevent the establishment of a permanent secondary labour market, where employment depends on one or another form of public subsidies, as well as an opportunistic shifting of burdens between local authorities and the labour market administration. In autumn 2000 a new law regulating the co-operation between local social administrations and labour offices came into force. Furthermore, the Federal Government sponsors several model projects in this field. Nevertheless, a better integration of unemployment assistance with social assistance and an improved co-ordination of activation policies is called for.

Enhancing active labour market provisions and re-entry into employment

The government attaches a higher priority for fostering the reintegration of the unemployed into the primary labour market than for measures that sustain the secondary labour market. Accordingly, several kinds of wage and labour cost subsidies have been introduced, and local labour offices have obtained more discretion in the use of funds. In August 1999, several minor modifications of active

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labour market measures came into operation, aiming at a better focus on targeted groups and a reduction of costs of administration (see 1999 *Survey on Germany*).[52] Some progress has been made recently in evaluating active labour market policies, although much remains to be done. Following the decentralisation of responsibilities for ALMPs, local labour offices are now obliged to give a detailed account once a year of their success in integrating unemployed into the primary labour market.[53] In autumn 1999, the first integration balances were published. These balances are supposed to signal the relative success of training, wage subsidies and employment creation schemes, broken down by instruments and regions. The main indicator is the proportion of former participants not registered as unemployed six months after the end of the measure. A further step will be the introduction of an indicator that displays the portion of participants that got a job in the primary labour market. Furthermore, the research institute of the Federal Labour Office, the Institute for Labour Market Research, is preparing a data set tracking individual participants in active labour market programmes which will allow an econometric assessment of the programmes' effects. Such research is necessary for proper evaluation of programmes. The integration balances suggest that ALMPs were more successful in 1999 than in the year before. However, as labour market conditions in general improved in 1999 the rising share of participants not unemployed six months after the end of the measure does not yet indicate that programme efficiency improved. However, even this rather crude indicator reveals a pattern identified in other studies: measures aiming at direct integration into the primary labour market tend to outperform other types of job creation measures, particularly those involving the secondary labour market (Table 15; see also Chapter IV).

Table 15. **Integration performance of active labour market policy measures**
Percentage share of former participants

Programme type	Not registered as unemployed six months after termination of measure	
	1998	1999
Temporary grant for the start of self-employment	89	91
Integration subsidies[1]	72	81
Subsidies for employment in newly founded enterprises	59	77
Further vocational training	66	68
Measures of structural adjustment for private enterprises[2]	49	66
Measures of structural adjustment[3]	48	63
Job creation measures	41	42

1. Wage cost subsidies to employers.
2. Wage cost subsidies to employers.
3. Job creation measures.

Source: Institute for Labour Market Research of the Federal Labour Office (2000), and OECD.

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Preliminary evaluation of the public scheme to reduce youth unemployment indicates that the achievements of this programme have been quite modest. In 1999, 176 100 persons entered the programme. Registered unemployment of young people dropped by nearly 9 per cent in the course of the year, as overall registered unemployment declined by about 4 per cent. The above-average reduction of youth unemployment might be understood as a consequence of the programme. However, given the structure of the German labour market, the cyclical volatility of employment prospects for young workers generally tends to be much higher than the volatility of average labour market prospects.[54] Thus, a similar development would have been expected even without a special programme. The employment effects of the programme for the youth are under investigation through an evaluation commissioned by the Ministry of Labour and Social Affairs.[55] Besides this special programme, traditional schemes for young persons carry on, with an increasing number of participants.

Two distinct comprehensive schemes to tackle the problem of low-skilled unemployment are being put to the test in four German States. One scheme attempts to strengthen the demand for low skilled labour, the other scheme tries to promote higher labour supply for low-wage jobs. In the West German State Saarland and in a local labour district in the East German State Saxony, employers' contributions to the social security funds are moderated by subsidies if companies employ low-skilled workers or long-term unemployed persons. The candidates receive some training before, during or after the spell of subsidised employment, which may last up to 18 months. The level of subsidies declines with the hourly wage rate.[56] In some labour market districts in the West German State Rhineland-Palatinate and in the East German State Brandenburg, unemployed or discouraged workers who take up a low-wage job receive additional benefits for a period up to 12 months. The level of the subsidy depends on income and family status. These experiments are restricted to a small number of participants in each state. The Federal Ministry for Labour is sponsoring research to evaluate a set of individual data of the participants and of a control group. Earlier studies had indicated that full-fledged subsidies either to low-wage earners or to wage costs for low productivity jobs would result in a heavy drain on public resources, but would not increase overall employment substantially.[57]

Improving skills and technological know-how

In autumn 2000 the Federal Government relaxed immigration laws and started issuing working permits ("green cards") for foreign computer experts.[58] This was done in response to complaints of private business about a tight labour market for of IT professionals. In a recent survey more than half of the companies providing IT services reported difficulties in hiring adequate personal.[59] With IT professionals appearing to be complementary to low-skilled or semi-skilled

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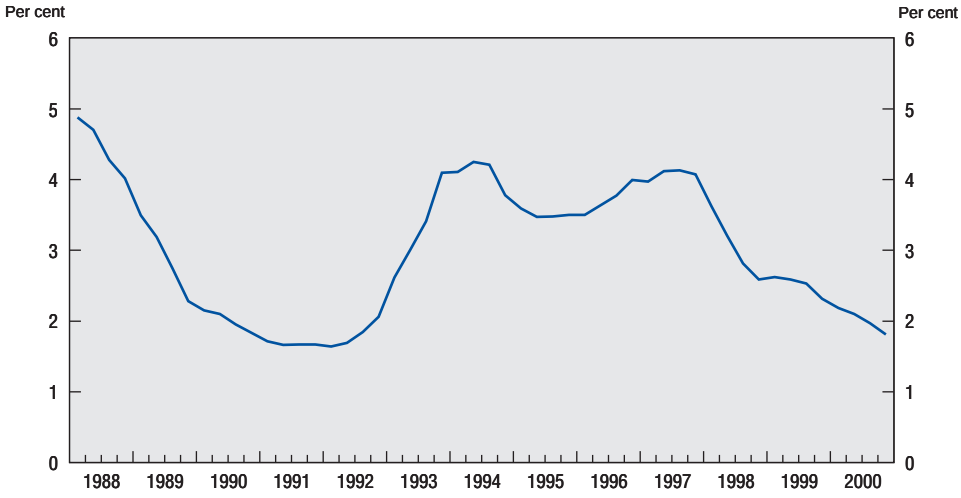
labour, the shortage of experts is perceived to slow down economic activity and employment growth. This development indicates that even with high overall unemployment, bottlenecks are emerging in the German labour market. This is also highlighted by the large and increasing number of vacancies. In autumn 1999 1.2 million job openings were not filled. The unemployed-to-vacancy ratio in Western Germany appears to be approaching the cyclical low observed at the height of the unification boom in 1991, which indicates a rising mismatch between labour supply and demand (Figure 13). These imbalances underline the need for an adjustment of vocational training and high-school and university education in Germany. Furthermore, lifelong learning needs to become more widespread as a measure to ensure employability.

The Alliance for Jobs agreed to build up additional training capacities in IT and to adapt training regulations expeditiously to changing needs. Furthermore, social partners reached an understanding on strengthening the provision of vocational training by private enterprises, as surrogate training in public institutions often had not delivered adequate results. Backed by improving economic conditions, at the end of September 2000 vocational training offers surpassed demand for the first time since the mid-1990s. However, there is still substantial regional and occupational mismatch. In southern Germany vacancies cannot be filled,

Figure 13. **The unemployed-vacancy ratio**

Unemployed persons per vacancy, western Germany



Source: Magvas, E. and Spitznagel (2000); Deutsche Bundesbank, and OECD.

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whereas in the new states private sector offers for training fall short of requirements. In the past, public sponsored vocational training moderated the mismatch. Now the Federal Government plans to spend 5 per cent of the funds allocated to the special scheme for the reduction of youth unemployment on additional mobility grants to apprentices who want to take up vocational training in enterprises away from home. Already in the past, the Labour Offices have given some grants to young persons for this purpose. However, the level of the aid was frequently considered inadequate in view of commuting expenses and cost-of-living differentials. As in the years before, the government has continued to adapt existing and create new vocational training curricula to new demands.

A comprehensive reform of public universities seems to be essential for improving the skills of the German workforce. While the output of German universities has increased in numbers since the 1970s, thus contributing to a higher qualification of the workforce on average, it has arguably declined in quality.[60] However, with the German working-age population projected to decline dramatically, the quality of education becomes even more important for the productive potential of the economy. Several explanations have been given for the decline in the quality of university education. As competition between universities is hindered by federal and state laws and regulations, which aim at equivalence of exams and diplomas, adjustment to changing needs typically takes place with considerable delay. Furthermore, public universities may not levy tuition fees (with some minor exceptions). Consequently, teaching obligations are often considered as a nuisance, while the students regard universities largely as a free resource. Moreover, the incentives for university teachers to deliver adequate teaching and research are weak once tenure has been granted.[61] For these reasons the German Monopolies Commission has recently proposed fundamental reform measures. Mainly, universities should be granted more independence with respect to introducing tuition fees and salary schemes for university teachers based on performance.[62] In a similar view, an expert group to the Federal Ministry of Education and Science also delivered detailed proposals for a reform of labour contracts for university teachers. It is therefore welcome that the Federal Government is preparing legislation in order to introduce performance-oriented elements into the remuneration scheme for university teachers and revise the system of tenure. Furthermore, universities have obtained new scope to introduce curricula designed to shorten studies and increase the international comparability of degrees.

Product market reforms to enhance competition

Product market reform has continued to progress in Germany. In the retailing sector the regulation restraining rebates and bonuses will be abolished, although further extension of shopping hours still has to be implemented. Local

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governments are facing increasing pressure to withdraw from banking activities, although progress is still to be made. Regulatory reforms in the network industries have progressed, although at a varying pace. The railway incumbent was incorporated in the mid-1990s, but progress with respect to competitiveness has been only modest. On the other hand, the incumbent in the postal service market was recently partly privatised and relatively large segments of the overall market have been opened up for outside competition. Progress has been quite pronounced in the telecommunication market, which has moved from being tightly regulated to be in the forefront in Europe as reflected in the rapid decline in most telecommunication prices, although they remain above the average of the countries with earlier liberalisation of telecommunication markets – Denmark, Finland, the Netherlands, Sweden, and the United Kingdom. Price indicators for the electricity market, on the other hand, indicate that despite recent improvements in the regulatory framework considerable scope exists for further price declines. A similar deregulatory approach was used in the natural gas market, but competitive market structures have not yet emerged and prices for both industry users and households are still above those in most other European countries.

Retailing

German regulations restraining the value of rebates and bonuses came under pressure from a new e-commerce directive of the EU. According to the directive, e-commerce trades are subject to the rebate legislation in their country of origin. Comparatively restrictive regulations in Germany would therefore have disadvantaged German traders. In response the government decided to abolish such regulation without replacement. Shop opening hours, on the other hand, are still relatively tightly regulated. Experiences from OECD countries[63] show that further liberalisation of opening hours would most likely increase the sector's overall employment as well as increase consumer choice, although first experiences at the firm level with recent liberalisation of opening hours in Germany show mixed results.[64] In summer 2000, several state governments pledged for an extension of shopping hours, but no new legislative initiatives have emerged yet.

Banking

A longstanding conflict between the European Commission and German authorities on the status of public credit institutions, which generate almost 40 per cent of the volume of banking business, extended into 2000.[65] Typically, local or state authorities hold ownership of these banks and act as guaranteeing authority. Therefore, the costs of capital are lower for public banks – in particular Länder owned banks – than for private banks, thus potentially distorting competition.[66] The European Commission classifies the resulting lessening of the costs of capital as unlawful subsidisation. Public ownership and guaranties are often rationalised

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by service duties imposed on public banks such as providing a basic financial infrastructure. However, in fully competitive markets, private banks would tend to provide area-wide basic financial services. If this failed, however, the government could tender for universal services provided by private banks. Thus, the idea that public financial institutions are needed to supply basic financial services is not convincing. Public financial institutions also often serve as vehicles for local or regional industrial policy by granting loans at reduced rates. Again, private banks, given adequate public funding, could perform these tasks, too, and this would also raise the transparency of industrial policies. While the Federal Government and the state governments maintain the position that German public banks are not under the jurisdiction of the European Commission, in autumn 2000 plans emerged to split up the biggest state bank into a private commercial bank and a public development bank. Furthermore, the Federal Government aims at adopting a market-oriented system of guarantees while preserving the role of public banks as providers of genuine public services. Public banks announced that they would forfeit explicit guaranties. Also, a big municipal savings bank is considering going public in the near future, and other commercially successful public banks might follow. While this would imply a loss of government guarantees, by changing their legal status the banks could get rid of certain restrictions on their business (notably the "territoriality principle" that allows savings banks to operate exclusively within the jurisdiction of the entity in which they were incorporated).

Railways

Up to now the effectiveness of regulatory reform of the German railways system (*Deutsche Bahn*, DB) with respect to competition has been relatively modest. The Deutsche Bahn was incorporated in 1994 but the Federal Government still holds all shares of the DB holding company. DB controls five separate companies serving the grid and the stations and delivering intra- and interregional passenger and cargo transportation. The productivity of DB, measured by transport units per unit of labour input, was increased by more than 100 per cent since 1993, by means of cutting down staff and rail connections. However, the overall profitability of DB is still low. Whereas interregional passenger transportation is lucrative, other fields either produce losses or just break even.

A major weakness of the regulatory design for the railway system is the lack of a true separation of track and stations from the provision of transport services. The conflict of interests naturally associated with common ownership of the railways infrastructure and utilities providing transport services has caused problems, which slowed down the emergence of competitive market structures.[67] Although private railways and utilities owned by local authorities have shown that they can provide transport services at terms accepted by customers on routes given up or neglected by the former incumbent,[68] the market share of the compet-

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itors of DB is still very low, amounting to some 6 per cent in intraregional passenger transport and 3 per cent in cargo transport. Entry into the railway service market is in principle free, but competitors to the incumbent have complained to the Federal Cartel Office that the structure of charges for the use of the track has been discriminating and preventing entry. Indeed, the grid-company of DB used to grant rebates to companies with large transport volumes, benefiting its sister companies. The Federal Cartel Office estimated that in regional passenger transport charges for the use of the track were on average 25 per cent lower for the DB Regio company than for its competitors.[69] In summer 2000, the Cartel Office asked DB to stop this practice. In response, in 2001 DB introduced a charge without rebates. Competitors also have complained to the regulatory authority for railways, (Eisenbahnbundesamt), that the track-company of DB fails to keep the track in good order. Allegedly, DB stops maintaining a track if it is no longer of use for its own business. Transport on such tracks had to slow down considerably, rendering it uncompetitive, or suspended. In individual cases, the regulatory office ordered DB to resume the maintenance of tracks. In autumn 2000 it became known that DB would incur losses of several DM billion because it had neither dealt adequately with the attrition of the existing track nor managed the construction of new networks properly.

DB and the union of railway workers defend the vertically integrated structure of DB, arguing that there exist "economies of scope" between the provision of the network and the transportation services. However, as the German constitution requires that the Federal Government maintains a majority stake in the track company, preserving this structure would imply that DB would continue to be a publicly-owned company with a dominant market position.[70] Indeed, a recent report forwarded by an expert commission attached to the Federal Ministry for Construction and Traffic recommends a separation of track and services.[71] This would allow public ownership of the "natural" monopoly of tracks and foster competition within railway services. The expert commission also proposed to establish a strong independent regulatory authority to prevent DB from abusing its dominant position in the provision of railway services.

Postal services

Liberalisation of postal services is relatively advanced in comparison with many other EU countries. In 1999 already two-thirds of the turnover in the market for postal services was generated in areas open to competition.[72] Only the segment delivery of letters is still tightly regulated and in part exclusively reserved for the incumbent monopolist post office, *Deutsche Post* AG (DP).[73] DP holds a share of some two-thirds in the overall market for postal services, with some 50 per cent in those segments open to competition, but 98 per cent in the only marginally liberalised market for letters.[74]

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Box 4. Is the tariff for letters in Germany high by international standards?

Three studies on the adequacy of the tariff for the delivery of letters in Germany have been published recently:

- The *Deutsche Post* released a report claiming that prices are not excessively high in international comparison.[1] In this study the price for delivering standard letters in Germany is compared to that for priority mail in other countries. However, no evidence is given that standard mail delivery in Germany is equivalent to priority mail delivery in other countries. Furthermore, prices are converted using purchasing power parities, and in addition they are adjusted for differences in wage costs. This procedure obviously results in an excessive adjustment since high wages are closely related to high costs of living.[2] However, insofar as wage costs reflect the sharing of rent arising from market power, such an adjustment may not be warranted.
- The German Association of Post Customers (DVPT) compared tariffs for standard mail, converted by purchasing power parities, but without any adjustment for quality of delivery. According to this study, the Deutsche Post charges the highest price among industrialised countries.[3]
- The Regulatory Authority for Telecommunications and Post tried to cope with the problem of quality heterogeneity by choosing similar products for comparison. Furthermore, it compared prices for a basket of services, covering not only standard letters but the whole range of service products in the market for the delivery of letters. This is preferable over the more narrow approach when relative prices for different modes of delivering letters differ between countries. Expenditures for this basket of services were converted via purchasing power parities into a common denominator. This study finds that there is just one country (Italy) for which the resulting price index exceeds the German level.[4]

While the method used by DP implies an over-adjustment of prices in favour of DP and the study by the customers association is biased in the opposite direction, the approach applied by the regulatory authority appears to be best suited to the different measurement problems. The evidence presented in this study clearly indicates that the Deutsche Post charges high prices in its protected domain.[5]

1. Research Services International (2000).
2. This is the core of the Balassa-Samuelson theorem, which is supported by empirical evidence. For recent evidence on EMU countries and further references, see de Grauwe and Skudelny (2000).
3. DVPT (1999).
4. Regulierungsbehörde für Telekommunikation und Post (2000) *op. cit.*
5. In autumn 2000, the Deutsche Post finally stated that the price for the conveyance of a standard letter contains a profit margin of about 10 per cent and that letter-related services generate 35 per cent of the revenues and more than 75 per cent of overall profits.

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DP recently got involved in a series of legal and regulatory battles concerning market access and cross-subsidisation. In more than 600 cases DP took legal action against the regulatory authority, which had granted restricted licences for the conveyance of letters to new market entrants. DP also sued competitors directly. In summer 2000, courts ruled that the claims of DP were mainly unfounded. However, the legal action by DP prevented potential competitors from entering the market, which should explain to some degree why the market share of DP for these services is still close to 100 per cent. An investigation by the EU Commission indicated that DP refused to deliver letters posted in foreign countries for cheaper delivery within Germany, potentially contradicting EU remailing agreements.[75] Moreover, the Commission claimed that DP used excessive profits originating from its protected domain to fund prices below cost in areas open to competition.[76] In October 2000, DP stopped these practices and announced that the dispute about international mail had been settled via a bilateral agreement with another post office.

DP was partly privatised (the initial public offer led to the sale of 25 per cent) in autumn 2000. While the present charges for mail delivery, which seem to be high in international comparison (see Box 4), were set to terminate in August 2000, they have been extended until the end of 2002, following an intervention by the government with the regulatory authority. The exclusive licence was originally scheduled to expire in 2002, but the government decided to prolong it until 2007. The regulatory authority has estimated that abolishing this licence could trigger price reductions of about 30 per cent on average, an increase in product variety and a better adaptation to customers' needs.[77] The view that fully opening the market for letter delivery to competition could generate substantial benefits is also shared by the German Monopolies Commission, which calls for the exclusive licence not to be extended beyond 2002.[78] Although the Federal Government shares the view that abolishing the exclusive licence is a necessary condition for the development of competition,[79] it has declared that a further opening of markets in Germany would not occur without similar steps in other European countries. A prolongation of DP's monopoly position in letter delivery will imply, however, higher costs for consumers, reduced competitiveness for enterprises and adverse effects on competition in other fields on account of cross-subsidisation.

Telecommunication

So far, the regulatory reform of the telecommunication sector has proved to be a major success, in terms of lower prices, increased product variety as well as improved quality. Competition has evolved rapidly, and its viability is being supported by a rapidly growing stock of telecommunication infrastructure (Table 16).[80]

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Table 16. **The market for telecommunication services**

	1998	1999	2000
Licences[1]			
Transmission path licences[2]	190	245	265
Voice telephony licences[2]	79	147	175
Service providers[2]	1 222	1 593	1 780
Local loop access agreements (end of year)	22	57	95[3]
Internet link-ups			
(change from previous end of year, per cent)	40.4	56.0	60.0[3]
Volume of traffic			
(call minutes, change from previous year, per cent)	10.5	17.4	26.0[3]
Share of the DTAG (per cent)	94.1	81.8	76.0[3]
Revenues (change from previous year, per cent)		11.0	10.0[3]
of which:			
Fixed line services		–8.5	
Mobile telephony		33.9	
Other services (including Internet access)		25.2	
Employment			
(change from previous end of year, per cent)		3.0	6.1[3]
Share of the DTAG (per cent)	81.6	76.1	75.0[3]

1. Including holder of licences which do not yet provide services.
2. Estimates.
3. As of late summer.

Source: Regulatory Authority for Telecommunications and Posts (2000); Regulatory Authority for Telecommunications and Posts Website and OECD.

Consumers directly benefit from declining prices. According to the Federal Statistical Office, consumer prices for fixed-line telecommunication services declined on average by 17 per cent since December 1997 (Table 17).[81] The largest price reductions were recorded for international calls, followed by domestic long-distance calls. At the end of 2000, minimum prices for the most common types of national and international long-distance calls were typically up to 90 per cent below the level of prices of 1997, reflecting, among other things, high initial prices. Prices for mobile phone services, where liberalisation began in 1992, have dropped by more than 60 per cent since the beginning of 1995,[82] and prices for terminal equipment have declined by nearly 50 per cent.[83] There is some evidence that the "true" decline of prices might have been even more pronounced than captured by the price index.[84] However, compared to other countries with earlier liberalised markets, average prices of the incumbent are still high in Germany (Figure 14). This indicates that in spite of the progress made fully competitive market structures have not yet emerged.[85]

As the local loop is not contestable in the short run, the incumbent provider, *Deutsche Telekom* AG (DTAG) so far has not cut fixed charges for final consumers and

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Table 17. **Deregulation induced changes in telecommunication prices for private households**

Percentage change from December 1997 (1995[1])

	Telecommunication services in the fixed-line network				Mobile phone services[2]	Equipment[3]
	Basic charges	Local calls	Long-distance calls	Foreign calls		
1998						
March	0.0	0.0	0.6	–13.8	–38.2	–37.9
June	0.8	0.7	1.3	–13.1	–38.2	–38.2
September	0.8	0.7	1.3	–13.1	–43.2	–38.2
December	0.8	0.7	1.3	–13.1	–51.3	–38.2
1999						
March	0.8	8.1	–37.5	–13.8	–51.7	–38.2
June	0.8	8.1	–42.1	–31.0	–53.9	–41.4
September	0.8	8.1	–42.5	–31.0	–55.0	–41.4
December	0.8	8.1	–42.5	–31.0	–55.3	–43.0
2000						
March	0.8	8.1	–45.8	–54.0	–57.5	–43.0
June	0.8	7.3	–47.0	–55.6	–59.9	–48.4
September	0.8	7.3	–47.1	–55.6	–60.9	–48.4
December	0.8	7.3	–47.3	–55.6	–61.5	–48.7

1. Mobile calls and equipment.
2. Mobil call services already have been partially opened to competition in 1992.
3. The market for telecommunications terminal equipment has already been liberalised in 1989.

Source: Federal Statistical Office and OECD.

there have been no major changes in prices for local calls. There was no regulatory pressure to proceed in a different way: while DTAG is obliged to provide its competitors unbundled access to the local loop, the price that it may charge its competitors for access is – in accordance with a ruling of the regulatory authority – higher than DTAG's fixed charge for its own final customers. Hence, new providers need to charge their customers with analogue service subscriptions higher fixed prices than DTAG, though they offer local calls and special services, such as high-speed data transfer, at lower rates.[86] This implies that a full change-over to a new provider, including the local loop, is sensible only for customers with a very large volume of local calls or with special needs. Consequently, the share of DTAG in local markets is still well above 95 per cent and only slowly decreasing. To promote the development of competition the regulatory authority, in winter 1999, did not approve a planned drastic reduction of the incumbent's consumer prices for local calls, and conditioned such a reduction on preceding cuts in interconnection charges.[87] In early 2001, DTAG requested permission for an even higher charge for the unbundling of the local loop. On 30 March 2001 the regulator lowered the charges for access to the local loop by 5 per cent.

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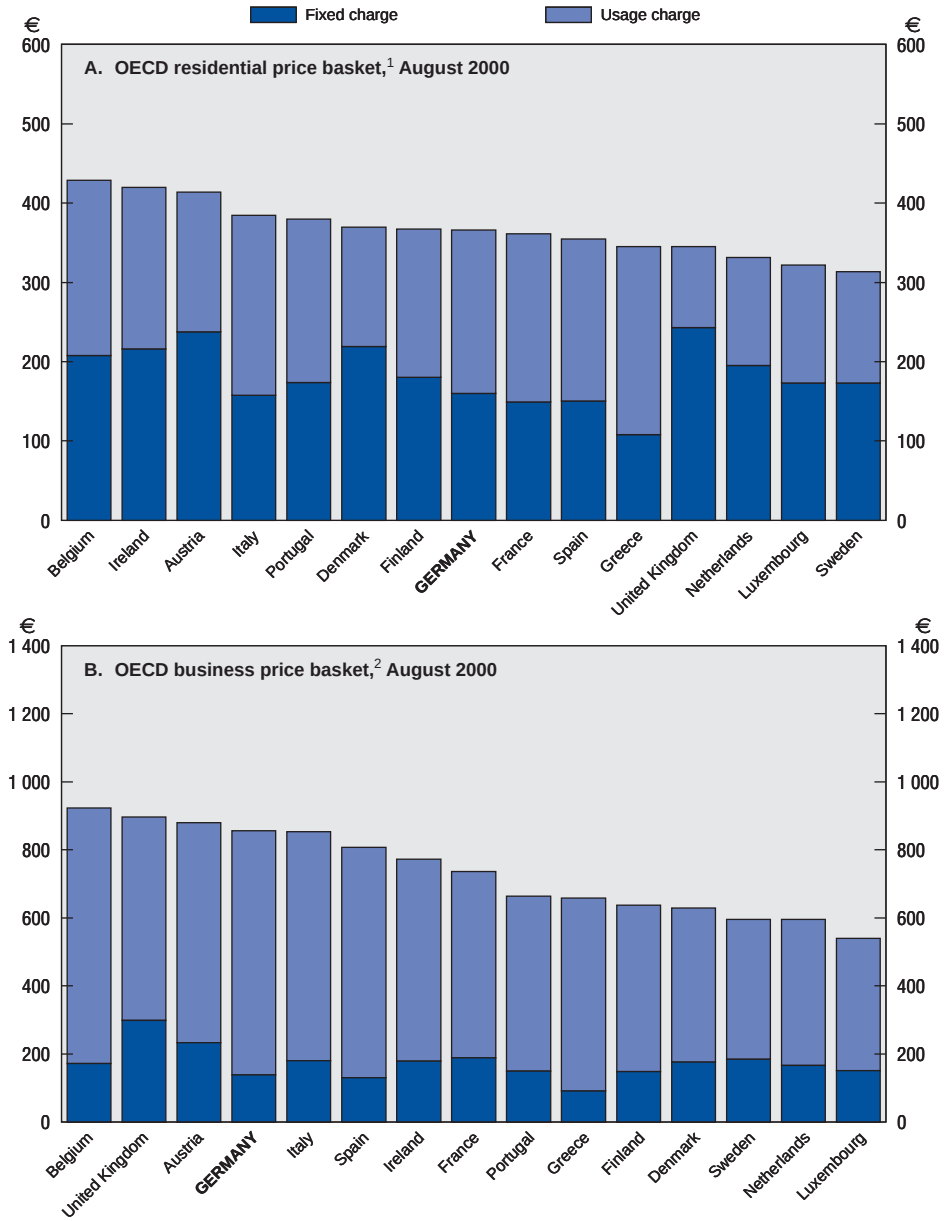
Figure 14. **Telecommunication prices in EU countries**



1. Composite basket including international calls and calls to mobile networks. Prices include VAT.
2. Composite basket including international calls and calls to mobile networks. Prices exclude VAT.

Source: OECD, *Communications Outlook 2001.*

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Nevertheless, with the number of contracts unbundling the local loop increasing steadily there are signs that competitive market structures for the local loop – and hence for local calls – are beginning to emerge (Table 16, fourth row).[88] However, new providers recently complained to the regulatory authority that DTAG often obstructs the entrance into local markets by delaying the provision of collocation rooms and the installation of interconnection hardware.[89] Techniques for bypassing the fixed network have made some progress recently as the regulatory authority has allocated spectrum required for wireless local loop systems and as DTAG partially sold regional cable TV companies. However, in all of these cable TV companies DTAG retained a stake, and arguably preferred selling blocks of shares to investors that had not yet provided telecommunication services.

Prices for Internet access have come down rapidly, and this has induced a massive increase in the use of the Internet.[90] There are indications that the number of link-ups to the Internet has risen by more than 50 per cent in two consecutive years, and that the number of Internet users increased by nearly 75 per cent in the course of 2000. However, similar to the development in other segments of the telecommunication market, DTAG hampered competition for the provision of Internet services. In 2000, the Internet subsidiary of DTAG introduced flat-rates for Internet access for final customers.[91] Competing providers also offered flat-rates for internet access, but since DTAG charged them, like its own internet subsidiary, with per minute prices for interconnections, most of them became uncompetitive and had to withdraw their flat-rate. In response the regulatory authority ruled that DTAG has to offer its competitors a wholesale flat-rate for the standard line. Furthermore, the regulatory authority decided that DTAG was no longer allowed to grant volume discounts for Internet providers, which effectively only benefited the Internet subsidiary of DTAG itself, given its dominating position in the market. The Internet subsidiary of DTAG meanwhile has withdrawn its flat-rate offer for the standard line. DTAG also obtained a ruling by a commercial court that it is no longer obliged to offer its competitors a wholesale flat-rate.

In the market for mobile phone services, competition has evolved vigorously, with DTAG's market share having dropped to less than 40 per cent. Overall, the structure of tariffs is still characterised by peak-load pricing resulting in high charges for daytime calls during the week and much lower charges for calls in the evening and at the weekend. The degree of price discrimination indicates a relative shortage of capacities, caused by the vigorous growth of traffic (Figure 15). The regulatory authority auctioned ten additional blocks of the spectrum in autumn 1999, which will become operational in 2001. Furthermore, in summer 2000 an auction for G3 licences took place (see Box 5).

The regulation of telecommunications is gradually putting more emphasis on fostering the development of competing infrastructures. Interconnection tariffs,

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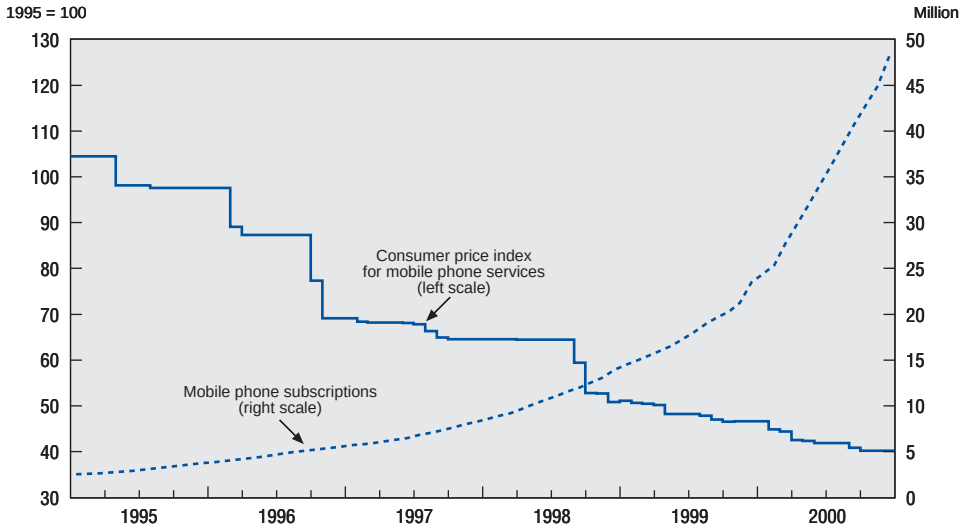
Figure 15. **Mobile telecommunication prices and market penetration**



Source: Federal Statistical Office and OECD.

which up to now are distance-related, have been ruled to relate exclusively to the number of network elements used from June 2001 on.[92] This would imply a reduction in interconnection charges on average by 23 per cent, and likely further price cuts for domestic and international long-distance calls. However, DTAG has challenged the ruling of the regulator in a lawsuit. After an initial ruling of the court in favour of DTAG the process is now pending. With the changeover to a competitive-based network having been delayed the private competitors are now likely to hold back investment, and further price cuts are endangered.

An important issue for the further evolution of the telecommunication market concerns the timing by which the present regulation of DTAG will be relaxed. The comprehensive *ex ante* regulation[93] of DTAG is motivated by the fact that markets are not yet fully competitive. As the *ex ante* regulation is planned to be withdrawn once sustainable competition has been reached, the regulatory authority has to report every two years on the progress achieved. The German Monopolies Commission adds to the assessment and makes proposals about the future regulatory design. In its first biannual report, the regulatory authority arrived at the conclusion that competition in fixed-line telecommunication still depends on tight *ex ante* regulation.[94] Whereas the German Monopolies Commission largely shares this view,[95]

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Box 5. The design and outcome of the German UMTS auction

In summer 2000, the Regulatory Authority for Post and Telekom put up for sale licences for the Universal Mobile Telecommunication System (UMTS or G3: third generation cellular services). Competitors were allowed to bid for up to three out of twelve blocks of spectrum. The acquisition of two blocks would qualify for a small licence, three blocks for a big one. Most experts agreed that two blocks of spectrum would allow the provision of basic UMTS services, but that further blocks would be necessary for proper provision of additional services. Therefore, the regulatory authority decided to let the auction determine the endowments of the licences. Thus, allocations of four, five or six licences were feasible outcomes of the process. After some screening of applicants, seven independent bidders took part in the auction. Four bidders already held licences for G2 networks, two bidders were reselling G2 capacities, and one bidder was a newcomer to the German market.

Prior to the start of the auction, most observers expected an allocation of four licences as the most likely outcome. Indeed, to avoid pressure on margins in the G2 business the incumbent operators of G2 networks had strong reasons for holding the number of UMTS licences down to the minimum of four. Moreover, the incumbents were likely to incur less expenses for setting up UMTS services and might economise on marketing outlays by transferring some share of their clients from G2 to G3 services. Whereas many observers criticised the design of the auction as anti-competitive on these grounds, the regulatory authority motivated its strategy by maximising the potential for technological innovation.

After one of the bidders, a reseller of G2 capacities, had exited the auction, an allocation of six licences associated with two blocks of spectrum was temporarily reached, at an average price of DM 11.4 billion per licence. However, the two market leaders in the G2 business continued bidding for three blocks of spectrum. When they stopped the average price of a UMTS licence had been bid up to nearly DM 16.5 billion, but none of the remaining competitors had quit the auction. Consequently, the final allocation consisted of six licences associated with two spectrum blocks, enabling the provision of basic UMTS services. Following the closing of the auction, the bidders complained about the auction design. The two market leaders claimed that a beauty contest would have been more adequate. Their competitors criticised that the two market leaders, notably DTAG, had driven up prices in an attempt to restrict competition.

the Federal Government believes that viable competition in several market segments[96] has evolved to an extent that allows abolishing *ex ante* regulation in the medium term, at least with respect to business customers. *ex ante* regulation could then be replaced by sector-specific regulation, under the responsibility of the regulatory authority. Caution is warranted in that a strictly separate assessment of sub-

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markets risks neglecting the interrelations between sub-markets that are critical for the sustainability of competition. As long as DTAG dominates the telecommunication infrastructure and especially the local loop, relaxing the regulation runs the risk that cross-subsidisation by the incumbent might distort competition in retail markets.[97] As DTAG has started offering combined products, which bundle services provided in competitive market segments with services provided in monopolised market segments, this aspect gains importance.[98]

Electricity

The electricity market is dominated by vertically-integrated conglomerates in generation and distribution as well as more than 500 publicly-owned local monopolies ("*Stadtwerke*") in local distribution (and to a lesser degree generation) with limited flexibility due to tight regulation on activities, although no sector specific regulator is in place. As has been highlighted in the last *Survey*, electricity network owners have often refused third-party access to the grid or to cut transmission fees widely judged to be too high. But in autumn 1999, the German Federal Cartel Office reinforced a ruling that local distributors had to open their grids to competitors. Following this decision, prices for industrial clients came down very rapidly. In some cases large-scale consumers were able to contract prices that cut in half previous charges. Medium-sized enterprises began setting up purchasing pools, thereby bundling demand and negotiating volume discounts. Under these pressures local distributors started to cut tariffs for industrial clients more broadly. Between December 1998 and December 2000 prices for industrial clients on average dropped by between 15 and 23 per cent (Table 18).[99] Despite this progress, prices for industrial clients in Germany are still above those in many other European countries (net of taxes), and in particular for smaller clients (Table 19).

Table 18. **Cumulated deregulation-induced drop of electricity prices for industry**
Percentage change from September 1998[1]

	Electricity prices based on:		
	Tariffs	Special contracts:	
		Low voltage	High voltage
December 1998	0.0	–0.1	–1.4
March 1999	–0.5	–0.9	–5.9
September 1999	–3.7	–6.9	–14.7
December 1999	–7.5	–9.4	–18.4
March 2000	–14.0	–15.4	–22.7
December 2000	–15.9	–14.7	–22.5

1. Without tax induced price-hump from April to June 1999.
Source: Federal Statistical Office and OECD.

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Table 19. **Electricity prices in selected countries**[1]
Germany = 100

	Industry prices			Consumer prices		
	Level of consumption					
	Small	Medium	Large	Small	Medium	Large
Belgium	107	112	92	80	96	99
Spain	71	93	96	61	74	75
France	62	80	84	68	75	80
Ireland	92	97	96	73	66	70
Italy	93	115	112	37	133	34
Luxembourg	95	102	79	112	87	92
Netherlands[2]	57	100	–	79	75	77
Austria	92	–	–	63	79	87
Portugal	76	94	84	64	99	97
Finland	40	55	48	62	53	49
Denmark	40	73	–	84	59	56
Sweden	39	57	55	88	54	55
United Kingdom	84	103	–	100	85	86
Greece	60	83	79	37	46	57

1. Without taxes, July 2000.
2. January 2000.
Source: L. de la Fuente (2001*a*, 2001*b*), and OECD.

Shrinking margins for electricity producers have accelerated structural adjustment in the electricity sector: generators are merging and local distributors are either forming partnerships, merging with other utilities or are being sold by their municipal owners. With excess capacity being reduced, productivity growth is accelerating and employment in the sector is falling. Manufacturing enterprises have considerably reduced electricity generation, which indicates that deregulation is increasing efficiency by deepening specialisation. With inefficiencies of electricity production being reduced, concerns that price declines originate solely from "unhealthy" squeezes in profit margins and prices will have to increase again in the future appear to be unfounded.

Developing competition in the electricity market for private households appears to be much slower. Prices declined in autumn 1999 when new supra-regional suppliers advertised favourable terms. Local distributors responded by cutting standard tariffs and offering special rates for higher energy consumption. According to consumer price statistics, prices for private households declined on average by 6.0 per cent in the period August 1999 to December 2000, the reduction being greater for households with higher consumption (Table 20). In summer 2000, prices started climbing again, reflecting the second stage of the ecological tax reform and "green interventions" (see Box 6). Notwithstanding the

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Table 20. **Cumulated deregulation-induced drop of electricity prices for private households**

Percentage change from August 1999

	Level of consumption			
	Small	Medium	Large	Large Including night-time tariff
December 1999	–5.9	–7.0	–8.3	–6.1
March 2000	–6.5	–7.5	–9.0	–7.0
December 2000	–6.0	–7.2	–8.4	–5.9

Source: Federal Statistical Office and OECD.

Box 6. Opening network industries to competition: two different reform strategies

The regulations set up for the telecom sector on the one hand and the electricity sector on the other hand highlight two different reform strategies to introducing competition in network industries. The opening of the telecommunication markets was largely based on establishing an independent regulator endowed with wide powers to promote sustainable competition. The stage for the regulator was set by a newly-created telecommunication law that permits comprehensive *ex ante* and *ex post* regulation of the incumbent's behaviour, including third-party access to essential inputs and the pricing of the incumbent's final consumer products. New market entrants are only examined if they achieve a dominant position in a particular market.

By contrast, the market for electricity was opened via a modification of existing laws, the cartel law (Act against Restraints of Competition) and the law regulating the supply of electricity and gas (Energy Industry Law), and no independent regulator was established. The revised cartel law subjects the energy sector to the general rules of competition and stipulates that dominating enterprises have to allow access to facilities essential for the provision of services at reasonable terms ("essential facility" doctrine). The revised energy law stipulates access to electricity networks at non-discriminatory prices. The terms of access need to be negotiated individually, although the Federal Ministry of Economics can issue a pertinent ordinance. So far, the Minister has not issued such an ordinance, but relied on sponsoring two agreements between associations representing generators, distributors and industrial consumers of electricity.[1] The two agreements are much less detailed than the rulings of the regulatory authority in telecommunication. In particular, there is no *ex ante* regulation of the level of transmission charges. Both regulatory approaches have provoked declining prices and more efficient provision of services. However,

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Box 6. **Opening network industries to competition: two different reform strategies** (*cont.*)

the two sectors differ in the degree to which utilities have the power to constrain competition by controlling facilities – such as networks – that are essential for market access of third parties.[2] This, in turn, has implications for the necessary degree of market regulation. In telecommunication, competition between utilities which provide their own infrastructure – notably networks – is largely viable. This is true in the first place in the segments for mobile phones and long-distance communication. However, the situation is quite different in the electricity sector and in other network industries such as gas supply and railways. In these sectors establishing competing networks is feasible only to a limited degree. This suggests a relatively tight regulatory approach with a strong independent regulatory authority to secure market access and prevent rent-seeking behaviour of the incumbents. For the same reason complete unbundling of vertically-integrated utilities – that is a separation of network, transmission and distribution services – appears necessary.[3] The regulatory framework would have to be even tighter – including *ex ante* price controls for final consumption – if the latter condition is not met, as is the case in the electricity and gas sectors and in railways.

On the other hand, there are some rationales for a more light-handed approach in the electricity sector and in other network branches. Numerous sector-specific regulators may hinder a uniform approach to the promotion of competition, and the risk of becoming captured by special interests is probably higher for sector-specific regulatory authorities than for general competition authorities. An integration of regulatory authorities into the competition authority or extending the jurisdiction of the latter to cover sector-specific issues might alleviate this problem. Moreover, dissolving sector-specific bureaucracies at a later stage may be difficult, and as bureaucracies might justify their existence by activism, this might result in excessive regulation. The German Telecommunications Act deals with these potential problems by providing an automatic cutback of regulation once competitive market structures have evolved.

Overall, most arguments favour relatively tight regulation of recently liberalised network industries combined with the establishment of regulatory authorities or an expansion of the powers of the competition authority. In this respect, the chosen approach should take into account existing market structures as well as the degree of potential competition within each sector. If regulatory authorities are established, however, they need to be independent and equipped with powers for investigation and sanctions without running the risk of getting into conflict with other regulators.

1. The legal quality of the associations' agreements is ambiguous. First, there is a potential conflict with the general cartel law, as there are similarities to a cartel contract. Second, it is still unclear whether these agreements are legally binding for individual network operators.
2. On this topic see also OECD (2000) and Monopolkommission (2000*b*).
3. For cross-country evidence on the efficiency advantages of unbundling of electricity transmission and generation use, see F. Steiner (2000).

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rate declines, pre-tax electricity prices for private households were for several reasons still higher in Germany than in other European countries (Table 19). Bureaucratic hurdles rendered it difficult for private households to change suppliers. Therefore, at the end of 1999, the share of new competitors in the market for small-scale electricity consumption was still close to zero. It increased over the last year, but it is estimated that at the end of 2000 only 2 to 3 per cent of private households were supplied by a competitor of the local utility.

The different regulatory approach chosen for reforming the electricity market as opposed to telecommunications may partly explain the more retarded development of competition in the electricity sector although other factors also play a role (Box 7). In telecommunication the regulatory framework is designed so as to prevent the incumbent from exploiting its dominant market position. The authorisation procedure for final consumer prices of DTAG slows down the reaction of the incumbent, giving competitors a first mover market advantage for their pricing strategies. This is different in electricity markets, where dominating utilities may cut prices without authorisation,[100] enabling the incumbents to pursue an entry-prohibiting pricing strategy.

There are a number of developments that will help increase the transparency in the electricity market and balance supply and demand more effectively. In 2000 power trading houses were established in Leipzig and Frankfurt. To ease the establishment of the houses, operators of electricity grids agreed to abolish distance related transmission charges and publish tariffs for third-party access.[101] Furthermore, the settlement of accounts with small-scale users of electricity has been based on synthesised consumption profiles, rendering real-time power measurement with expensive metering equipment unnecessary. Also, in summer 2000, a clearing-centre for facilitating the exchange of customer-related information was founded. Overall the German trading practice, while still based on the principle of negotiated access, thus moves towards the regulated access mode where tariffs are published and where there is no obligation to renegotiate prices and general terms of access each time a contract terminates.

However, with the high degree of vertical integration and concentration in the sector, non-discriminatory access to networks is still under threat, pointing again to the benefits of establishing a powerful independent regulatory authority (see Box 6). Although operators of networks are obliged by law to keep separately their activities in generation, distribution and transmission, and to publish separate accounts, further unbundling is necessary to remove incentives to discriminate against competitors. Research by the association of industrial energy users (VIK) indicates that tariffs for third-party access are indeed excessive in many regions, and there is an increasing number of complaints that grid operators obstruct access to networks by bureaucratic hurdles. In the absence of a regulatory authority fixing access prices and enforcing transmission, customers can only

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Box 7. "Green" price fixings, regulations and taxes in the German electricity market

Besides engaging in regulatory reform to foster competition, the Federal Government has intervened in various ways to meet environmental objectives:

- Nuclear energy, which currently generates almost a third of electricity supply in Germany, is to be phased out as agreed in the "consensus talks" between the Federal Government and leading power companies. The operating lifespan of existing nuclear plants will be limited and no new permits for new nuclear facilities will be issued. For each plant, an overall generation allotment equivalent to an average operating period of 32 calendar years has been fixed, from which the years a plant already has been in operation will be deducted to determine the remaining generation of each nuclear power plant to be allowed. These net-allotments may be transferred between plants. Undisturbed operation of nuclear power plants and disposal of nuclear waste will be ensured in the remaining period of nuclear power use. On the basis of this agreement the Federal Government will draw up a draft amendment to the Nuclear Energy Act. Final signing of the agreement by the Federal Government and the power companies will take place after the legislative procedures have been completed.
- Environmental taxes were introduced to curb the demand for electricity. Since April 1999, electricity consumption is taxed at DM 0.02 per kWh. The rate is being increased by DM 0.005 per kWh in January of each year until 2003. Large-scale industrial energy consumers are subject to a lower tax burden (approximately 20 per cent of the standard tariff).
- Energy generated from renewable sources for own-consumption is exempted from the electricity tax. Furthermore, power stations are obliged to buy energy generated from renewable sources at fixed minimum prices to encourage the generation of electricity from these sources.[1] The extra-costs are disseminated to all electricity retailers.
- Electricity generated by small and medium combined heat and electrical power stations (typically block-type thermal stations) remains untaxed. Similarly, natural gas power plants for combined heat and electricity generation with a high level of efficiency receive a refund of taxes on natural gas used as an input. Furthermore, grid operators under certain conditions are obliged to buy electricity generated by combined heat and electricity systems at fixed minimum prices.[2] The extra-costs resulting from this measure are disseminated to electricity retailers.

The tax-rebates, subsidies and minimum price fixing shelter facilities considered to be environmentally friendly from the consequences of the opening of markets. In the past, public authorities had supported generously the development and the usage of new power generation techniques which, under changed market conditions, came under pressure when traditional power generators reduced prices. Such pressure appears to be economically efficient: capacities should be employed such that prices charged equal marginal operating costs and

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Box 7. **"Green" price fixings, regulations and taxes in the German electricity market** (*cont.*)

no investment in additional capacity should be made unless it will recover full cost. However, the measures of the Federal Government should be seen in the wider context of environmental policies, which aim at reducing greenhouse gas emissions. The phasing out of nuclear energy will increase the need to establish low-emission plants. Thus, some preference for new techniques is reasonable. But policies supporting such techniques should be consistent with the overall regulation of energy markets. Commitment to purchase quotas of electricity generated by specific techniques combined with price fixing above market clearing levels will not result in efficient outcomes. Rather, these would require subsidies to be given to low emission plants, unrelated to the technique applied, or true CO_2 taxes. This would give clear signals to investors without distorting the choice of techniques.[3]

1. The minimum rate for photovoltaic energy is DM/kWh 0.99, for wind energy DM/kWh 0.121 to DM/kWh 0.178, for geothermal energy DM/kWh 0.14 to DM/kWh 0.175, for biomass energy DM/kWh 0.17 to DM/kWh 0.20, for small hydro energy and industrial gas DM/kWh 0.13 to DM/kWh 0.15. Some of the minimum prices will reduce progressively in consecutive years.
2. The minimum rate is 0.09 DM/kWh. It will be reduced by DM 0.005 at 1 January in each consecutive year. The duration of the law is limited to end of December 2004.
3. See *Economic Survey of Germany* 1999.

appeal to a sector-specific mediation agency, to the courts or the Federal Cartel Office. However, the potentially protracted legal procedure this implies may cause substantial hold-up costs for potential market entrants.[102]

Gas

In the gas sector, the same procedure for regulatory reform was applied as for electricity, with one major exception: The opening of gas markets was based on the "essential facility" clause in the Act against Restraints of Competition. Under the supervision of the Federal Minister of Economics, in July 2000 seller and buyer associations reached a provisional agreement about general conditions for third-party access for large-scale industrial customers and wholesale traders, which will be valid until October 2001. A successor agreement is planned to also cover private households and small-scale industrial consumers. The preliminary agreement is based on the principle of negotiated access and does not specify the level of transmission charges. Grid operators, however, agreed to publish tariffs. Interregional transmission charges shall depend on distance, but charges for the use of regional or local grids shall be unrelated to distance. The preliminary regulations for the gas market

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imply complex procedures for third-party access, and they ignore important aspects which are essential for competition to develop further, like access to storage facilities.[103] The fact that there is no comprehensive and binding fixing of rules by a public authority has already caused problems. In one case where the price for third-party access was controversial the German Cartel Office brokered a preliminary arrangement in which parties agreed to apply a preliminary tariff and to proceed with the transmission. The competition authority promised to estimate the appropriate cost-based price according to the competition law.

In winter 2000 the Federal Government decided to modify the Energy Industry Law in order to foster third-party access to gas networks.[104] However, it is unlikely to induce sufficient competition as the revised draft law neither demands full unbundling of activities nor installs a regulator. On the other hand, the seller and buyer associations reached an agreement concerning large-scale consumers to remove some barriers to third-party access and facilitate commercial access to storage.

It is difficult to assess the consequences of the opening of markets in advance. In contrast to electricity generation, there does not seem to be excess capacity in gas supply. Thus price reductions for gas would primarily result from shrinkage of margins and a reduction of retail costs. Still the price decline might be substantial. According to a survey by Eurostat, in July 2000 gas prices in Germany were higher than in many other European countries (see Table 21). To what extent the potential price

Table 21. **Natural gas prices in selected countries**[1]
Germany = 100

	Industry prices			Consumer prices		
	Level of consumption					
	Small	Medium	Large	Small	Medium	Large
Belgium	94	85	84	92	100	97
Spain	80	80	90	92	124	101
France	81	81	92	79	90	87
Ireland	88	65	76	108	88	96
Italy	105	87	85	74	119	139
Luxembourg	97	101	115	85	83	97
Netherlands[2]	82	78	–	79	73	83
Austria	92	68	74	71	96	112
Portugal	232	122	–	94	182	–
Finland	110	88	83	–	–	–
Denmark	265	247	–	216	272	226
Sweden	116	113	120	72	94	101
United Kingdom	67	58	60	77	77	82

1. Without taxes, July 2000.
2. January 2000 for consumer prices.
Source: L. de la Fuente (2001*c*, 2001*d*) and OECD.

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cuts will materialise depends on further progress in opening up the markets. The development of sustainable competition might require further steps such as the unbundling of utilities and the establishment of a regulatory authority.

Assessment and scope for further action

As has been outlined in the previous S*urvey*, an important element in the government's economic and social policy approach has been establishing round table talks between the social partners and the government for the purpose of developing a joint strategy for employment and growth. In important respects these talks appear to have produced positive results. In 1999 the Alliance had reached a broad understanding that wage policies should primarily be geared towards increasing employment. In 2000 relatively moderate wage agreements, mostly extending over two years, validated this agreement. It is remarkable that these settlements came about in a period of rapidly rising oil and energy prices, and that unions did not demand additional compensation when prices accelerated. It is important, however, that policy makers do not trade off wage moderation by the social partners against more regulation in other fields. The preferencial role the pension reform assigns to collective agreements when it comes to determining funded pension plans with tax deductible contributions indicates that such a risk might exist. The wage negotiations confirmed the trend towards more flexibility on the company level as opening clauses often were renewed. Labour market institutions are adjusting, but the degree of adjustment is not sufficient to cope with the high levels of unemployment still persisting in Germany, particularly in the new states. The authorities should support the process of introducing higher flexibility in the wage bargaining process. As has been outlined in the 1998 S*urvey*, important in this respect would be defining more realistically the legal principle that collective agreements can only be altered at the company level in the worker's favour. At present this is taken to mean higher wages. This is not appropriate under current conditions in which wage moderation has often been agreed in return for employment security. Compulsory *extensions of wage agreements* to non-contracting parties by the minister of labour restrain outsider competition, and should therefore be abolished entirely.

The trend towards more flexible working time arrangements has continued. The benchmarking group of the Alliance for Jobs has made it clear that future working time arrangements should benefit both the preferences of the workers and the needs of the employers, and must not increase costs or reinforce skills imbalances. It would be welcome if the Alliance endorsed this assessment. Progress is mixed with respect to new legislation on part-time employment and fixed-term contracts. The newly introduced legal right to transform full-time employment into part-time employment runs the risk of introducing legal uncertainty about the terms of such conversions, and this would hamper the potential

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for job creation. The new restraint on the renewability of fixed-term contracts might reduce employment flexibility at the margin, although fixed term contracts not based on formal motives played only a subdued role in the past. In both fields the impact of the new regulation should be monitored, and legislation modified if necessary. On the other hand, the relaxation of fixed-term employment conditions for older employees is a valuable step in the direction of re-evaluating the labour utilisation of older people. Over many years labour force participation by older workers has been discouraged by several measures, and this trend needs to be reversed. The re-introduction of the obligation for older recipients of unemployment benefits to search for a job – which becomes effective in 2001 because the exempting legislation expires – is another important step in this direction. But more needs to be done to eliminate the bias inherent in the pension system and other social transfers in favour of early retirements.

Perceived shortages in the supply of skilled IT personnel points to the mismatch in qualifications that has developed and appears to be developing further in certain segments of the labour market. Continued efforts need to be made to adapt Germany's training system, notably by defining new curricula and exploiting the scope for broadening existing ones. Training in public sector institutions is no efficient substitute for training provided in private enterprises. Extending mobility grants to apprentices, as is being done recently on an increased scale, is therefore worth pursuing further. As has been outlined in previous OECD *Surveys on Germany*, there is also an urgent need to reform the system of tertiary education. First steps in this direction are presently being made. Mainly, what is needed is a higher degree of competition between universities, and this encompasses the right to levy tuition fees as well as performance related remuneration for university teachers.

Lowering labour taxation and non-wage labour costs remains an issue, despite the progress associated with the phased reduction in income tax rates that are part of the income tax reform. Effective labour taxation is particularly high for recipients of social assistance due to high benefit withdrawal rates, as outlined in the 1996 *Survey*. Experiments with reducing effective labour taxation by providing in-work benefits are currently under way. But such measures would not suffice to comprehensively tackle the unemployment problem for the low qualified. Job placement for social assistance recipients is not yet efficient enough. Hence, reform is necessary to integrate the provision of social assistance benefits with the general system of unemployment benefit provision and workplacement. Careful relaxation of the restrictions imposed on temporary help agencies is worth trying and monitoring. But most importantly, the wage distribution needs to be spread out for labour demand for the low-qualified to increase. It is not helpful with respect to raising work incentives that unemployment benefits will rise due to the inclusion of bonuses and gratifications into the wage base. Active labour market measures need to be reduced and more narrowly targeted on problem groups tak-

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ing into account the outcomes of evaluations that are necessary to improve the efficiency of programme design.

Considerable progress has been made in recent years in liberalising network industries. In telecommunication regulatory reform has proved to be a major success reflected in sizeable price cuts, increased product variety, and improved quality of services. Liberalisation of postal services is relatively advanced in comparison with many other EU countries. Substantial price cuts also have been observed in the electricity sector, although less than in telecommunication. Regulatory reform in the railways system has been only modest. All network industries have in common that the incumbents attempt to exploit their market power, thus preventing the evolution of competitive markets. Often, incumbents appear to adopt a strategy where obstruction of competition instigates legal action. In such cases incumbents benefit from the fact that legal clarification is time consuming. Policy makers and regulators need to firmly counter attempts to obstruct competition. Indeed, further policy initiatives involving changes in the regulatory framework are necessary for competition to evolve. Policy actions in these fields are warranted by potentially very substantial welfare improvements for consumers and can be expected to contribute to raising productivity growth in the economy. Major outstanding issues regarding competition in network industries are:

- In telecommunication, competition is most underdeveloped with respect to the local loop, as is the case in many other countries with liberalised telecommunication markets. To enhance competition it is essential that the incumbent Deutsche Telekom, sells its remaining shares in the cable TV network. More generally, sector specific *ex ante* regulation of the Deutsche Telekom should not be relaxed as long as the incumbent dominates the telecommunication infrastructure. Otherwise there is the risk that cross-subsidisation by the incumbent might distort competition in retail markets.
- Regarding the Post Office, delivery of letters is still tightly regulated and in part exclusively reserved for the incumbent monopolist *Deutsche Post* AG. While the present charges for mail delivery, which seem to be high in international comparison, were originally set to terminate in 2000, they have been extended until the end of 2002, following an intervention by the government with the regulatory authority. The prolongation implies higher costs for consumers and reduced competitiveness for enterprises. Hence, the present charges for mail delivery should not be extended beyond 2002, and the monopoly rights need to be terminated. Moreover, the Post Office needs to be fully privatised to exclude future conflicts of interest between ownership and regulation.
- A major weakness of the present regulatory design in railways is the lack of a true separation of the track and the stations from the provision of

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transport services allowing the incumbent to exercise its market power to prevent competition to develop. The provision of transport services should therefore be unbundled from the operation of the grid. Further consideration also needs to be given to establish a strong independent regulatory authority for the railways system.

– With vertically integrated conglomerates dominating electricity markets, non-discriminatory access to networks is still under threat. Unbundling of electricity generation, distribution and transmission is therefore required. This holds analogously for the gas sector. In both sectors there is no comprehensive and binding fixing of rules by a public authority. Establishing an independent regulatory authority should therefore be considered.

Important regulatory issues also arise with respect to non-network sectors:

– Community or state governments typically act as guaranteeing authority for public sector banks, effectively lowering their refinancing costs in comparison to private sector banks. This has lead to a long-standing conflict between the German authorities and the EU commission. There are hardly convincing reasons, however, why banks should be government owned at all. Therefore the state owned banks should be privatised.

– In retailing, comparatively restrictive regulations on rebates and bonuses will be abolished without replacement. This will put German retailers on a more equal footing with international competitors in e-commerce, and should strengthen retailing competition more generally. In addition, the regulation on shop opening hours should be liberalised further, and several state governments pledged for such a move. But no new legislative initiative has emerged yet.

Box 8 summarises OECD policy recommendations for structural reform in labour and product markets based on past jobs strategy reviews and the policy initiatives which have followed.

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Box 8. A review of progress on structural reform

Since the last review a number of policy measures have been implemented. This box summarises actions taken over the entire period since the original review in the OECD Economic Survey of 1996.

Previous proposal	Action taken	OECD assessment/ recommendations
A. Progress in labour market reform		
I. Increase wage and labour cost flexibility		
• Encourage wage differentiation, greater plant-level bargaining and opening clauses.	Minimum wage in construction was declared binding in mid-1997. The scope for the government to extend wage agreements to non-contracting parties was widened in 1999.	Wage bargains should incorporate greater flexibility; widen the legal framework covering wage negotiations; eliminate administrative extension of wage agreements.
• Facilitate the employment of older workers and reduce incentives for early retirement.	Early retirement age to be increased; part-time work of older workers is subsidised under some conditions; eligibility conditions for invalidity pensions tightened. Actions reversed in 1999. Preferential job-search requirement for older unemployed was terminated.	Consider further means to raise the retirement age (stricter eligibility criteria, actuarial discounts for pension benefits) and keep transitional arrangements to a minimum.
II. Increase working-time flexibility and ease employment security provisions		
• Reform regulations underpinning inflexible working practices.		Review effects of regulations.
• Liberalise terms for renewing fixed-term contracts.	Renewal period and admissible frequency of renewals increased; scope for secondments extended. In 2000 liberalisation of renewals for older workers, but tightening otherwise.	Monitor employment effects and continue with reforms.
• Facilitate part-time work.	More flexible work arrangements in the public sector; subsidies to part-time work for older workers. In 2000 workers given legal right to switch to part time.	Avoid further subsidisation of part-time work for older workers. Monitor right for part-time employment and abolish if necessary.

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Box 8. **A review of progress on structural reform** (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Reform dismissal protection.	Exemption from general dismissal law widened; social criteria for dismissals relaxed. Both reversed in 1999.	Liberalise dismissal protection. Relax restrictions on temporary help agencies and monitor outcome.
III. Reduce the distortions arising from unemployment insurance and related benefits		
• Improve the incentive structure of benefits.	Minimum ages for receiving unemployment benefits for more than one year increased by three years; redundancy payments credited against unemployment benefits, reversed in 1999. In 2001 more generous unemployment benefits for older unemployed were not renewed.	Increase the gap between social assistance payments and the earnings of low-income families.
• Restrict the criteria for not accepting job offers, strengthen penalties, tighten controls.	Qualification no longer reason for rejecting job offers; definition of acceptable wage narrowed; internal auditing introduced of criteria; controls tightened, but reversed in 1999.	Monitor criteria for refusing job offers or taking up employment and narrow if necessary.
• Lower social security contribution rates.	Temporary curbing of certain social transfers; increase in VAT to avoid further rises in rates; pension reform legislation passed (reversed in 1999); pension contribution rates lowered by introducing new eco-taxes; obligation to pay contributions extended to casual employment and "apparently" self-employed. In 2000 experimental schemes for low-qualified workers introduced.	Curb social spending; consider further measures to secure financing of the pension system (retirement age, basic pension, funding). Keep the flexibility of casual employment and self-employment by redefining the pertinent rules for paying contributions
• Simplify the personal tax system and lower marginal taxes.	Income tax reform laws implemented in 1999, 2000 and 2001.	Simplify the tax system and monitor the tax reform.

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Box 8. **A review of progress on structural reform** (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Reduce the generosity of sickness benefits and strengthen incentives to avoid abuse.	Statutory sick pay lowered from 100 per cent to 80 per cent, reversed in 1999.	Reduce the generosity of minimum sickness benefits.
IV. Enhance active labour market provisions		
• Exclude measures from the requalifying period for unemployment benefits.	Training periods will not re-establish eligibility for another entitlement period.	Continue reform.
• Target ALMPs at at-risk groups.	More flexible work contracts for formerly unemployed. Communes given greater powers to introduce job-creation programmes and new loan facility given for associated non-wage costs; ALMPs extended (both training and work provision) and eligibility conditions relaxed.	Monitor the efficiency of work contracts for the unemployed, in particular in relation to employment provision measures. Tighten new powers of communes to organise job-creation programmes to avoid simply transferring benefit recipients from one level of government to another. Target ALMPs on at-risk groups and reallocate funds in favour of training and other measures aiming at integration in the primary labour market. Evaluate all programmes.
• Base employment subsidies on wages below market rates.	Admissible wage base lowered; special conditions for the new Länder phased out.	
B. Improving skills, entrepreneurial dynamism and competition		
V. Improve labour force skills		
• Preserve and restore the attractiveness of the dual vocational training system, clarify its relationship to higher education.	Curricula for apprenticeships revised and new occupations introduced. Public support scheme to create additional training introduced. In 2000 mobility grants for apprentices widened.	Continue to revise vocational training schemes and preserve benefits from training apprentices. Do not introduce a training levy; monitor the efficiency of public support schemes; avoid attaching subsidies to training.

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Box 8. **A review of progress on structural reform** (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Shorten higher education and focus it on more occupational-oriented studies.	Universities have been given new scope to introduce shorter studies leading to degrees that are internationally comparable.	Shorten higher education and make it more occupationally oriented. Make universities more performance-oriented. Greater use of fees or student loans.
VI. Enhance creation and diffusion of technological know-how		
• Deepen financial markets, foster venture capital and reduce regulatory barriers.	Laws implemented to improve flexibility in financial markets, ease access of companies to equity capital and change corporate governance.	Monitor development of risk capital; introduce pension funds.
VII. Support an entrepreneurial climate		
• Facilitate the establishment of new enterprises.	Accelerated depreciation allowance for newly-founded enterprises; wage subsidies for small new-founded firms.	Continue to reduce regulatory impediments.
• Increase the knowledge of the new owners of enterprises in the new Länder.	No major changes.	Support projects to diffuse knowledge.
• Reduce the overall tax burden.	Abolition of wealth tax and the business capital tax; reduction of the inheritance tax; reduction of household tax but temporary increase in business tax. In 2001 major income and business tax reform.	Monitor the tax reform.
VIII. Increasing product market competition		
• Simplify planning approvals.	Approval procedures simplified and entry into trades made somewhat easier.	Monitor the effectiveness of the new regulations; continue reform and modernisation of law on trades.
• Liberalise shop-closing hours.	Statutory shop opening hours extended in 1997.	Liberalise further statutory shop opening hours.

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Box 8. **A review of progress on structural reform** (*cont.*)

Previous proposal	Action taken	OECD assessment/ recommendations
• Encourage competition in the network sectors.	Energy, telecommunication and postal service sectors liberalised, and competition in the first two sectors is developing rapidly. Competition law widened to cover network sectors and transport. Telecom regulation established.	Monitor whether regulations to enforce the opening up of networks are effective and promote contestable markets. Consider establishing strong independent regulatory authorities for electricity, gas and railways. Support restructuring of community generators and resist demands to protect them. Resist demands to protect the interests of the incumbent provider.
• Services by communities need to be more closely market-tested.	No major changes.	Abolish preferential tax treatment of public utilities.
• Pursue privatisation.	Telekom and Deutsche Post partly privatised.	Step up privatisation.
• Reduce subsidies.	Subsidies benefiting construction reduced overall but reduction of coal subsidies slowed. Subsidies to east placed on new basis.	Avoid new industrial subsidies; set up schedule for reducing existing subsidies; close tax expenditures. Reform regional support.
C. Enhancing sustainable development		
• Integrating sectoral policies within an economy-wide framework with clearly defined environmental targets		Same as previous *Survey*.
• Increase the cost-effectiveness of environmental measures by greater use of economic instruments.		Same as previous *Survey*.
• Set energy taxes so as to reflect relative emission intensities and equalise rates across energy sources and economic activities.		Same as previous *Survey*.
• Establishing a timetable for the phasing out of nuclear power stations.	Done.	

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IV. The economic integration of the new Länder

Introduction

The economic development of eastern Germany over the ten years that have passed since the wall came down has been largely shaped by the full integration of the new states into the western German market economy. At the time of reunification analysts pointed to the benefits to be expected from taking over western Germany's legal and economic system and from receiving assistance from the west both financially and in terms of human capital. But concerns were also voiced that completely integrated markets for goods and factors of production could over-stretch the adaptability of the new states and instigate demands for transfers to compensate for the weak position of the east in such integrated markets. In the event, both the positive expectations and the warnings were warranted. Unification extended – virtually in one sweep – the western German legal, economic and social system to the new states. Accordingly, the new states were exposed on impact to a highly productive competition-based environment in which their enterprises faced enormous difficulty to compete at the wage levels that were established. On the other hand the new states were instantly embraced by the western German solidarity principle.[105] This implied *inter alia* that the elaborate western German social security system was fully and rapidly extended to the population in the eastern part of the country, that work provision measures were boosted to cushion labour shedding in the east, that the new states and communities received massive budgetary transfers from both the Federal Government and the old states, largely equalising per capita tax revenues across the federation, and that new and privatised enterprises obtained substantial financial aid to cope with the competitive disadvantages associated with transition.

In several respects the achievements of restructuring the eastern German economy are impressive. Viable market structures have largely been established with formerly state-run enterprises having been privatised and newly founded enterprises accounting for a large share of production. Earnings of the employed and transfer incomes of the non-employed have risen fast. A large part of the housing stock has largely been renewed and extended, new infrastructure has been established on a large scale, and tremendous environmental damages inher-

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ited from the GDR have been remedied. However, in the second half of the 1990's the economic convergence process stalled. Economic growth in the east has stagnated at relatively low levels, a sizeable productivity gap with the west persists and unemployment rates are twice as high as in the old states. These setbacks have occurred against the background of an almost unchanged high level of west-to-east transfers over the 1990s, amounting to an annual average of some 4½ per cent of western German GDP on a net basis, excluding tax concessions and certain debt take-overs by federal institutions.

The choice of appropriate policy instruments to foster rapid convergence of the eastern German economy with that of the west was difficult. Blueprints based on past experiences were not available, and policymakers had to act quickly in a great variety of fields under conditions of severely limited information. Beginning in 1990 the OECD *Surveys on Germany* have contributed to the policy debate on the economic integration of the new states. The stance taken by the OECD in the early 1990s was that convergence to the western German economy should be market driven. Key recommendations at the time were that the restructuring of the eastern economy and social support would require sizeable real transfers from the western public budgets, and these were assessed to pose no major problem for the west. Simultaneously, the OECD warned that setting wages in the new states above market clearing levels would jeopardise convergence because of adverse effects on investment and job creation. Massive and sustained levels of financial support were judged to risk severe distortions in eastern Germany's economic structure. Aid would therefore have to be only temporary, tapering off over time.

The developments that occurred within roughly ten years after currency union and reunification underline the relevance of a fundamental stock taking with respect to the effectiveness of integration policies and the identification of measures to improve them. The plan of this chapter is as follows: Main features characterising the macroeconomic and structural evolution of eastern Germany are highlighted in the next section. The following section seeks to identify major forces shaping the economic performance of the new states. Based on this analysis the fourth section derives a number of policy options that should foster further economic convergence. Finally, the last section concludes and summarises.

The evolution of production, employment and incomes

After a buoyant start growth has levelled off at low levels…

The sectoral contribution to growth and employment in the new states changed markedly over the last decade, shaping economic activity in the aggregate (Figure 16):

– The manufacturing sector collapsed in 1990 and 1991, accounting for the bulk of the contraction in aggregate production at the beginning of the

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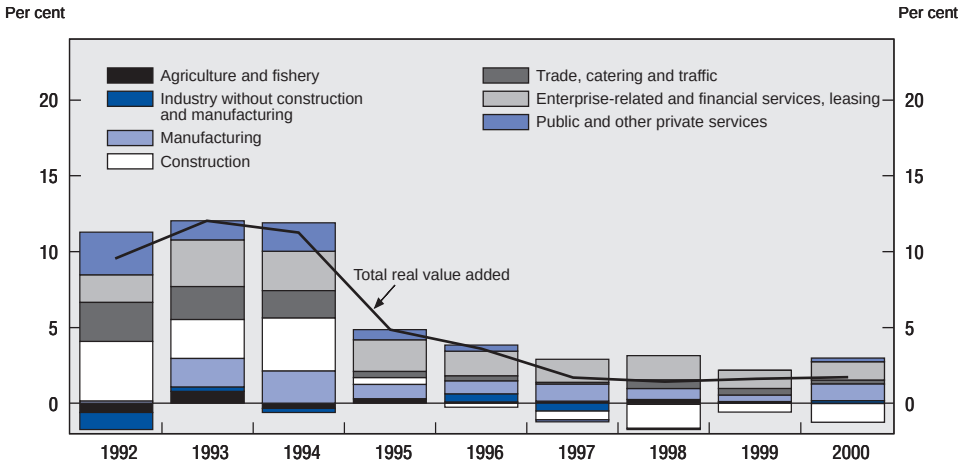
Figure 16. **Contributions to value added growth by sector**[1]



1. Value added in 1995 prices. Total value added growth is decomposed by a weighted sum of value added growth by sector, with the weights being the share of sectoral value added in the total recalculated each year. Excluding Berlin.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder" and OECD.

decade. The sector started growing slowly in 1992, but expanded at rates of around 20 per cent in the two years thereafter. From the middle of the 1990s on, manufacturing value added expanded at annual rates above 7 per cent on average, becoming a main contributor to aggregate growth. While the export share still remains substantially below the level of western Germany, in recent years high-growth branches within the manufacturing sector experienced marked improvements of their export performance. But labour shedding continued (Figure 17).[106]

– In the first half of the decade the construction sector was the main contributor to growth. The early lead of construction mainly reflects high rates of public investment in infrastructure, high priorities by policy makers to upgrade and replace east Germany's housing stock as well as high investment by enterprises in business related buildings. Private-sector investment was spurred by high rates of subsidisation. By the middle of the decade the share of construction in east German value added totalled three times the size of the sector's share in the west. Construction also constituted the main source for employment growth in the first half of the 1990s. The picture changed substantially in the second half of the 1990s, with the role of construction twisting from the

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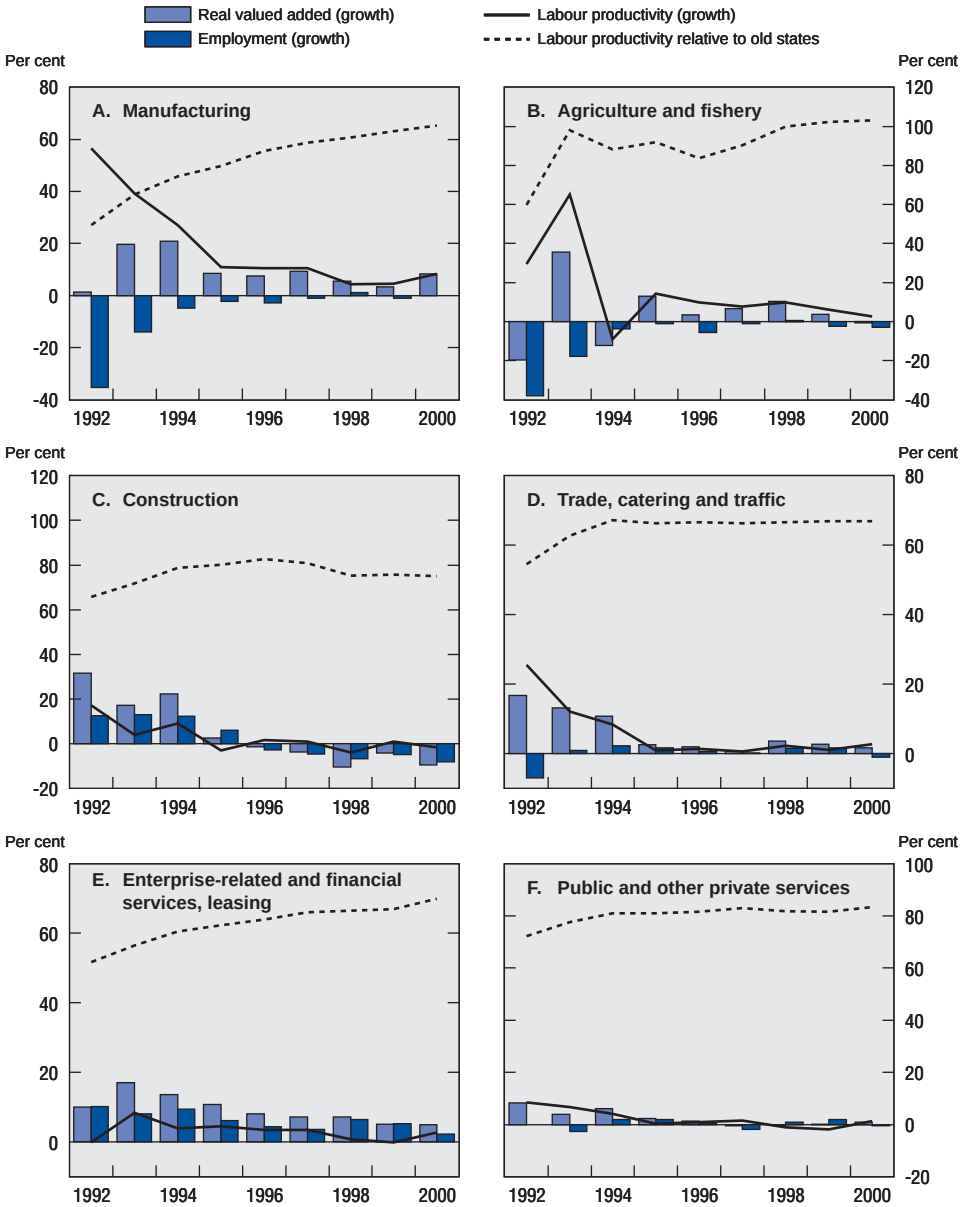
Figure 17. **Economic performance by sector**[1]



1. Excluding Berlin.

Source: Arbeitskreis "Volswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und Länder" and OECD.

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main contributor to aggregate growth to the main brake on further growth. This was related to over-capacity in both the housing sector and business construction, largely created by special depreciation allowances (*Fördergebietsgesetz*) that were finally phased out by the end of 1988. With activity slowing employment creation also came to a halt and employment declined at increasing rates in the second half of the 1990s (Figure 17). Labour shedding in the second part of the 1990s is almost exclusively attributable to the downsizing in construction.

– Business related and financial services grew rapidly from a very small base, the low initial output level reflecting the irrelevance of these sectors in the GDR. While growth slowed down in the second half of the 1990s, it was higher in the second half than in all other sectors, but manufacturing. Employment continued to grow over the entire decade (Figure 17).

The sectoral composition of real value added reflects these developments. The largest sustainable gains have been made by the manufacturing sector and enterprise-related and financial services. The share of construction in overall value added is still very high, but after a further boost in the middle of the decade it is coming back down to the level that prevailed at the beginning of the 1990s. Despite the steady expansion of the value-added share in manufacturing, its share in employment is now some 11 percentage points lower than in 1991. Indeed, by the end of the decade manufacturing employment totalled about one fourth of what it was in the end phase of the GDR, reflecting the magnitude of the previous overmanning and slow employment creation after reunification. Overall, the sectoral composition of production and employment is gradually approaching a pattern that is normal in a market economy, although the value added share of construction – totalling two and a half times the share in the west – is still very large (Table 22).[107]

Following the collapse in manufacturing production and employment in 1990 and 1991, aggregate growth was buoyant in the first half of the decade, with real GDP growing at annual rates of around 10 per cent.[108] But activity decelerated markedly from the middle of the decade onwards, and for the last four years growth even fell short of that in the west, at rates of below 2 per cent. While the slowdown occurred in all major sectors, aggregate growth suffered most from the recession in construction. In contrast, those segments of the economy with relatively strong growth – notably manufacturing and business-related services – are still too small to produce robust growth in the aggregate. Overall, real GDP in the new Länder grew by about 54½ per cent cumulatively between 1991 and 2000 (as opposed to some 11½ per cent in the west), with the eastern share in total Germany's real GDP rising from 8½ to 11½ per cent (both regions excluding Berlin). Measured on a per capita basis real GDP in the new states reached 60½ per cent of the western level in 2000, falling short of the EU average by some 27 per cent. While the average gap in economic activity between the new and the old states is still large, some regions in the

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Table 22. **Value added and employment by sector**[1]

Share of total, per cent

		1991	1995	2000	2000 Old states
Agriculture and fishing	Value added	3.1	2.3	2.6	1.2
	Employment	7.3	4.0	3.6	2.4
Industry, excluding construction and manufacturing	Value added	6.7	3.9	3.7	2.2
	Employment	3.5	1.9	1.3	1.1
Manufacturing	Value added	10.4	11.5	14.4	23.5
	Employment	25.8	15.0	14.8	22.9
Construction	Value added	12.5	16.9	11.2	4.5
	Employment	10.3	17.4	13.4	5.9
Trade, catering and traffic	Value added	15.5	16.3	16.3	18.1
	Employment	20.5	22.4	23.7	25.5
Enterprise-related and financial services, leasing	Value added	18.1	20.4	25.3	31.3
	Employment	6.0	9.2	11.8	14.8
Public and private services	Value added	33.8	28.8	26.5	19.1
	Employment	26.6	30.1	31.3	27.4
Total	Value added	100.0	100.0	100.0	100.0
	Employment	100.0	100.0	100.0	100.0

1. Value added in 1995 prices.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und der Länder"; and OECD.

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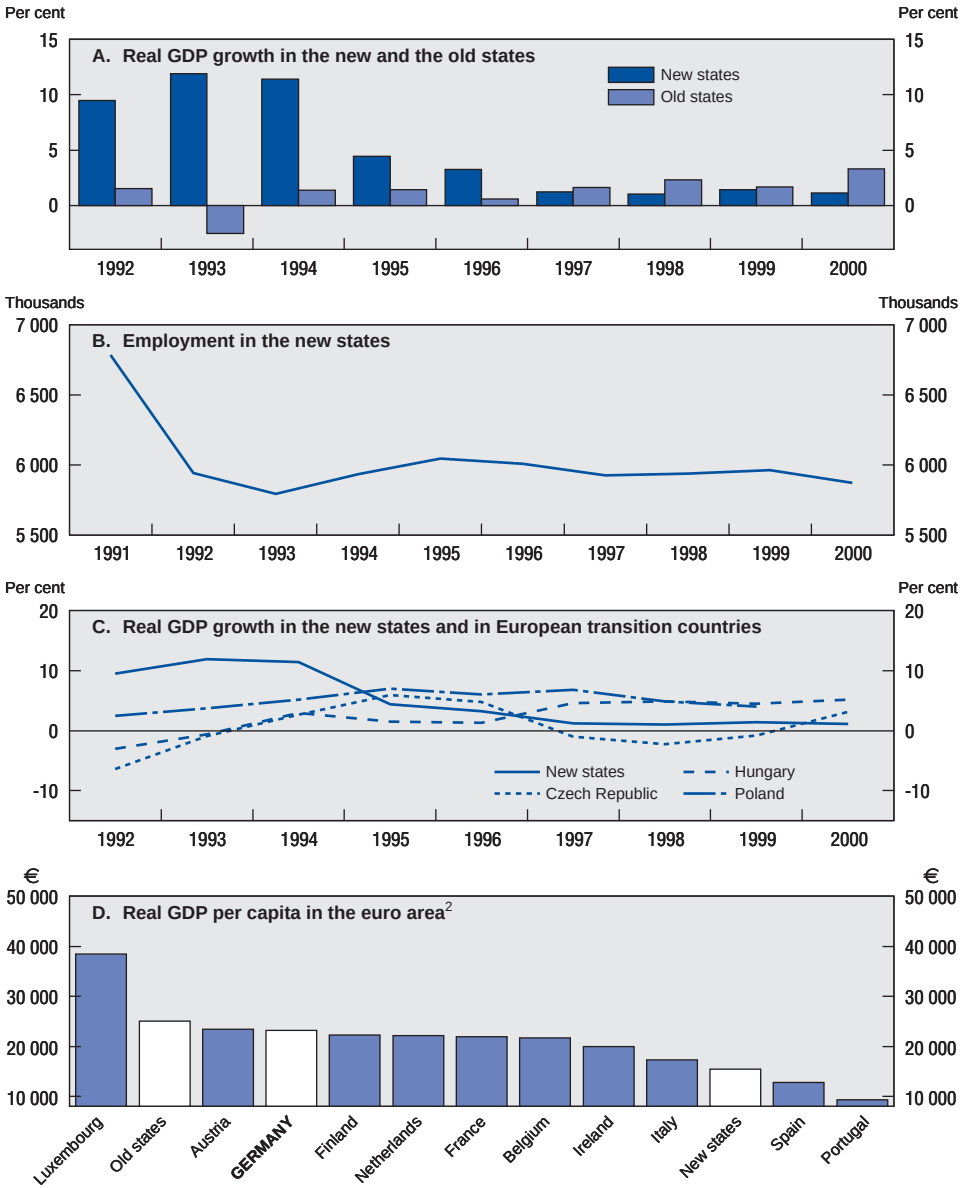
Figure 18. **GDP and employment**[1]



1. Excluding Berlin.
2. In 1999.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und Länder" and OECD.

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east have surpassed weaker regions in western Germany.[109] Despite output growing rapidly from 1991, mainly due to the labour shedding in manufacturing, employment continued to decline in the first years of the 1990s. Sizeable employment gains occurred temporarily in the mid-1990s, helped by massive employment creation programmes of the Federal Labour Office. In the second half of the decade employment fell again, however, and in 2000 employment undercut the 1991 level by 13½ per cent (Figure 18).[110]

... with a sizeable productivity gap persisting despite substantial labour shedding

In the first half of the 1990s, labour productivity grew rapidly, from roughly one-third of the level in the old states in 1991 to some 55 per cent in 1995.[111] This productivity growth, some 5 per cent per year, was higher than in any of the eastern European transition countries. Productivity gains were particularly high in manufacturing. Indeed, the pressure to adjust was higher in manufacturing than in the other sectors: initial productivity levels were very low, due to massive over-manning inherited from the old system, while exposure to competition from western German and foreign companies was higher. But similarly to the evolution of output, the catch up process with the old German states slowed in the second half of the decade, and in 2000 labour productivity totalled around 68 per cent of the western level for the economy overall and 65 per cent in manufacturing.[112] However, with the two economies being fully integrated with respect to the rules governing the economic and the social systems, similar productivity levels by economic sector between the new and the old states are a key condition for economic convergence of the new and the old Länder.

Real wages have risen strongly...

Between 1991 and 2000 gross compensation per dependent employee rose by some 91 per cent, raising its level relative to the west from 49 to 77 per cent. Rapid wage increases were mainly driven by the attempt to reach fast convergence with western wage levels. Already in 1991 the social partners agreed on a wage path that aimed at achieving wage equalisation with the west within five years. Wage increases were buoyant within the first three years after reunification, and today's wage levels relative to the west are still dominated by the early settlements. In recent years, wage increases have levelled off, the slow down reflecting not only more modest collective bargaining agreements but also a significant undercutting on the level of individual companies of collectively bargained wages (see further below) (Figure 19, left panel).

... and unemployment is high

When manufacturing employment collapsed at the beginning of the 1990s unemployment increased accordingly, although it was cushioned by active labour market measures and schemes to reduce labour force participation (Figure 20).

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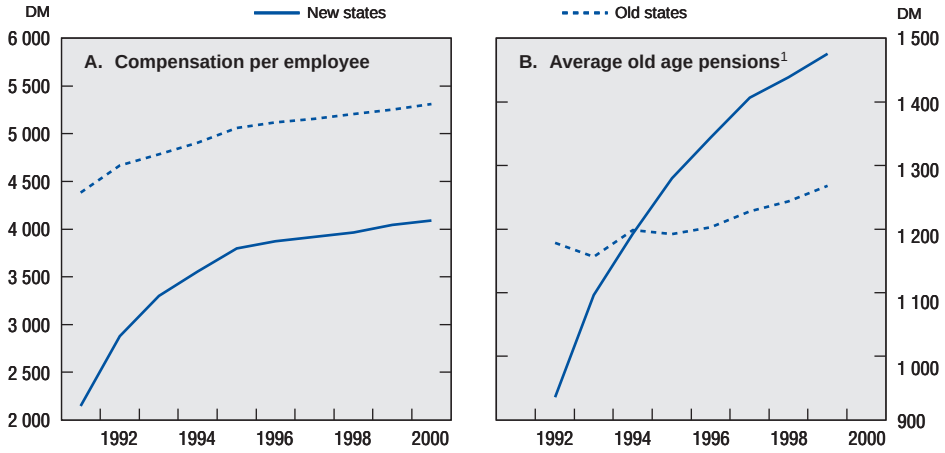
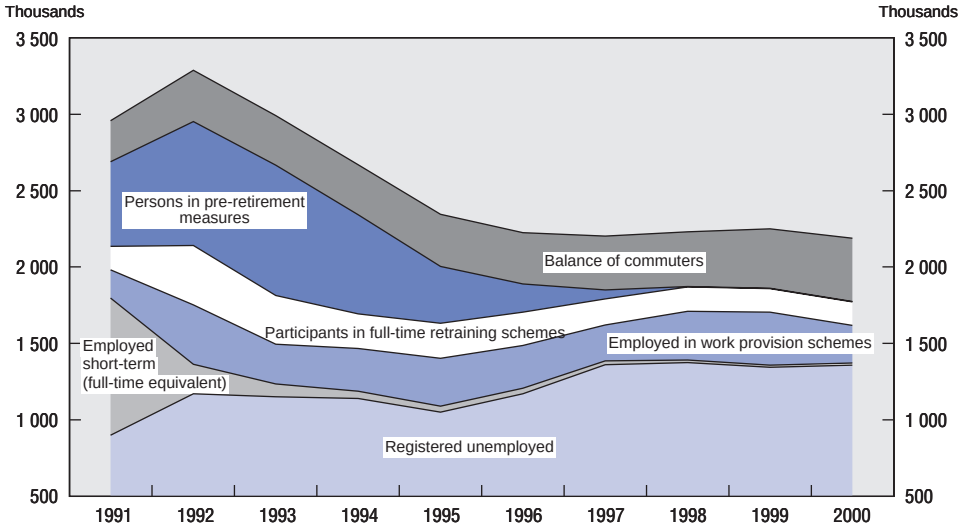
Figure 19. **The evolution of wages and pensions**
DM per month



1. Pension insurance for workers and salaried employees.
Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und Länder"; Pension Report 2000 of the Federal government and OECD.

Figure 20. **Unemployment and active labour market measures in the new states**



Source: Institute for Labour Market Research of the Federal Labour Office and OECD.

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Rapid structural change, as it occurs in the new German states, is normally associated with high levels of labour turnover between jobs and into and out of unemployment, implying higher levels of unemployment than would be observed otherwise.[113] Indeed, movements between jobs indicate a high dynamism in the eastern German labour market: In 1997 14 per cent of the employees changed their employer, as opposed to 9 per cent in the old states, and since unification some three quarters of employees have changed their working place.[114]

However, the persistence of high unemployment rates and the high level of long-term unemployment point to serious deficiencies in adjustment mechanisms on the labour market. According to official statistics, average unemployment duration totals some 30 weeks,[115] more than in the west, and about one-third of the registered unemployed are jobless for more than a year. With active labour market policies (ALMPs) – work provision and training schemes – mainly targeting the long-term unemployed, long-term unemployment would even be higher in the absence of such schemes. Moreover, unemployment is frequently re-entered immediately after completion of one or more consecutive spells of ALMPs, and it has been estimated that long-term unemployment may be twice as high as apparent in the official figures if such episodes were considered as one extended unemployment spell.[116] Econometric studies indicate that long-term unemployment is more likely for the less qualified, the older and for women.[117] Indeed, although the employment ratio of the low-qualified was already much lower than in the old states at the beginning of the 1990s, it continued to decline in the years thereafter.[118]

... despite efforts to reduce labour supply

Starting immediately after reunification, ALMPs were applied on a large scale, together with early retirement schemes that aim at reducing labour force participation of older people. While active labour market measures where meant to preserve and develop the human capital of those made redundant for later re-employment, they were also seen as social measures to avoid even higher levels of unemployment. The scale at which various ALMPs were applied was uneven over time (see 1999 *Survey*).[119] In 2000 people in full-time active labour market programmes and recipients of unemployment benefits without an obligation to accept job offers totalled 7.8 per cent in terms of employment (Figure 20). However, the total degree of under-utilisation of labour is even larger: early retirements on account of unemployment expanded rapidly and in 1999 totalled some 8 per cent in terms of employment.[120] Adverse employment prospects also account for part of the early retirements on account of incapability to work.

Despite these efforts the labour force participation rate in the east is still somewhat higher than the one in the west, higher female participation accounting for most of the difference.[121] In addition, the annual average working time of employees is higher than in the west (by more than 7 per cent).[122] Measured in

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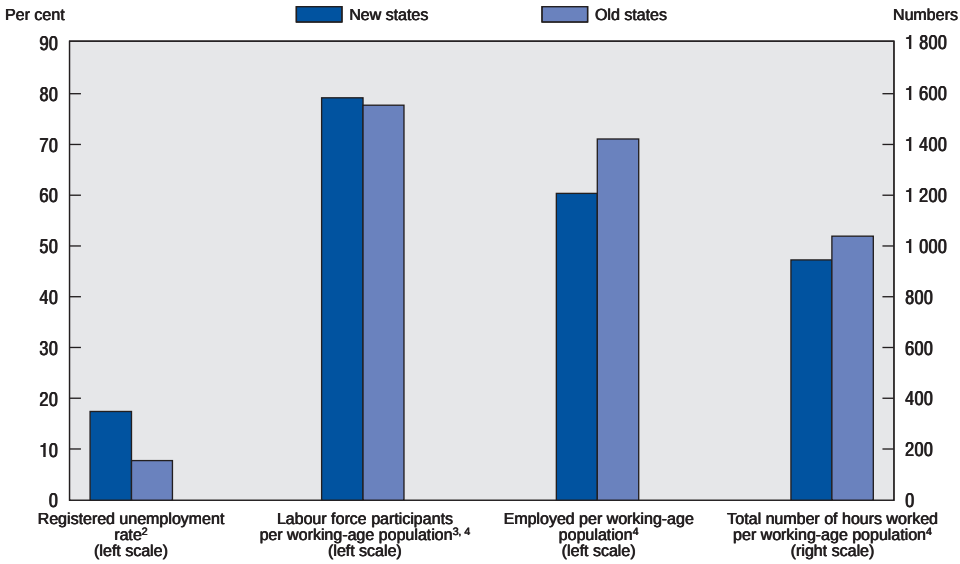
Figure 21. **Labour force participation and employment rates, 2000**[1]



1. Old states including Berlin, except for the unemployment rate.
2. Registered unemployment as a percentage of the civilian labour force.
3. Labour force participants comprise the employed, the unemployed, participants in training courses offered by the Federal Labour Office and persons in "pre-retirement" measures (taking into account commuters).
4. Working-age population: population aged from 16 to 65.

Source: Federal Statistical Office; Institute for Labour Market Research; Institut für Wirtschaftsforschung Halle and OECD.

hours per working age population the difference in the volume of employment – including work provision schemes – between east and west is relatively moderate (Figure 21). However, although labour supply in the new states might decline further towards western levels, labour market conditions need to adjust if the work preferences of the people in the east are to be accommodated.

Incomes have converged much more rapidly than output…

With transfer incomes growing strongly as well household incomes in the new states have largely approached those in the west. According to the 1998 Income and Consumption Survey, both average household gross and net incomes totalled some 75 per cent of the level in the old states, and information from other sources similarly suggests a net income ratio of around 80 per cent.[123] In real terms this would imply net household disposable income totalling around 85 per cent of the western level, given that consumer prices (including rents) in the new state fall

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short of those in the west. For both east and west the household net income distribution peaks in the same income bracket (between DM 2 000 and DM 3 000 per month). But in the east the distribution is more concentrated around lower income groups than in the west, and the opposite is true for households in the top income range. The net income distribution reflects a relatively high degree of income equality, which is partly attributable to the operation of the benefit system: Measured in terms of net equivalence incomes, where incomes of individual household members are weighted according to their position in the household, the income distribution in the new states has become more dispersed in the 1990s, but is still more equal than the one in the west (Figure 22).[124]

With the evolution of unemployment-related benefits and pensions being linked to the development of wages, large wage increases were also a main driving force behind rapid increases in transfer income of households from public sources.[125] Gains were particularly pronounced for retirees. With the western German rules of pension adjustment having been extended to the east, an initial boost in pensions received was succeeded by large annual adjustments driven by rising wages.[126] Between 1991 and 1999 average old-age pensions in the new states have increased by 140 per cent. On average public old-age pensions in the east exceed those in the west, and this is also true for the ratio of average old-age pensions to average compensation per capita. Mainly this is due to differing employment biographies: Given that the German pension system links the level of entry pensions to the length of the working life, high pensions in the new states are largely attributable to longer uninterrupted employment spells than those that occurred in the west. In the GDR a high level of labour input was needed to fulfil economic plans, making up for economic inefficiencies. However, for similar reasons eastern retirees often receive less additional pensions from other sources than those in the west. The difference between eastern and western employment biographies is particularly pronounced for women. In 1999 average old-age pensions paid for females exceeded the western level by 39½ per cent, as opposed to 7½ per cent for males (Figure 19, right panel, and Table 23).

The composition of household income by source reflects the high transfer dependency of the eastern economy. Public transfers account for 32 per cent of average gross income, 10 percentage points more than in the old states. Between 1993 and 1998 the share of transfers has increased by 2½ percentage points. The rise in pensions and increases in child benefits, which occurred in Germany overall, are main factors behind this increase. Apart from pensions, the level of unemployment-related transfers and child benefits relative to the income of salaried employees also exceeds the western levels. Wages account for the same income share as in the old Länder while the contribution of income from self-employment and property income falls short of the one in the west (Figure 23).

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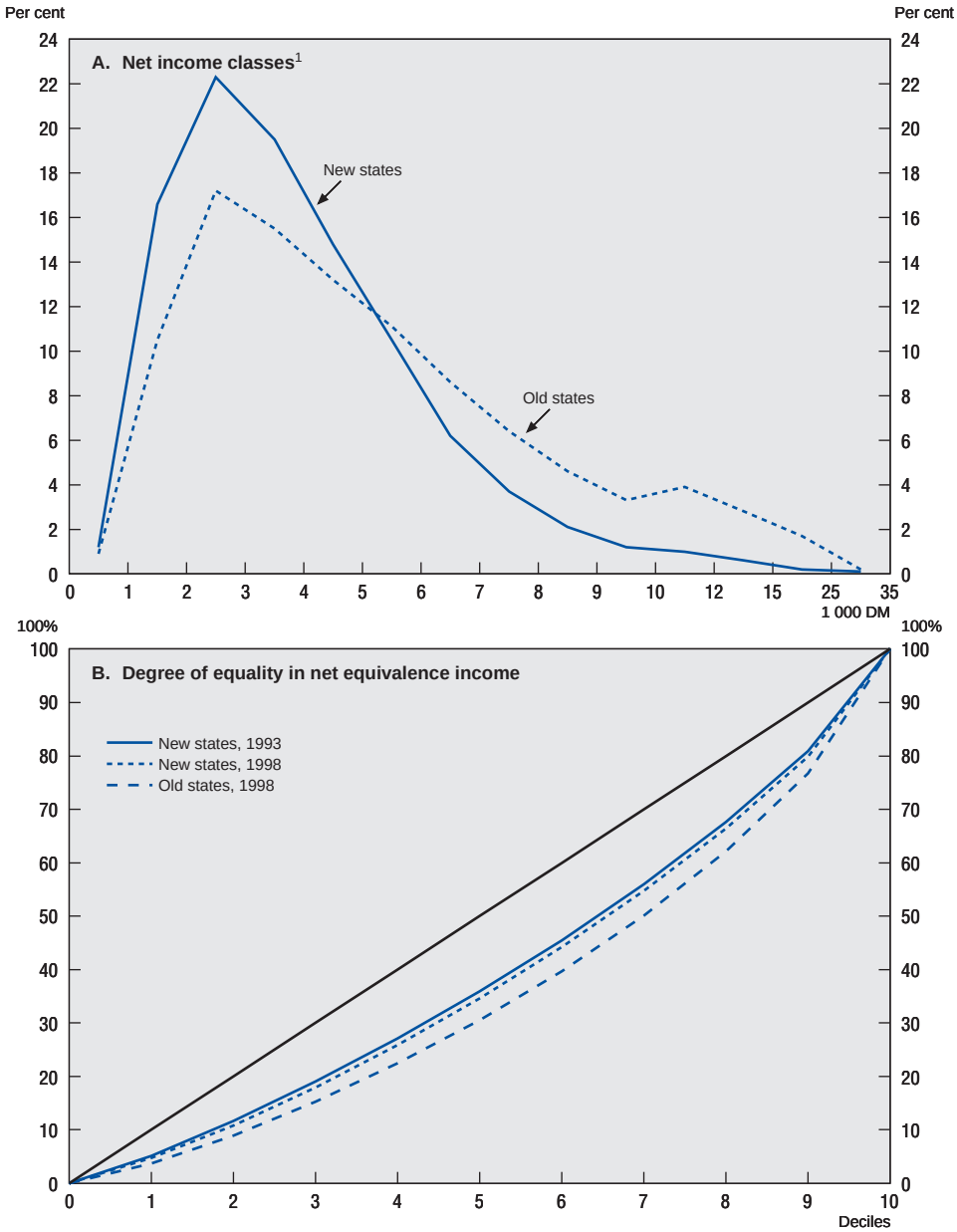
Figure 22. **The income distribution of households**



1. In 1998. Percentage of households with net income within the respective bracket.
Source: Federal Statistical Office, 1998.

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Table 23. **The size of old age pensions**[1]

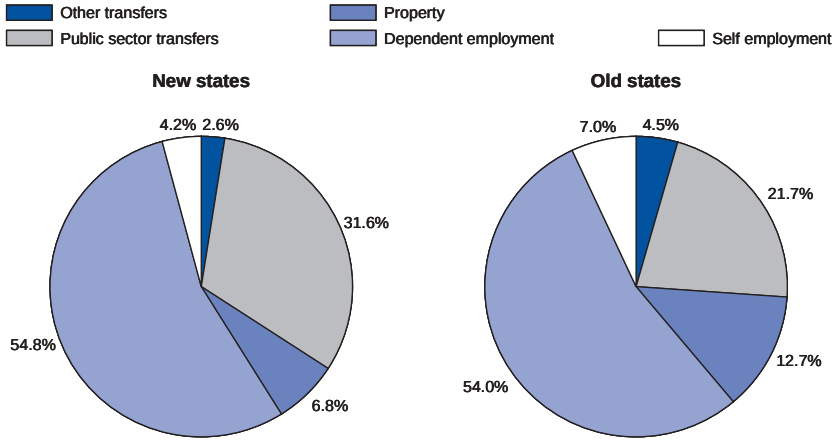
Per cent

	Percentage deviation of average pensions in the new states from average pensions in the old states[2]				Average pensions as a fraction of average compensation	
	Males		Females		New states	Old states
	Stock	Inflows	Stock	Inflows	Stock, males and females	
Total	+7.4	+2.8	+39.6	+51.0	37.4	24.9
Statutory retirement age (65)	+28.5	+135.8	+60.7	+63.9	35.6	20.2
Early retirement						
On account of unemployment	–13.3	–15.5	+3.4	+7.2	46.2	39.9
Invalidity	–13.0	–16.3	+7.5	+0.2	44.5	38.6
For females	. .	. .	+5.9	+7.5	. .	. .
Long-term insured	–5.1	–6.9	+33.7	+33.7	51.6	38.9

1. In 1999; stocks at 31st December.
2. A + sign indicates that pensions in the new Länder exceed those in the old Länder.

Source: Verband Deutscher Rentenversicherungsträger Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und der Länder" and OECD.

Figure 23. **Composition of household income by source**



Source: Federal Statistical Office.

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Table 24. **Public transfers to the new states**

	1991	1992	1993	1994	1995	1996	1997	1998	1999
	A. Gross and net transfers by source[1] (DM million)								
Expenditures									
Federal government	75.1	90.0	115.7	115.9	136.7	136.7	129.6	130.8	140.0
Western states and communities	5.3	5.7	10.3	13.5	11.2	11.3	11.6	11.5	11.6
Germany Unity Funds	31.0	24.0	15.0	5.0	0.0	0.0	0.0	0.0	0.0
Social security system (net)[2]	18.7	34.2	23.0	29.8	33.3	30.9	34.7	31.9	36.0
EU	4.0	5.0	5.0	6.0	7.0	7.0	7.0	7.0	7.0
Treuhandanstalt	8.8	13.7	23.0	23.8	0.0	0.0	0.0	0.0	0.0
Revenues									
Federal government	33.0	39.1	41.4	45.2	46.8	48.2	47.8	48.6	50.6
Gross transfers	142.9	172.6	192.0	194.0	188.1	186.0	183.0	181.2	194.6
Net transfers	109.9	133.5	150.6	148.8	141.3	137.7	135.1	132.6	144.0
As per cent western German GDP[3]	4.2	4.8	5.4	5.2	4.7	4.5	4.4	4.1	4.4
	B. Gross transfers by function[1] (Per cent of total)								
Business-near infrastructure	12.4	9.9	8.6	10.1	13.0	13.3	13.2	12.9	12.6
Support to enterprises	2.5	4.7	7.6	7.5	8.0	7.0	6.3	6.4	5.8
Social transfers	45.4	54.1	54.3	54.4	49.5	49.7	49.7	49.1	51.4
Non-earmarked transfers	28.0	22.3	20.0	19.5	23.5	24.6	25.0	25.8	24.5
Not attributable	11.7	9.0	9.3	8.4	6.0	5.4	5.8	5.8	5.7
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

1. Excluding special depreciation allowances, debt servicing expenditure and credits.
2. Without the contributions by the federal and state governments.
3. GDP of western Germany excluding Berlin.

Source: Institut für Wirtschaftsforschung Halle and OECD.

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... and absorption exceeds production by a wide margin

A major characteristic of the eastern German economy is the huge gap between total absorption (private and government consumption and investment) and domestic production. In 1994, the last year for which separate national accounts statistics by east and west Germany are available, total absorption exceeded GDP by more than fifty per cent. This ratio has remained roughly unchanged: in 1999 net public transfers from the west accounted for roughly one-third of eastern GDP. Capital inflows – including credits by the public sector – have been estimated to total some 15 per cent of GDP.[127] Hence, about one-third of total absorption by private households, investors and the government in the new states was financed by transfers from the west or capital imports. Totalling some 50 per cent of GDP, net imports of the new German states (which equal net transfers and capital imports received) are extra-ordinarily high.[128]

Public sector transfers to the new states involve the Federal Government, the western states and communities, the social security system and the EU. Some 51½ per cent of the gross total are social transfers, allocated mainly to the unemployed and pensioners, 12½ per cent finance public sector investment, and some 6 per cent consist of direct subsidies, 24½ per cent are general budgetary transfers to public sector authorities, designed to compensate for the low revenue power of the eastern states (Table 24). About one-quarter of all public-sector transfers are based on specific (federal) legislation for the new states, with the remainder being implied by the rules governing Germany's federal fiscal and social system.[129]

Major forces shaping the economic performance of the new states

The macroeconomic indicators considered above clearly show that much has been achieved in raising eastern Germany's economic performance and increasing incomes in the new states. In part, the failure to achieve a self-sustaining strong growth path must be ascribed to the size and persistence of the initial transition challenge. Indeed, the exposure to competition of the eastern Germany economy revealed that its structural deficiencies were too severe to be remedied quickly. Consequently, a large part of eastern German industry had to be closed down. The reindustrialisation thereafter had to start almost from scratch. Its impact on aggregate performance is to some extent masked by the downsizing of construction since the middle of the 1990s. However, two characteristics of the eastern German economy also account for the fact that the actual growth and employment performance over the last decade has remained subdued below its potential:

- In the business sector labour productivity in eastern Germany falls short of the western level by a margin that is significantly larger than the east-west gap in the capital intensity (business capital stock per employed in the business sector) would suggest.

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– Unit labour costs on average in the eastern German economy have failed to reach competitive levels. Although declines in unit labour costs have been substantial – in particular in manufacturing – they are still above the level in western Germany.

The productivity of labour and capital inputs determines the size of output associated with any level of labour and capital employed. Hence, for self-sustaining growth in eastern Germany to be generated and economic convergence with the west to be achieved it is crucial that in the new states labour and capital are employed as productively as they are in the old states. Moreover, even if this condition were met, it would not suffice to establish the new states on a high growth path if eastern German products were still lacking competitiveness. Hence, it is crucial as well that unit labour costs not exceed western German levels and rather undercut them. This section highlights both of these aspects. Policy implications following from this analysis are presented in the next section.

The capital endowment has improved, but business sector productivity is lower than to be expected

The infrastructure has improved…

It became obvious immediately after the fall of the wall that the dramatic lack of infrastructure and its obsolete conditions would act as a major brake to economic competitiveness and growth. Indeed, with certain infrastructure being of particular importance for exporters, the slow expansion of the production of tradable goods is likely to be attributable to some degree to these adverse conditions. However, since the beginning of the 1990s a great deal has been achieved in building up a modern infrastructure, although deficiencies remain, mainly in local roads and sewerage systems.

The infrastructure endowment of the new Länder grew strongly over the last decade. According to recent estimates, by 1999 the per-capita public sector capital stock of the states and communities combined exceeded its 1992 level by some 70 per cent (respectively 53 per cent when a more narrow definition is applied that excludes public sector investment in areas where agents outside the general government also contribute to the infrastructure, such as energy supply). The rapid speed of instalment is also apparent from the fact that about half of the eastern capital stock dates from the 1990s, compared with somewhat more than 20 per cent in the old Länder. Despite this progress, on a per capita basis the capital stock of the new states and communities still falls substantially short of that in the old German states. Compared with the western "area states" (F*lächenländer*), *i.e.* excluding the city-states Hamburg, Bremen and Berlin, the gap amounts to 43 per cent in the wider and 30 per cent in the narrower definition. Compared to

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the financially weaker western area states, the gap narrows somewhat to 37½ and 25½ per cent, respectively (Table 25).[130] These relations might look different if other bases were chosen for comparison. For example, relative to the number of students the east-west gap in the university sector is smaller than on a per capita basis, whereas the gap for the roads system is larger if related to the area. Nevertheless, the conclusion would remain that on average the western level is not yet reached.

Infrastructure investment by the states and communities appears to be biased, however, in favour of culture, social security, hospitals, housing and area development, and the central administration. The estimates suggest that the per capita capital stock associated with cultural and social security services exceeded the western average already in 1992 by a significant margin. This is largely attributable to the fact that these services have been provided to a

Table 25. **Infrastructure capital stock per inhabitant of the states and communities**

	In per cent of western German area states[1]		In per cent of financially weak western German area states[2]	
	1992	1999	1992	1999
Infrastructure in a narrow sense				
Central administration	35.9	103.8	39.6	112.8
Public security and order	58.2	98.5	60.3	105.4
Juridiction	59.1	82.4	69.5	102.6
Schools	47.4	61.1	45.9	60.3
Universities and other research	35.0	56.1	46.4	74.9
Culture	122.2	146.9	186.2	220.3
Social security	116.8	119.4	139.6	146.6
Sports and recreation	62.4	76.6	71.7	88.0
Hospitals	71.2	110.9	81.2	131.1
Housing and area development	8.4	92.5	10.1	109.3
Nutrition, agriculture and forests	96.9	79.7	77.8	64.1
Traffic and communication service	44.5	53.2	45.0	53.4
of which: Roads	38.1	49.3	38.1	49.2
Total infrastructure in a narrow sense	50.1	69.9	52.9	74.3
Further infrastructure				
Communal joint tasks	5.6	20.4	6.1	23.5
of which: Sewerage	5.2	19.8	5.8	23.5
Energy, water, services	12.8	73.8	11.3	65.1
Companies	2.2	16.9	3.3	26.8
Real estate, special assets	1.3	46.1	1.4	50.1
Total infrastructure	37.1	57.1	39.8	62.4

1. The area states do not comprise the city-states Berlin, Bremen and Hamburg.
2. Western German states receiving transfers in the inter-state fiscal equalisation system.
Source: Seidel, B. and D. Vesper (2000).

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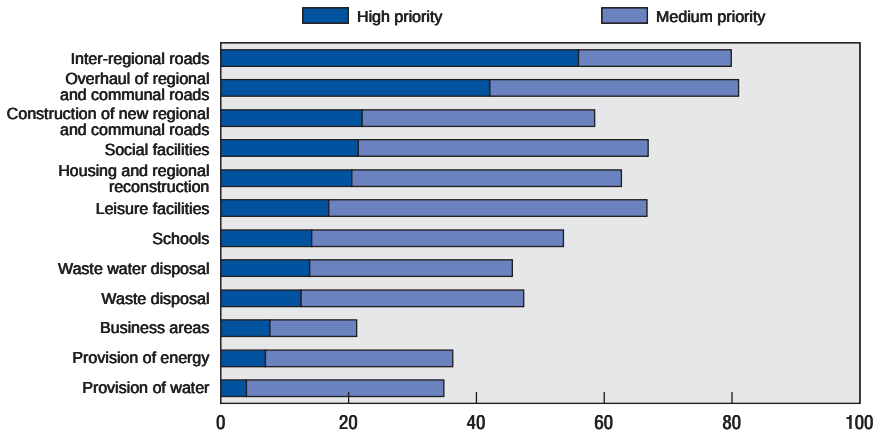
larger extent within the government sector than in the west. By 1999 the "overhang" had increased further, totalling more than 100 per cent and almost 50 per cent of the endowment of the financially weaker western states, respectively. In the other three fields, the per capita endowment also has surpassed the western levels by a significant margin. Although desirable for various reasons, these investments are not at the core for raising the growth potential of the economy. It is also worth noting that more than one-quarter of the total infrastructure gap to the west (in the more comprehensive definition) is attributable to the sewerage systems of the communities, which appear to be less important as a bottleneck to growth.

However, in fields, which appear to be more important for economic growth – notably road construction – the gap still appears to be sizeable. Indeed, surveys indicate that enterprises attach the highest priority to improving the network of inter-regional roads, to construction of new roads, and the overhaul of obsolete regional roads (Figure 24). The infrastructure gap in these fields should account to some extent for the productivity gap between the eastern and the western states, and the sluggish improvement in competitiveness more generally.

Figure 24. **Deficits in infrastructure perceived by enterprises**[1]

Per cent of enterprises sampled



1. The survey was conducted among industrial enterprises by the Institut für Wirtschaftsforschung Halle in January 1998. The question asked was: In what fields of public infrastructure should investments be made, seen from the perspectives of your company?

Source: Snelting, M., C. Schumacher, W. Komar, P. Franz (1998).

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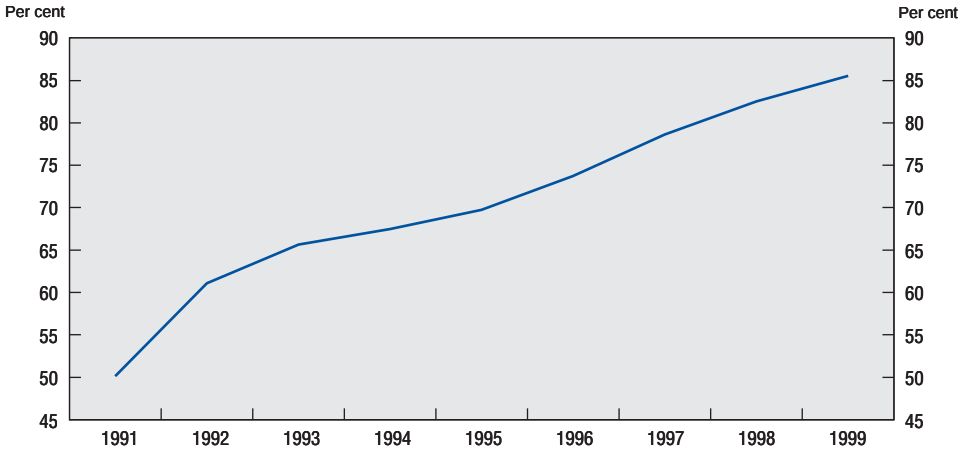
... and the business capital stock has been expanded and modernised

The capital stock of eastern German business increased rapidly as well. At present, the average capital intensity in the eastern German business sector (gross fixed capital stock per employed in the business sector) is estimated to total 85½ per cent of the western level (up from 50 per cent in 1991), and in manufacturing it has been estimated to fall short of the western level by just 8 per cent.[131] Disaggregated data for manufacturing indicate large differences in capital intensities between sectors, however, with the intensity in final goods-producing sectors on average falling substantially short of the level in the west. Moreover, with fixed capital formation growing at rates substantially higher than those in the west, the present average age of the business capital stock in the new states should roughly equal the one in the old Länder (Figure 25). This is confirmed by surveys on the average age of equipment in eastern and western enterprises. The firms' own judgement also supports the assessment that the technical state of equipment has reached the western level. In the 1999 "Enterprise Establishment Survey" only 4 per cent of the eastern firms judged their equipment to be old or very obsolete. This share remained almost unchanged from the first wave of the panel conducted in 1996 and equals the corresponding share amongst western German establishments (5 per cent).[132]

Figure 25. **Capital intensity in the business sector**[1]
In per cent of the level in the old states[2]



1. Capital stock per employed.
2. New states excluding Berlin; old states including Berlin.
Source: Müller, A. (2001).

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Although the building up of the business capital stock coincided with a large increase in labour productivity, empirical investigations point to the fact that productivity levels in the business sector relative to the west should be substantially larger than those actually observed. This is especially visible in manufacturing: in 1999 labour productivity in manufacturing equalled some 65 per cent of the western level, more than 25 percentage points less than what the relative size of the capital stock would suggest. Various factors influencing the productivity gap have been identified and to some extent they reflect distortions in the market process, although it is difficult to specify their relative importance.[133]

Subsidies helped renew the capital stock, but had significant distortionary effects

The rapid growth of the capital stock was supported by massive state aid to enterprises. With infrastructure in the new states initially lacking to a substantial degree, industrial subsidies appeared to be justified to compensate for competitive disadvantages. The bulk of this aid subsidises investment, but there is also financial support for research personnel to foster the development of research and development in the east (Box 9). Subsidies to the new states peaked in 1995 but declined since then by some 20 per cent until 1999.[134] But the intensity of subsidisation remains much higher in the east than in the west, with the volume of subsidies relative to GDP having been estimated to total 3½ times the western level in1999[135] While financial support to eastern German enterprises is likely to have generated positive output and employment effects, available evidence also indicates that it implied distortions in the production structure which should have retarded productivity growth. Mainly, investment support and special depreciation allowances have biased resource allocation in favour of capital intensive industries and construction in particular.[136] In the process, to some extent investment was pushed into low-productivity sectors, and in some sectors the capital intensity of production was driven to high levels relative to the west while associated gains in relative productivity were only modest.[137] The development of a competitive export base was hampered by the over-expansion of the construction sector which was associated with profit rates that proved to be unsustainable. Other features of the eastern German economy with adverse effects on productivity growth, such as a lack in product marketing and unfavourable terms of trades, are also likely to be related to distortions associated with high levels of financial aid. In view of such developments the Federal Government phased out special depreciation tax allowances for investment in 1998 and restructured support schemes in favour of small and medium-sized enterprises and capacity widening investments.

Indeed, while overall fixed investment per capita substantially exceeds the level observed in the old states, this is essentially accounted for by the abnormally high level of construction investment. Per capita investment in machinery

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Box 9. State aid for private enterprises in the new Länder

To foster the reconstruction of the eastern German economy, beginning in 1990 the Federal Government and the governments of the new states as well as state-owned banks and the EU set up several support programmes for private enterprises, designed to foster investment, business start-ups, research and development, and exports. Additional funds were utilised to rescue former state-owned enterprises held by the *Treuhandanstalt* and its successor organisation (*Bundesanstalt für Vereinigungsbedingte Sonderaufgaben*). Counselling and training was also funded, mainly by the Federal Labour Office.

With the states having established their own programmes, a multitude of individual support programmes is still in operation (several hundred), but the bulk of the aid extended (some 90 per cent of the funds) has always been concentrated on a handful of instruments highlighted below (excluding training and active labour market measures). Eligibility conditions, volumes extended and rates of subsidisation of these instruments have repeatedly been subject to change. Initially, the Federal Government envisaged a phased reduction of the volume of aid after 1998, when the first phase of funding agreed between the Bund and the states expired. But with the reconstruction of the new states progressing slower than expected main programmes of special support for eastern Germany have been fixed until 2004. For the follow-up phase after 1998 tax concessions were reshaped: Special depreciation allowances for the new *Länder* are no longer available, and tax-free investment grants are the only tax instruments applied. There has also been a restructuring of aid in favour of small and medium-sized companies in manufacturing and business-related services. The modified design has implied a substantial reduction in tax expenditures for the new states. On average, annual tax revenues foregone on account of special allowances for eastern Germany are projected to total between 1999 and 2004 roughly 35 per cent of the level prevailing between 1991 and 1998. Regional aid has also been reduced over the 1990s. Overall subsidies in the new states – in the definition of the Federal Government's Subsidy Report – declined by some 20 per cent between 1995 (the year where subsidies peaked) and 1999.* But the support intensity overall in the new states remains much higher than in the west. Relative to GDP subsidies have been estimated to total 3½ times the western level for 1999, based on the information of the Federal Government's Subsidy Report (see Ragnitz, 2000*a*). Subsidies extended by the states and communities account for some 40 per cent of the total. Maximum support rates vary between 20 and 50 per cent and must be in line with the EU regional support regime (EU-*Fördergebietsrahmen für Ziel*-1-*Gebiete*). Accumulation of different support schemes is possible (grants, bonuses and credits), but in total have to respect maximum subsidisation limits. The major instruments presently applied are:

- Investment grant (*Investitionszuschuss*). This is a taxable cash payment co-financed by the federal and state governments. Support is extended on a case-by-case basis without a legal right for funding. It is designed to help economically disadvantaged regions ("Improvement of Regional Economic Structure") by fostering investment that is expected to raise employment

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Box 9. **State aid for private enterprises in the new Länder** (*cont.*)

and training or to strengthen the innovation process, or which is associated with enterprise foundation. Support rates amount to up to 50 per cent for small and medium-sized enterprises and up to 35 per cent for other enterprises.

– Investment bonus (*Investitionszulage*). Tax-free cash payment after completion of investment. Those eligible have a legal right for funding. Bonuses are given for investment in machinery and equipment as well as construction investment by manufacturing enterprises, providers of enterprise-near services, and small and medium-sized craftsmenships. Bonuses are also granted for rental apartment construction and the renovation of rental houses in inner city districts. Support rates vary between 5 and 27.5 per cent, depending on the type of investment, the size of the firm and the region. Maximum support rates are only applicable for investment by small and medium-sized enterprises in manufacturing and business-related services enlarging their production capacities and being located along border regions to Poland and the Czech Republic. Subsidies for replacement investments in larger companies presently have a support rate of 5 per cent; they will expire by the end of 2001.

– Credit Programmes. On the federal level, preferential credits are extended by the German European Recovery Programme Funds (ERP) and federal development banks (KfW, DtA). Aid is given for investment projects and company acquisitions by small and medium sized enterprises as well as to personal entrepreneurs for the purpose of fostering start-ups, enterprise growth and reorganisation. The credits offer preferential interest rates and special debt redemption conditions. They can be designed to function as surrogates for equity capital. Besides direct support, credits are also granted to investment firms for the purpose of refinancing equity participation in small and medium-sized enterprises. In most cases credits are mediated by private sector banks, which then become liable to the public banks. The private banks, in turn, normally must participate in the crediting with their own funds.

– Support programmes for research and development, consisting of aid to small and medium-sized enterprises for research projects and research personnel, as well as support to research networks.

* See Bundesministerium der Finanzen (1999).

and equipment has continued to grow strongly but stands below the western German level (Figure 26).[138] Subsidy-induced misallocations of investment – which adversely affect the productivity of the economy's capital stock – are perhaps best visible in residential construction. In 2000 almost one million flats in eastern Germany stood empty, twice as many as in 1993. Flat vacancy rates ranged

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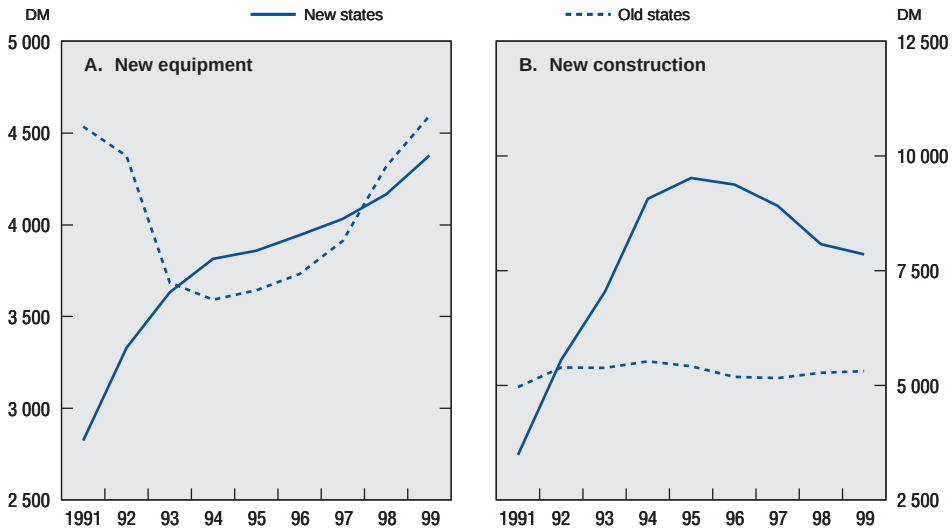
Figure 26. **Fixed investment per capita**



Source: Ifo Institut für Wirtschaftsforschung; Federal Statistical Office and OECD.

from 17 per cent in Saxony to 9 per cent in Thuringia. They reflect migration from old homes to – subsidised – new residences, and – to a lesser extent – migration to the west. Prior to desertion, a large share of the old flats was renovated with the help of public sector funds. Moreover, to reduce mounting over-capacity, old homes – mainly tiles' constructions (*Plattenbauten*) – are now being dismantled with the Länder governments covering the cost of demolition.[139] In addition, there are demands that the Federal Government and the eastern states should financially support the rehabilitation of such deserted areas. The total implications for resource allocation of this phenomenon could even extend further in that associated regional shifts might necessitate the establishment of new infrastructure, such as public transport, waste disposal and water supply and disposal.

It is striking that within the manufacturing sector – the only sector where disaggregated information on both capital endowment and productivity is available on the branch level – labour productivity relative to the west does not positively correlate with relative capital intensity. Such a positive correlation is to be expected, however, if distortions in the production structure in the east did not exceed those in the west. Within the group of branches, whose capital intensity has expanded above the corresponding levels in the old states, relative productivity and capital intensity even appear to be negatively related, although the correlation is not significant (Figure 27). This group represents 51.8 per cent of the

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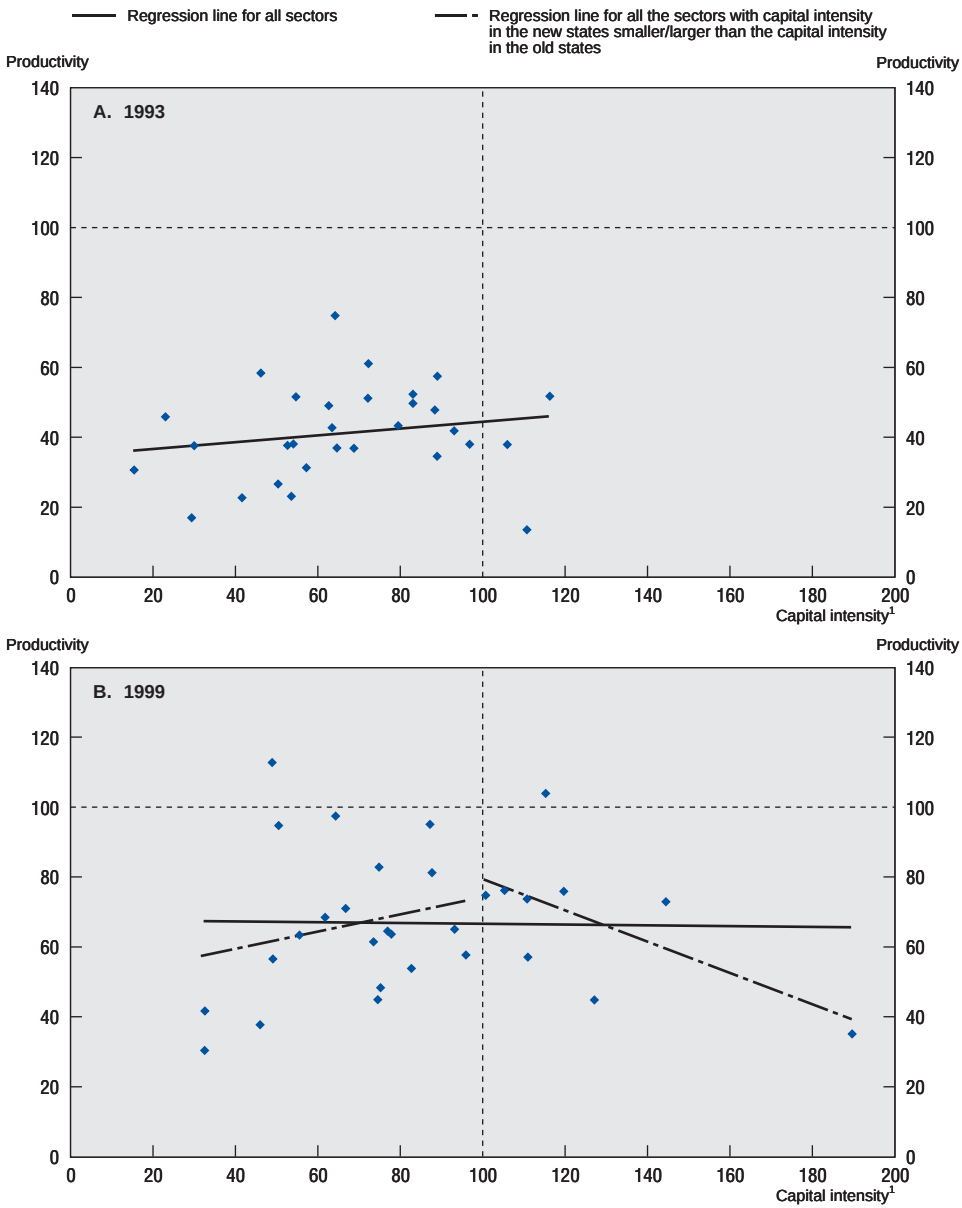
Figure 27. **Capital intensity and productivity in manufacturing**

In per cent of the levels in the old states



1. Business gross fixed capital stock per employed.

Source: Deutsches Institut für Wirtschaftsforschung and OECD.

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total manufacturing gross fixed capital stock in the new states (1999). In contrast, the same branches account for 32.3 per cent of the capital stock in the old states.[140] The mismatch between capital intensity and productivity appears to be largely concentrated in basic goods industries and mainly reflects action during the 1990s. In mining, the capital intensity exceeds the western level by 50 per cent while productivity undercuts that in the west. Cases where the mismatch between capital intensity and productivity is very high if measured against the western standard raise questions whether investment would be profitable at all without subsidies.[141]

At the same time, the strong incentives for investment and capital intensive sectors are likely to have caused a substitution of capital for labour and adversely affected the development of labour intensive sectors that benefited less from financial support, such as the services sector.[142] However, human capital intensive products and service activities are increasingly important for designing and marketing industrial products and for generating high overall productivity growth. Indeed, various investigations suggest that the competitiveness of eastern German products is hampered by a lack of marketing and organisational labour input.[143] This deficiency should also explain to some extent the observation that eastern German companies are able to charge only relatively low output prices for their products.[144]

A *lack of "net-working" adversely affects productivity*

Part of the productivity gap to the west has also been attributed to a lower density of enterprises than in the old states. Networks of enterprises can produce positive scale and spill-over effects, and these in turn may contribute to lowering input prices. To some extent the existence of networks is likely to be related to population density within the regions, and therefore only to a lesser degree influenceable by economic policy. For western Germany empirical investigations indicate significant positive links between productivity and agglomeration.[145] In the new states both population density by state and the number of high-density regions within states are significantly lower in the east than in the west (Figure 28).[146]

Moreover, the share of young and smaller companies is larger in the new Länder than in the old ones, and the difference in the age and size distribution might also explain some portion of the productivity gap to the west.[147] This is also true for the rate of capacity utilisation. In 1999 and 2000 the capacity utilisation of eastern German manufacturing was found to fall short of the western rate by only between 2 and 3 percentage points.[148] Similarly, according to other survey information average capacity utilisation in all sectors combined was lower in the east than in the west by 3 percentage points.[149]

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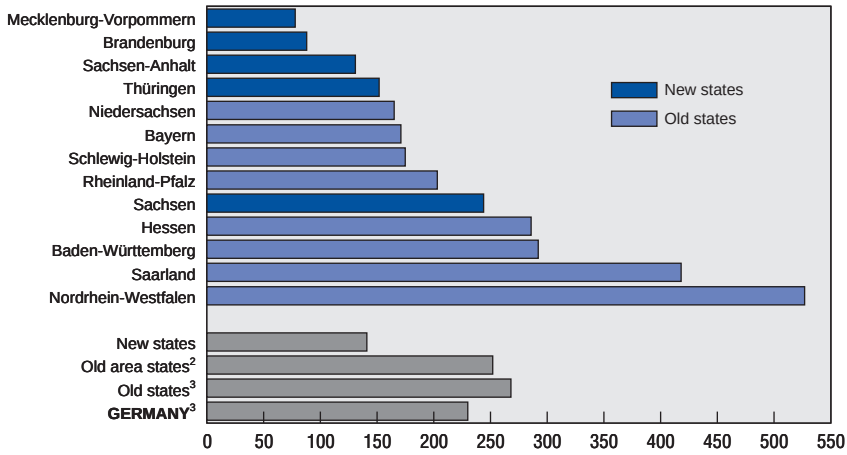
Figure 28. **Population density in the new and the old states**[1]
Inhabitants per km²



1. In 1998.
2. Excluding the city states Berlin, Bremen and Hamburg.
3. Including the city states.

Source: Federal Statistical Office and OECD.

... but skill levels are high, and migration does not appear to constrain economic growth

The eastern German labour force is generally considered to be well qualified. Indeed, in manufacturing establishments the proportion of skilled workers is much higher in the east than in the west. To a large extent this is still attributable to the high level of formal training provided in the former GDR, although little can be said about actual demands on these skills.[150] Similarly, while companies in the east spot a "shortage" of skilled labour, it falls short of the one recorded by western firms: according to the 1997 enterprise panel of the German Institute of Employment research, 11.1 per cent of all sampled firms in the east notified a shortage of qualified labour, as opposed to 17.0 per cent of the firms in the old states. On the other hand, the employment share of qualified white-collar workers has been found to be smaller than in the west, and this seems to hold in the first place for smaller firms.

Nevertheless, the public debate about east-to-west migration within Germany is still dominated by the concern the new states might suffer a drain of qualified labour, hampering economic development in the east and raising excess

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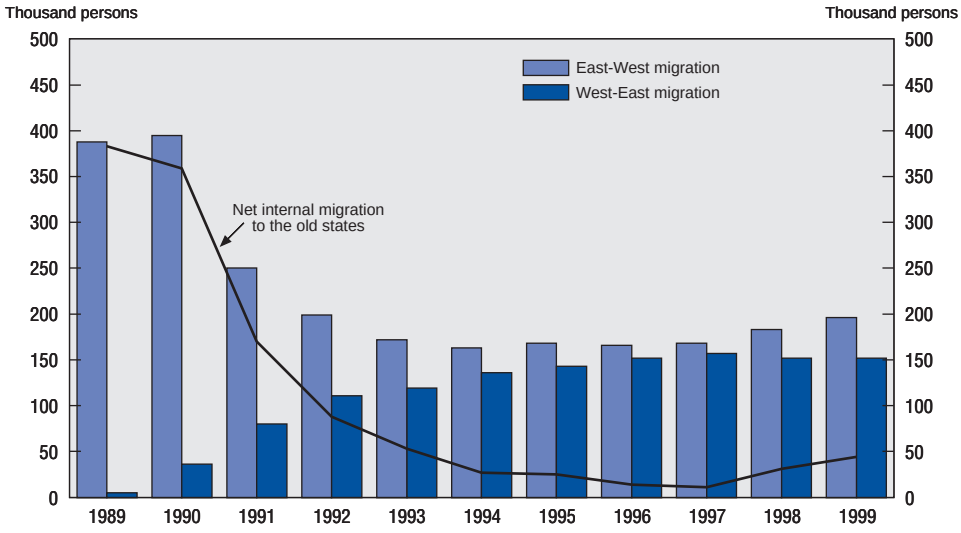
labour supply in the west. Indeed, in 1989 and 1990 almost 800 000 persons migrated to the west, some 5 per cent of the eastern German population. This high outflow, reflecting to a large extent uncertainty about the political future of eastern Germany at the time, had a major impact on the decision to establish the "currency, economic and social union" between the two parts of Germany in July 1990 – prior to political unification. After the currency union was established east-west migration ebbed down and was increasingly matched by migration in the opposite direction. However, expectations of potentially detrimental effects of east-to-west migration continued to rank high among policy makers. In the early 1990s they were an important motivation for the high wage settlements in the new states, and they also motivate high levels of financial transfers to eastern Germany and large-scale provisions of active labour market measures (Figure 29).

Empirical evidence points to differentials in earnings and job availability as significant explanatory factors for intra-German east-west migration.[151] Seen in isolation, both the steep increase in eastern wages relative to the west and the high level of ALMPs should have contributed to the drop in east-west migration in the first half of the 1990s. However, the overall effect of these measures on migration appears less clear taking into consideration that high-wage policies have retarded job creation in the primary labour market.

Figure 29. **Intra-German migration**



Source: Institut für Wirtschaftsforschung Halle.

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Table 26. **Intra-German migration by qualification**[1]

Qualification	East-West migrants		*For comparison*: total population new states (1997) in per cent	West-East migrants		*For comparison*: total population old states (1997) in per cent	Net intra-German migration to the new states (absolute in 1 000)
	Absolute in 1 000	In per cent		Absolute in 1 000	In per cent		
Primary education							
Low	116	26.4	29.4	105	25.0	48.3	–11
Medium	232	52.7	54.3	180	43.1	32.1	–52
High	92	20.9	16.3	134	31.9	20.6	+42
Total	440	100.0	100.0	419	100.0	100.0	–21
Vocational/higher education							
Low	78	17.6	12.5	66	15.3	26.1	–12
Medium	312	70.4	77.3	284	66.3	61.4	–28
High	53	12.0	10.2	79	18.4	12.5	+26
Total	443	100.0	100.0	429	100.0	100.0	–14

1. Between 1992 and 1997. Projections of the total based on sample outcome of the Socio-Economic Panel.
Source: Kempe, W. (1999).

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While east-west migration in 1989 and 1990 was associated with a considerable outflow of skilled labour, there appears to be little evidence that the new states continued to suffer from such a net drain. Relative to their weight in the total population, young people and those with higher education have been found to be more inclined to migrate than others.[152] However, evidence from the German Socio-Economic Panel shows that among the highly qualified, west German residents are more inclined to move to the east than east German residents to the west, and that the new states experienced net migration gains of highly qualified labour.[153] Net outflows to the west were confined to lower and medium qualifications (Table 26). In addition, about half of the west-to-east migration consists of re-migration of people who initially lived in the new states but had emigrated to the west. Most of them got formal or informal training in the west before they re-migrated to the east.

Unit labour costs have not adjusted to competitive levels

High labour costs retarded growth and employment

Unit labour costs in the total economy came down from 117½ per cent of the western German level in 1991 to 111½ per cent in 2000, with the development differing across sectors.[154] In the construction sector, unit labour costs have temporarily declined to below the level in the old states. But the largest reduction occurred in manufacturing – which is more exposed to competitive pressures than other sectors – where unit labour costs fell by 110 percentage points (Figure 30).[155] The decline in unit labour costs has largely been achieved by employment reductions, notably in the first half of the 1990s, while wages increased throughout the 1990s. However, this implied that adjustment was not designed to improve the competitiveness of the new states so as to support high levels of sustainable growth and rapid employment creation.

Rapid wage increases were the major factor depressing the development of eastern German competitiveness. Effective wages were already boosted prior to reunification by the 1:1 currency conversion of Deutschemarks for East Marks. The impact of the conversion on labour costs has been substantial, although a concise assessment of the revaluation is hardly possible.[156] Neutralisation of the implied cost push for labour input would have required, at the minimum, a considerable delay of further nominal wage increases (which still would have preserved the increased purchasing power of wages in comparison to the state prior to currency conversion). Such an adjustment did not materialise, however. Instead, the social partners agreed in 1991 on a predetermined wage path targeting wage convergence with the west within five years. Although this agreement was abandoned in 1993, after it proved to be unsustainable, it lead to very high wage increases in the first years after unification and continuing increases thereafter.

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Figure 30. **Unit labour costs by sector**[1]

Unit labour costs relative to the old states (percentage point difference)

1. Unit labour costs have been computed as compensation per dependent employed divided by gross value added in 1995 prices per employed. Excluding Berlin.

Source: Arbeitskreis "Volkswirtschaftliche Gesamtrechnungen der Länder"; Arbeitskreis "Erwerbstätigenrechnung des Bundes und Länder" and OECD.

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Indeed, the parties involved in wage negotiations at the beginning of the 1990s had little incentives to agree on a low wage policy. Employers were represented by managers of former state-owned firms and representatives of western employers' associations. The employees' side was dominated by the west German Trade Union Federation (*Deutscher Gewerkschaftsbund*) which rapidly established branch-wide unions after the breakdown of the socialist regime. Total labour costs were further boosted by rising social security charges, which increased from 37.3 per cent of the gross wage base in 1990 to 41.7 per cent in 1999, despite successive broadenings of the revenue base.[157] In sum, between 1991 and 2000 hourly compensation in manufacturing increased by almost 85 per cent, following sizeable wage increases that had already occurred in 1990.

Obsolete production structures – and the breakdown of eastern Germany's traditional eastern European export markets in the first half of the 1990s – inevitably implied a massive re-allocation of labour accompanied by large employment losses over a transitionary period. However, imposing a strongly accelerating wage path on this process retarded the capacity of the economy to generate new competitive working places in the primary labour market. This policy also negatively influenced the terms of trade, which should have contributed to the sluggish development of eastern Germany's export share (presently amounting to about 55 per cent of that in the west). A high degree of competitiveness is crucial, however, for generating growth. This is evidenced by the fact that all those branches in manufacturing which experienced very high growth rates over the last couple of years, are export oriented. A state characterised by high wage increases relative to productivity, very high and persistent unemployment, and relatively modest net migration to the west would not have been sustainable without massive transfers from western Germany.[158] Industrial subsidies effectively helped safeguarding the viability of enterprises despite the push in labour costs, and unemployment related benefits provided income support to replaced workers.[159] Simultaneously, high and persistent unemployment was cushioned by work provision and training schemes, often in the public sector, while early retirement schemes reduced labour force participation.

While wage differentials across industrial sectors have increased in the 1990s they appear to be smaller than in western Germany or other developed market economies. In particular, in sectors which are little exposed to competition and where coverage with sector-wide wage bargainings is high – in the first place the (oversized) government sector and heavy industries – little pressure on wages has occurred.[160] Flexible relative wages are important, however, to support structural change.

Surveys indicate that firms consider high labour costs still as a major impediment for higher competitiveness and more employment.[161] This is in line with empirical evidence on the regional level suggesting that employment is

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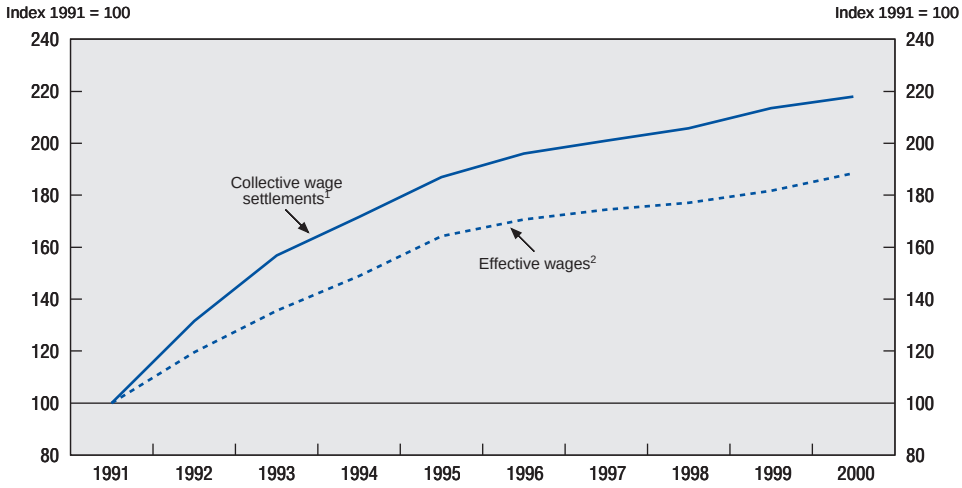
Figure 31. **Effective wages and wage settlements**



1. Collectively agreed hourly wages.
2. Compensation per hour worked.

Source: Deutsche Bundesbank; Institute for Labour Market Research and OECD.

higher when past wage increases were subdued.[162] Given that area-wide agreed wage settlements proved unsustainable for a sizeable share of enterprises, wage agreements on the company level, to some extent based on opening clauses in collective contracts and often undercutting collectively-agreed wages, have gained importance. This is reflected in the wage drift becoming increasingly negative (Figure 31).[163] In the process, labour cost flexibility has improved in recent years, notably in manufacturing, which is particularly exposed to competition.

... and large-scale active labour market measures have been largely ineffective

While active labour market policies (ALMPs) – work provision and training schemes – have helped to cushion the displacement of workers, the available evidence suggests that they have done little to increase employment prospects in the primary labour market. As has been discussed in last year's *Survey on Germany*, several studies indicate that work provision and training schemes in eastern Germany have not been effective in raising re-employment prospects of the unemployed in comparison to "passive" unemployment benefits. New empirical work supports the conclusion that work provision schemes in the east either have no effect at all or even reduce re-employment prospects (Box 10). There is also a high likelihood of circular participation in such schemes. According to longitudinal information

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Box 10. The effectiveness of active labour market measures in eastern Germany

Active labour market measures (ALMPs) have been used extensively in the new states, whereas in the old states they have been applied on a much more limited scale.

The Box Table summarises the findings of micro-econometric studies on the transition into regular employment for eastern Germany. They are all based on longitudinal data sets of individual labour market transitions. The methodological framework rests either on an approach, which models conditional exit probabilities from non-employment for the sample of participants in ALMPs and non-participants (given the length of time that passed in the respective state already), such as hazard rate models, or is a statistical matching approach that constructs a comparison group of non-participants in ALMPs and then compares average

Microeconometric studies on the effectiveness of ALMPs in eastern Germany

Author	Time span/data set	Impact on employment	
	Job-creation schemes		
Eichler, Lechner, 1999	1991-97/LMMSA		negative
Hübler, 1997	1990-94/LMME	Males: Females:	negative insignificant
Kraus, Puhani, Steiner, 1998	1990-94/LMME		negative
Schneider, Bergemann, Fuchs *et al.*, 2000	1990-98/LMMSA	Over large range:	negative
Steiner, Kraus, 1995	1990-92/LMME	Males: Females:	insignificant negative
	Training schemes		
Fitzenberger, Prey, 1998,	1990-94/LMME		negative, insignifiant, positive
Hübler, 1997	1993-94/LMME	Males: Females	positive insignificant
Kraus, Puhani, Steiner 1999	1990-94/LMME	1990-92: 1992-94:	negative positive
Lechner, 1999	1990-96/SOEP		negative, insignifiant
Pannenberg, 1996	1990-94/SOEP		positive
Schneider, Bergemann, Fuchs *et al.*, 2000	1990-98/LMMSA	Over large range:	negative
Staat, 1997	1992-94/SOEP		insignificant

LMME: Labour market monitor eastern Germany (*Arbeitsmarktmonitor Ostdeutschland*).
LMMSA: Labour market monitor Saxony-Anhalt (*Arbeitsmarktmonitor Sachsen-Anhalt*).
SOEP: Socio-economic panel (*Sozio-ökonomisches Panel*).
For similar summaries see also: Hagen and Steiner (2000) and Fitzenberger and Speckesser (2000).

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Box 10. The effectiveness of active labour market measures in eastern Germany (*cont.*)

outcomes for both groups. Various approaches are applied to control the potential bias on estimates associated with unobserved sample heterogeneity. While the results of these studies differ, most of them do not find significant positive employment effects of participation in job-creation or training schemes. In some cases, participation in such schemes is estimated to even reduce transition probab ilities into regular employment.

Aggregate effects and efficiency costs

A full evaluation of ALMPs would also need to take into account potential deadweight costs and worker displacement effects (dead-weight costs concern expenditures which are actually not necessary to achieve a given labour-market outcome, while displacement effects refer to the possibility that there will be only a substitution between individuals, leaving aggregate unemployment unaffected). Moreover, there are other effects apart from workers' displacement (notably the adverse impact of financing ALMPs) which may render ALMPs ineffective in the aggregate, even if they had a positive impact on individual re-employment probabilities. Econometric work assessing the overall effect of work provision and training schemes for the unemployed on aggregate unemployment in eastern Germany – rather than the impact on individual re-employment probabilities – indicates that both types of programmes increased unemployment rather than reducing it.[1] In part, this might also be attributable to higher rates of labour force participation induced by ALMPs.

Research in other countries suggests that job-creation/work-provision schemes are associated with sizeable deadweight costs and displacement effects (Fay, 1996; Martin, 1998). Little is known on such effects in Germany, however, due to a lack of suitable statistical data. The responses from a firm sample by the Institute for Labour Market and Occupational Research of the Federal Labour Office (*Institut für Arbeitsmarkt- und Berufsforschung der Bundesanstalt für Arbeit*, IAB) indicates that the deadweight costs associated with wage subsidies may be substantial. 54 companies in the new states were asked about the impact of wage subsidies on hirings, and 22 per cent answered that the wage subsidies they receive over-compensated for the lower productivity of the supported workers they had hired. 30 per cent replied that they would have hired the same worker without financial support. Another 22 per cent answered that without support for the person hired they would have employed another worker. But this response does not necessarily indicate the size of the displacement effect, because in some cases companies would have hired another financially supported person. Attempts to econometrically quantify dead weight costs based on aggregate data failed to produce statistically significant outcomes, however, and this reinforces the need to make available micro-data sets on job mediation that are sufficiently rich to allow for cost-benefit analyses by programme type.[2]

1. See Hagen and Steiner (2000).
2. See Schneider, Hilmar *et al* (2000).

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Table 27. **Repetitiveness of participation in active labour market measures (ALMPs)**[1]

From first participation in ALMP in 1990:	Into second participation in ALMP between 1990 and 1998 in: (per cent)			
	Work provision scheme	Training scheme	Other state	Total
Work provision scheme	26	30	44	100
Training scheme	23	21	56	100

1. Sample of the Labour Market Monitor Saxony-Anhalt (*Arbeitsmarkt-Monitor Sachsen-Anhalt*).
Source: S *ource*: Institut für Wirtschaftsforschung Halle.

on participation histories in Saxony, between 1990 and 1998 56 per cent of participants in a work provision scheme in the first year re-entered a work provision or a training programme at least one more time. The same is true for 44 per cent of the participants in a training scheme (Table 27). However, the limited availability of job mediation data for empirical research is preventing more detailed analysis of the effectiveness of different programme designs and associated deadweight costs.[164] Recent econometric work on the effects of ALMPs on aggregate unemployment suggests that work provision and training schemes even increase the stock of unemployment in eastern Germany rather than reducing it. This may be attributable to displacement effects and induced labour supply, which would not have come about without these measures.[165]

Policy implications

From the beginning on financial support policies for the new states were meant to serve two purposes: They should compensate for various disadvantages the new states had inherited, and they should help to overcome these adverse framework conditions paving the way to self-sustainable growth in the eastern German economy. As the analysis above has shown, economic conditions in eastern Germany have improved strongly over the last decade. However, the evidence suggests that subsidisation has biased resource allocation in favour of capital intensive industries and construction with negative implications for factor productivity. Moreover, high wage policies of the social partners, which have been introduced in the early 1990s, are still keeping wages at levels too high relative to productivity and have rendered much of eastern German industry uncompetitive. Such wage settlements, in effect would not have been sustainable without massive transfers from the west. Under such conditions persistently lower per-capita GDP levels and higher levels of unemployment than in western Germany are likely, and this induces migration to the west of eastern German employees who want to improve their employment prospects. In the future, structural and fiscal policies therefore need to rely much more on market forces to bring about economic convergence with the west than they did in the past. The dynamic develop-

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ments within the manufacturing sector indicate the growth potential of the eastern German economy. In fact, the OECD recommended early after reunification that convergence should be market driven.[166] The restructuring of industrial support policies in the second half of the 1990s – notably the phasing out of special depreciation allowances (*Investitionszuschüsse*) – has been a significant step in this direction. The business and income tax reform, highlighted in Chapter II above, also contributes to strengthening market forces. But more needs to be done. Industrial subsidies need to be further reduced and restructured, and resources should be redirected in favour of abolishing bottlenecks in infrastructure to the extent these still exist. Reallocation of resources also requires further attempts by the governments in the new states to streamline and reshape their spending. To reduce unemployment wage moderation is unavoidable and this needs to coincide with reforms in the social transfer system, notably with respect to active labour market measures. This section highlights the policy recommendations. In some cases, conditions observable in the new states reinforce the need for policy reform in Germany overall rather than requiring special action for the east.

Indeed, these policies are largely complementary. Lower wage settings relative to productivity developments facilitate reducing transfers and costly work provision schemes, which in turn eases the fiscal pressure on public-sector households and frees resources that could be used to abolish remaining infrastructure bottlenecks. As has been outlined in Chapter III above, the government has installed round table talks involving the government, industry and the unions, which aim at developing a joint strategy for fostering jobs and competitiveness. These "Alliance for Jobs" talks should be utilised to establish a policy framework that fosters market-driven convergence between eastern and western Germany.

Continue the development of the infrastructure

As has been highlighted in the third part of the chapter, infrastructure gaps in comparison to the west – notably with respect to roads – may account to some degree for the productivity gap. However, it would not be appropriate – as is often done in the public debate – to equate the projected value of the east-west infrastructure gap with the level of west-east transfers required for financing the catching-up of the east. Links between infrastructure and productivity depend on the projects chosen and their degree of complementarity to private-sector investment. Efficient infrastructure investment therefore requires that concrete bottlenecks hampering growth be identified. Hence, individual infrastructure investment projects should be subjected to cost-benefit analysis. While east-west comparisons of relative infrastructure endowments can give a rough picture of potential shortcomings they can be no substitute for project evaluation. Foregoing such evaluations risks a misleading assessment of requirements in the new states and a waste of capital, which in turn might instigate further demands for additional aid.

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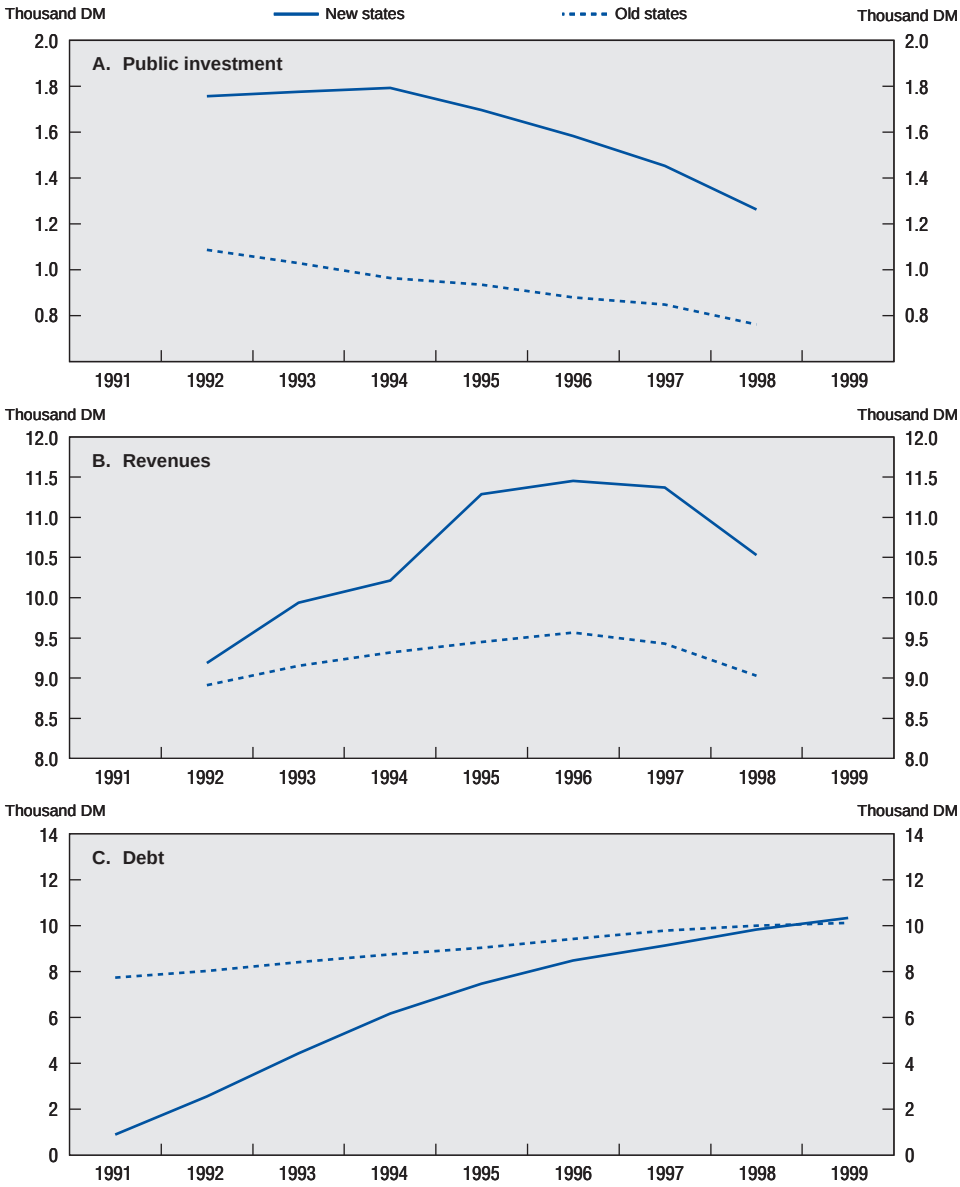
Figure 32. **Public investment, revenues and debt**
Per capita[1]



1. Excluding Berlin.
Source: Federal Statistical Office; Deutsche Bundesbank and OECD.

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Moreover, as is apparent from the analysis above, infrastructure spending by the new states has been biased in favour of culture, social services, housing, health and the central administration, and high endowments in these fields co-exist with gaps in other fields that are more important to foster the economic integration of the new states. These imbalances need to be abolished, requiring a reallocation of resources by the eastern states and communities.

The need to reallocate spending is reinforced by the fact that in the 1990s the new states and communities devoted a declining share of their overall resources to investment. Between 1992 and 1998 investment by the communities and the state governments declined by 30 per cent, and this trend continued thereafter. In contrast, revenues increased, although the development was uneven.[167] Moreover, the indebtedness of the new states and communities increased sharply over the last 10 years and by now has reached the western level on a per-capita basis (Figure 32).[168]

Streamline and restructure spending by states and communities

Government employment in the new states is still relatively high in relation to the west, despite massive reductions over the 1990s. Indeed, between 1991 and 1999 employment of the state and community governments in the new Länder was cut down by some 37 per cent, as opposed to 7½ per cent in the west. Nevertheless, government employment per inhabitant in the new states exceeds the western level by some 19 per cent, and the difference is particularly pronounced at the level of the communities (Figure 33). Moreover, with wages having risen faster than employment having been cut, total wages paid by state and local governments relative to the west increased over the 1990s.[169] Retirement outlays for civil servants, on the other hand, are nearly irrelevant. This is mainly attributable to the limited time that elapsed since re-unification, implying that retirement spending can be expected to rise substantially in the future when the stock of retirees is maturing.[170]

Government employment appears to be inefficiently high and needs to be scaled down. Indeed, it is already visible that those new states with relatively low per capita outlays for personnel tend to spend more on infrastructure.[171] The need to cut down government consumption is further reinforced by the fact that in some fields budgetary pressures are likely to increase. With unemployment benefits and active labour market measures financed by the Federal Labour Office and the Bund, social assistance outlays to be paid by the states could increase to the extent eligibility for other support is fading or the level of ALMPs were scaled down. The latter would also have a direct impact on personnel outlays of regional governments, which employ a significant share of participants in such schemes. In addition, there may be hidden budgetary risks associated with the Länders' financial support for enterprises, notably in terms of government guaranties.

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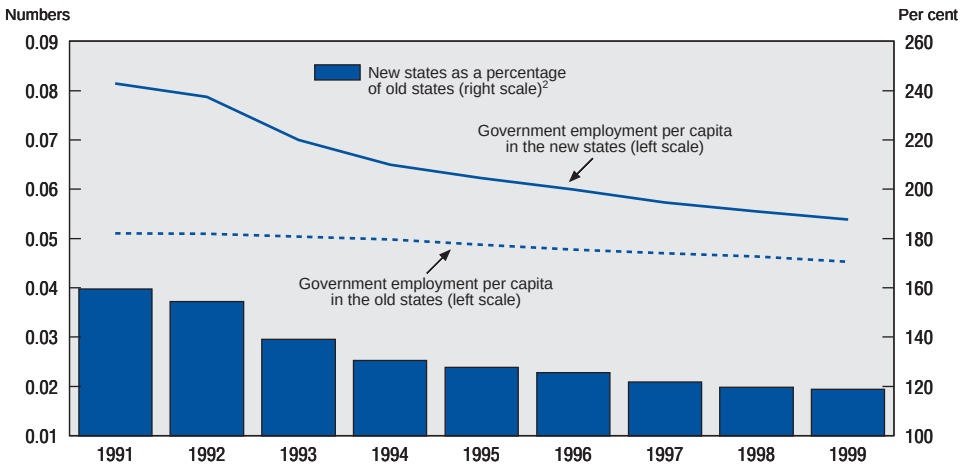
Figure 33. **Employment in the government sector**[1]



1. Government employment refers to employment in state and community governments combined.
2. Government employment per capita in the new states as a percentage of government employment per capita in the old states.

Source: Federal Statistical Office and OECD.

Hence, the downsizing of government employment needs to be continued. This is facilitated by the fact that the share of tenured civil servants in the new states is relatively low (18.5 per cent as opposed to 42.2 per cent in the old states in 1999). But adjustments of relative wages within the public sector should be considered as well. East-to-west wage convergence is most advanced in the public sector. This observation coincides with the fact that the system governing the remuneration of civil servants precludes any regional wage differentiation, apart from the east-west differential of 13.5 per cent. This restriction should be abolished and wage differentiation be allowed. Such reform would contribute to relieving the budgets of regional governments and raising the allocative efficiency within the public sector. In addition, it would support a larger degree of wage differentiation in the private sector.

Regarding the functional distribution of wage consumption, in most fields outlays in the east per inhabitant appear high in relation to the west (Table 28, lower panel). Moreover, the structure of consumption largely mirrors the one that has already been ascertained with respect to capital spending. In particular, public sector employment in the east exceeds western levels in those fields where the endowment with infrastructure capital, controlled by the states and communities, has already been found to surpass the western levels: the central administration,

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Table 28. **Spending patterns of the states and communities**

Per capita outlays, percentage of spending in the old states[1] (1999)

Current expenditures	100.0	**Capital expenditures**	211.2
of which:		*of which*: Investment	161.2
Personnel	96.1	*of which*:	
Operating expenses	117.0	Construction	192.0
		Equipment	86.7
Interest	97.5	Capital transfers to private sector	330.7
Current transfers to private sector	97.3	Loans to private sector	169.0
		Purchase of equity participations	84.0

Per capita employment and outlays for employees in the government sector, percentage of magnitude in the old states

	Outlays (1998)	Employment (1999)
Central administration	122.7	122.2
Public security and order	113.2	113.6
Jurisdiction	86.1	84.3
Schools	134.7	131.6
Universities and other research	93.9	88.7
Culture	154.0	174.6
Social security	101.9	121.8
Sports and recreation	144.5	120.2
Housing and area development	116.2	111.9
Nutrition, agriculture and forests	–	179.3
Traffic and communication services	103.4	104.5
of which: Roads	106.1	–

Source: Federal Statistical Office and OECD.

culture and social security (Table 25 above). Once capacities have been created, the size of staff employed appears to be complementary.

Given the budgetary constraints in the new states and in Germany overall and the need to improve the frameword conditions for self-sustainable growth (in particular with respect to continuing the modernisation of the infrastructure), it is necessary to restructure the Länder budgets. Public day care services for children (*Kindertagesstätten*) are a major example for social services provided on terms much more generous than in the west. The availability of substantial day care is a positive achievement. However, the demand for such services by private households is a matter of preferences – which might differ between east and west – and the provision of the services has important private goods features and should therefore be adequately priced. With respect to schooling, higher employment-to-population ratios than in the west may be justified given the higher share of young people in the new states. However, demographic developments will dramatically

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reduce the demand for teachers in the near future. Some new states have already reacted to this prospect by offering conditional part-time contracts to teachers, but a more general adoption of such schemes would be appropriate (see the 1998 *Survey*). More generally, surveys indicate that enterprises attach a high value to reducing bureaucratic hurdles to entrepreneurial activity and improving the efficiency of services by state and community governments to foster business activity.[172] Progress in this field appears to be an important contribution to strengthening the framework conditions for self-sustainable growth.

Efforts should also be made to engage more non-governmental bodies and associations in the provision of social and cultural services that up to now are supplied directly by the government. Although of importance for Germany overall, this point is of particular relevance for the eastern states, where the need to mobilise funds to support economic restructuring coincides with above-average government activity in these fields.[173] With a higher weight on subsidiary the extent and quality of the supply of such services could be better tested. Although such action would not fully translate into budgetary savings because public outlays for personnel would be replaced to some degree by transfers to non-governmental organisations, savings could be generated.[174]

Further reduce and restructure subsidies

Subsidies to enterprises in the new states have originally been motivated by the intention to temporarily compensate enterprises for comparative disadvantages originating from the time of the GDR, notably a rotten infrastructure. More then ten years after reunification this rationale is vanishing. Nevertheless, the subsidy intensity has remained very high over the last decade, in contrast to original intentions. To avoid further distortions of the type highlighted above, aid needs to be further scaled down, and support rates need to be reduced, taking into account the accumulation of different support schemes as well.[175] Indeed, the more the economic framework conditions in the east are becoming similar to those in the west the higher will be the damage implied by the distortionary effects of large-scale subsidisation, and the risk that enterprises and activities are supported that would not stand the market test without aid. Moreover, empirical work suggests that the investment volume triggered by subsidies falls short of the financial aid extended if the rate of subsidisation exceeds a certain threshold (an estimated 33 per cent). This threshold is significantly lower than the accumulated maximum support rates applied in the new states. The average rate of investment support in manufacturing has been estimated to total 14 per cent (geometric mean), with about one-third of the supported enterprises exhibiting support rates between 20 and 50 per cent.[176] The resources freed by reducing subsidies would be available for other purposes supporting restructuring and growth, such as the abolishment of still existing bottlenecks in the infrastructure of the new states.

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Care needs also to be taken to avoid that aid for structurally weak regions results in supporting geographically concentrated declining industries.[177]

Consequently, the recent trend towards applying the same set of rules for subsidisation for Germany overall should be enforced and continued.[178] This proposal is reinforced by the fact that the economic performance of some regions in the new states has surpassed the performance of weaker regions in the old states.[179] To account for this fact, the regional support schemes (*Gemeinschaftsaufgabe "Verbesserung der regionaler Wirtschaftsstruktur"*) should be based on a unified scheme for Germany overall. Implementation of this policy is restrained by the fact that the present regional support scheme has been fixed until 2003, tax subsidies until 2004. However, a phasing out or restructuring of subsidies should be decided and announced soon to trigger the necessary adjustments on the side of the enterprises. Since the new Länder are still structurally weaker on average than the old ones a unified subsidy regime for Germany overall would imply that the new states would continue to obtain the larger share of aid per capita.

As highlighted above, empirical research suggests that a significant part of the eastern German productivity gap is attributable to a lack of management and marketing skills. More resources should therefore be devoted to providing such skills. The share of financial aid for training the management and the workforce should be increased at the expense of investment subsidies, reflecting the fact that relatively high degrees of labour turn-over in a transition period make it more difficult for enterprises to finance such training. The propensity of the trained workforce to stay in the east will increase to the extent a more efficient policy mix is contributing to the upswing of the eastern economy.

It has been estimated that there are more than 400 support programmes for eastern German enterprises, including financial aid extended by the state governments. Often the programmes by the states – accounting for some 40 per cent of the total volume of subsidies – are designed so as to fill "support gaps" left in federal programmes.[180] This heterogeneity is likely to increase the distortionary risks of state aid and favours "subsidy competition" between the states. Efforts should therefore be made to co-ordinate and streamline support programmes between the governments involved.

Reform the inter-governmental Revenues Equalisation System

Given that the per capita taxing power of the new states totals only some 40 per cent of the western level, west-to east transfers on a high level will continue to be important. But as has been argued in Chapter II of this Survey, high effective taxation of additional income tax revenues implied by the inter-governmental fiscal equalisation system produces few incentives for the states to support the development of taxable activity. This is particularly true for the new Länder, whose loss rates of between 82½ and 85 per cent are near the top among the federal

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states. The fact that tax auditing rates in the new states fall substantially short of those observed in the old states – as outlined in Chapter II above – supports the view that incentives are ill-defined. Given that the revenues base of the new states falls substantially short of that of the western states, reforming the transfer mechanism of the system, as outlined in Chapter II, is of particular importance for the east (notwithstanding the fact that there are other incentives – outside the sphere of fiscal redistribution – to develop economic activity within the states).

Also, discrimination of the equalisation system in favour of regions with a high degree of agglomeration (see Chapter II above) is of special relevance for the new Länder, two of which stand out with a particularly low degree of agglomeration. This re-enforces the need for adapting the population based rules in the inter-governmental equalisation system to the extent such discrimination is not justified.

Better control social security transfers

As is the case in other segments of the general government budget, in the social insurance system the revenues generated in the eastern Länder do not suffice to finance current spending. West-east transfers via the social insurance system are predominantly financed via taxation of labour (contributions from the wage base). Relieving the system from spending pressures and lowering the tax burden on labour associated with income redistribution would be a potentially important step to reduce adverse effects on employment.

High levels of pensions relative to earnings and a rapidly increasing stock of early retirements, encouraged by financial incentives, put pension finances in the east under particular stress.[181] The special conditions associated with German re-unification therefore reinforce the need for effective pension reform along the lines discussed in Chapter II of this Survey and in the 1999 OECD *Germany Survey*. Moreover, they point to the benefits of introducing a tax-financed layer into the pension system as suggested in the 1999 Survey. If high pension incomes in the east are welcomed as an element of social policies, general tax revenues appear to be better suited than payroll contributions for financing income redistribution because they are more neutral with respect to employment (although they still may imply distortions). The future evolution of pension adjustments in the new states relative to the west might be worth considering as well. While the pension system is unified across the federal states and governed by a common set of rules, eastern pensions are still indexed to the evolution of wages in the east accounting for the fact that full convergence of wage levels has not yet been reached. But with eastern wages still rising faster than those in the west, pensions in the new states also continue to grow faster than in the old ones.

For several years in a row, health care spending in the east exceeded the revenues of the system, and as a consequence, the debt of the health funds is

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mounting. Indeed, health care spending is not only expanding faster than the eastern German wage base but also than spending in the west. Again, these findings reinforce the necessity for a general reform of the German health care system, as it has been addressed in the special chapter of the 1997 Economic Survey on Germany.

To enable the eastern health funds to reduce its accumulated debt, since 1999 transfers are paid from the western to the eastern health funds, with the transfer volume increasing (by about DM 1 billion between 1999 and 2000). While this has helped reduce the debt and the contribution rates of the health funds in the new states, such transfers across the boundaries of individual health insurers do not appear to be compatible with the government's plan to introduce a higher degree of competition between health insurers. They are also at risk to boost spending further. Moreover, at the margin the implied contribution burden in western Germany provides an incentive for the insured in the west to exit the public health insurance system in favour of private insurance, which would have adverse effects on the wage base for the public system overall. Similar problems could arise when the risk equalisation system (*Risikostrukturausgleich*) is being extended to the east. It is therefore welcome that the government is phasing in the extension of the pan-German risk equalisation system over a longer horizon only (between 2001 and 2007), rather than introducing it quickly. However, conditioning the introduction of this system on further progress in the economic integration of the new Länder would be preferable.

Allow for wages undercutting productivity growth and reduce effective labour taxation

As has been argued above, excessive wage settlements in the 1990s relative to the evolution of productivity have diminished east Germany's competitiveness and hampered employment creation. Wage settlements undercutting the evolution of productivity are required to improve competitiveness and employment prospects. While plant level bargaining and opening clauses in collective contracts allowing for plant level agreements have substantially increased at the end of the 1990s, the pace of adjustment appears insufficient given the scale of the unemployment problem. Shrinking membership in employment associations – mainly due to exits by small and medium-sized enterprises – low and declining union membership rates, and a significant share of non-observance of collective agreements indicate that collective outcomes are still largely considered to be at variance with the requirements of individual companies (see Chapter III above and the 1999 *Survey*). Furthermore, the wage distribution in the new states needs to become broader with respect to economic activity to better accommodate structural change and it needs to be sufficiently wide across qualifications to attract the highly qualified while simultaneously supporting employment of those at the low end of the skill distribution. To achieve these goals, the scope for plant level agreements should be widened further.

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For labour demand in low wage segments to develop, supplementary reform in the tax and transfer system is necessary that reduces the effective taxation of labour. This is particularly true for social assistance payments to prevent them from acting as effective minimum wages (see the 1996 *and* 1999 *Surveys*). While reform in this sphere is appropriate for Germany overall, its urgency is re-enforced by the particularly high levels of unemployment in the new states.

Scale-down and revise active labour market measures

While the labour market after unification was in a very difficult condition, and keeping unemployed in work programmes under such exceptional circumstances may have had a value in itself, it cannot be ignored that work provision and training schemes for the unemployed (ALMPs) have not proven to be effective in eastern Germany, might be associated with substantial displacement effects, and are therefore inappropriate in their present form. Work provision and training schemes might be useful tools to foster the re-employment chances of a narrowly-focused group of persons with particularly unfavourable socio-economic characteristics. But extended on a massive scale, such measures give incentives to resist necessary wage adjustments in the primary labour market, and act as an expensive means of social policy, which is also burdensome for the budgets of regional governments.[182]

ALMPs have also been motivated by the desire to reduce east-west migration. However, as has been outlined above, eastern Germany does not appear to suffer from a "brain drain" to the west, and re-migration of qualified labour to the east is occurring at a significant rate. Given the available evidence, a policy is fruitless that aims at keeping young people in the east by means of ALMPs extended on a large scale. Commuting and migration to western Germany should be seen as a chance to improve employment prospects – and preserve and develop human capital in the primary labour market – at a time when the overall capacity of the east to absorb "domestic" labour supply falls short of that in the west.[183]

Hence, the scale of work provision and training schemes in eastern Germany should be reduced considerably, to levels where the programmes and subsequent placement are better manageable than in the present system. With such schemes being concentrated on narrowly defined problem groups, the content of market relevant training should be raised via close consultation with employers on programme design. Programme evaluation by independent bodies should be institutionalised to widen the scope for further corrective actions if necessary. Indeed, the experience in eastern Germany reinforces the need for more efficient job mediation (see Chapter III above). Given the high costs of work provision and training schemes in comparison to unemployment benefits the savings generated by such a policy would be very large.

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Conclusions

In major fields the economic integration of the eastern German states has progressed rapidly. The infrastructure has been built up and modernised at a high speed. Although deficiencies in infrastructure remain, major barriers to growth have been removed. Aided by financial assistance from the west, a strong trend to modernise the business capital stock has been established. The indicators suggest that the age and quality of equipment have reached the west German standard, although the capital intensity, especially in final goods manufacturing, still falls short of the western level. Already at the beginning of the 1990s the elaborate western German social security system has been extended to the new states. Incomes of both the employed and the non-employed, in particular the retirees, have risen fast, and largely approached west German levels. Also, up to now there is a high degree of structural change, as witnessed by high growth rates in manufacturing, increasing export shares, the rapid expansion of the service sector and the down-sizing of the construction sector after very high – and largely policy induced – growth rates in the first half of the 1990s.

However, in the second half of the 1990s economic growth in the east decelerated, and on the aggregate level income convergence has stalled and employment stagnated. Absorption in the new states persistently exceeds eastern production by some 50 per cent, and average productivity has levelled off at two-third of the western level only (slightly less if measured in terms of hours worked) although the size and quality of the production factors would suggest significantly higher levels. These backdrops took place against the background of a high level of west-to-east transfers, which changed little, amounting to some 4½ per cent of western GDP on a net basis annually (tax concessions and debt take-overs not included).

Various factors are associated with this finding. In part, the failure to achieve a self-sustaining strong growth path must be ascribed to the size and persistence of the initial transition challenge. Indeed, the exposure to competition of the eastern German economy revealed that its structural deficiencies were too severe to be remedied quickly by restructuring or wage moderation. Consequently, a large part of the eastern German industry had to be closed down. The industrialisation thereafter had to start almost from scratch and its impact on performance is to some extent masked by the simultaneous downsizing of construction. However, distortions in the economic structure of eastern Germany and high labour costs play a major role. While industrial subsidies helped establishing a modern capital stock, associated distortions of the production structure appear to have reduced productivity growth. In manufacturing, distortions appear to be largely concentrated in basic goods industries. Rapid wage increases, that were not warranted in terms of productivity improvements, hampered the economic expansion and biased it in the direction of non-tradable goods. Such wage

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increases have been a major factor instigating massive transfers from the west, but simultaneously would not have been sustainable without such transfers.

In the future structural and fiscal policies need to rely much more on market forces to bring about economic convergence with the west than they did in the past. The dynamic developments within the manufacturing sector, seen in recent years, indicate the growth potential of the eastern German economy which is to be broadened by such an approach. The OECD Secretariat's policy recommendations, designed to aid the further economic integration of the new states, are summarised in Box 11 below.

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Box 11. Policy recommendations

Continue the development of the infrastructure

Continue preferential infrastructure funding via transfers to the east.

Eastern states and communities should reallocate resources from social services, housing, health and the central administration to other fields being more important for economic convergence.

Revise government spending preferences in favour of investment and at the expense of consumption.

Infrastructure investment projects should be subjected to cost-benefit analysis.

Streamline and restructure spending by states and communities

Public wage consumption needs to be scaled down to free resources for investment.

The downsizing of government employment needs to be continued.

The efficiency of business-related services by state and community government should be increased.

The system governing the remuneration of civil servants should be reformed so as to allow for more regional wage differentiation.

The provision of the marketable public services (like child care services) should be adequately priced.

A more general adoption of part-time contracts (as currently offered in some states to teachers) would be adequate.

Efforts should also be made to engage more non-governmental bodies and associations in the provision of social and cultural services that up to now are supplied directly by the government.

In supplying public services the subsidiary principle should be more often followed.

Further reduce and restructure subsidies

The total volume of subsidies should be substantially further reduced.

Investment support rates – in some cases amounting to 50 per cent in eastern Germany – should be reduced.

The same set of rules – both with respect to eligibility and support rates – for subsidisation should be applied for Germany overall as a principle.

All special programmes for the New States should be associated with a pre-announced time path specifying their phasing out. To the extent aid has been fixed for several years this policy should be decided and announced soon.

Efforts should be made to co-ordinate and streamline the overwhelming number (about 400) of support programmes between the governments involved in order to avoid duplication and support accumulation.

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Box 11. **Policy recommendations** (*cont.*)

More resources should be devoted to training the management and the workforce, at the expense of investment subsidies.

Reform the inter-governmental Revenues Equalisation System

Reform the transfer mechanism of the system (to reduce the disincentive effects of high marginal effective tax rates of the intergovernmental revenues equalisation system).

Abolish the discrimination in the inter-governmental equalisation system against regions with a low population density to the extent such discrimination is not justified.

Better control social security transfers

Relieve the system from excessive social spending pressures and lower the tax burden on labour associated with income redistribution in order to reduce adverse effects on employment.

Finance income redistribution (within the pension system) through general tax revenues, because they are more neutral with respect to employment.

Revisit the future evolution of pension adjustment in the new states.

The financing problems of health insurance funds in the eastern states reinforce the need to continue with health reform.

Phase out transfers for the purpose of debt servicing from the western to the eastern health funds.

The extension of the risk equalisation system (R*isikostrukturausgleich*) to the eastern German health funds should be conditioned on further significant progress in the economic integration of the new Länder.

Allow for wages undercutting productivity growth and reduce effective labour taxation

Almost all of these recommendations have been made for Germany overall, but they are reinforced by the situation in the new states.

Wage moderation should reduce labour cost growth relative to the evolution of productivity in order to improve competitiveness and employment prospects.

The wage distribution in the new states needs to become broader to better accommodate structural change and to attract the highly qualified while simultaneously supporting employment of those at the low end of the skill distribution.

The (legal) scope for plant level agreements should be widened further.

Further reform in the tax and transfer system is necessary that reduces the effective taxation of labour.

Reforms are required that prevent social assistance payments from acting as effective minimum wages.

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Box 11. **Policy recommendations** (*cont.*)

Scale-down and revise active labour market policies

Utilise the "Alliance for jobs" talks to establish a policy framework that fosters market-driven economic convergence.

The provision of active labour market policies (ALMPs) – work provision and training schemes – should be reduced considerably to levels where the programmes and subsequent placement are better manageable than in the present system.

Work provision and training schemes should be concentrated on narrowly defined problem groups and the content of market relevant training should be raised via close consultation with employers on programme design.

Programme evaluation by independent bodies should be institutionalised to widen the scope for further corrective actions if necessary.

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Notes

1. Indeed, German exports appear to be less price sensitive than that of other countries. See, for example, Carlin *et al.* (1999). This study finds that the German export market performance is more relying on new technologies than on price competition.
2. Deutsche Bundesbank (2000) Monthly Report, October 2000.
3. Association of German Chambers of Industry and Commerce (DIHT) (2000).
4. For a more extensive discussion of the issues involved see S. Koch (2001) and H.-U. Bach (2001).
5. "Small hours" jobs are defined as less than 15 hours per week with a maximal monthly earning of DM 630, DM 590 for eastern Germany. For employers there has been little change: according to the new rules, employers have to pay social security contributions of 12 per cent for the pension system and 10 per cent for the health funds, replacing a previous 20 per cent lump sum income tax payment.
6. In 1999 the Federal budget took over the debt servicing of three hitherto off-budget funds: the Inherited Debt Fund, the Railways Fund and the Coal Fund. Associated additional budgetary outlays for debt servicing are matched by the abolishment of Federal Government transfers to these funds.
7. Total tax reductions, implied by the entire income and business tax reform legislation implemented since 1999, have been estimated by the government to amount to more than DM 90 billion (or about 2 per cent of GDP in 2000) annually when all reform steps are fully effective.
8. Federal privatisation receipts are expected to double in 2001 compared with the year before. Main projects are the sales of shares in the Post Office and in airports.
9. Boss *et al.* (2000).
10. The "solidarity" income tax surcharge will remain unchanged at 5.5 per cent of the income tax liability.
11. The German Council of Economic Experts, Sachverständigenrat (2000), points out that the local business tax should not be deductible from the income tax liability for reasons of taxation systematics.
12. Annual revenues foregone from extending the community allowance are estimated to total DM 1 billion.
13. This projection assumes a net immigration rate to Germany of 100 000 persons annually.
14. Originally, the government planned to index pensions to the inflation rate in both 2000 and 2001. While indexation to inflation was maintained for 2000, the government decided, to apply already in 2001 the new indexation formula of the pension reform law.

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15. The reference replacement ratio in the pay-as-you-go system is defined as the level of an old age pension received after 45 years of contributions out of average gross earnings relative to net average earnings.
16. See Sachverständigenrat (2000).
17. The envisaged reduction in replacement ratios of the pay-as-you-go pension system has been judged by the Scientific Council of the Ministry of Economics and Technology as insufficient to stabilise contribution rates at the levels targeted by the government. See Wissenschaftlicher Beirat beim Bundesministerium für Wirtschaft und Technologie (1998) and Wissenschaftlicher Beirat beim Bundesministerium für Wirtschaft und Technologie (2000).
18. Psychiatrics will be exempted from the new reimbursement system.
19. Lowering unemployment is one such incentive. However, similar to taxation of individuals, the equalisation system is associated with disincentives to raise the income base. The loss of a Land's additional tax revenues due to redistribution and transfer reductions can itself be considered as a "taxation" of Länder revenues, in analogy to effective taxation of personal income. For individuals, effective taxation of additional income also consists of charges to be paid by those receiving income above a certain level and benefit withdrawals for those with income below this level.
20. Following interventions by the Federal Ministry of Finance the number of tax auditors was raised, and tax audit coverage rates have improved from 1996 onwards. The number of tax audits of medium-sized companies increased from 51 240 in 1996 to 67 054 in 1999. Moreover, in 1999, the Federal Finance Office participated in a total of 1 374 audits conducted by the revenue authorities of the Länder. This figure includes 185 full audits of large-sized enterprises in the new states undertaken by the Federal Finance Office after consultation with the states' revenue authorities in the new states. In 1999 the tax audits generated almost 27 billion additional tax revenues in Germany overall.
21. See Baretti *et al.* (2000).
22. Bundesrechnungshof, Bemerkungen 2000 zur Haushalts- und Wirtschaftsführung des Bundes, Bonn, 2000.
23. Proposals for reform along these lines, taking into account the rulings of the Constitutional Court have are contained in Huber and Lichtblau (2000).
24. See Baretti *et al.* (2001). This study uses two approaches to assess the differences in costs between the city states and the area states. One approach compares the costs of input factors (personnel, equipment), identifying as determinants for the overall cost differentials both differences in input prices and in quantities necessary to produce a given output level. The other approach is output-oriented and aims at identifying additional public-sector services of the city states in comparison to the area states. Netting out additional services from the total allows estimating the city states' incremental costs of producing the same level of output as the area states. Furthermore, see Deutsches Institut für Wirtschaftsforschung (2001). The DIW argues that commuting of employees to and from city states implies major fiscal disadvantages for the city states which in combination with agglomeration-related factors require a high premium in the revenues redistribution system. Mainly, the commuters' income tax revenues do not accrue to the city states. On the other hand, the city states face extra outlays for providing infrastructure for the commuters.
25. The Centre for European Economic Research, Mannheim, found for the level of state governments that the per-capita costs of certain public sector activities are significantly positively related to population density while for other activities the opposite is

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true. Overall, the study finds that the degree of agglomeration has no statistically significant impact on total costs in the budgets of the state governments (see Büttner *et al.*, 2001).

26. Seitz (2000*b*) also considers the impact of population density on the costs of providing certain types of public-sector services. This study includes spending on the level of the communities as well, and arrives at the conclusion that additional per-capita costs on account of extremely low population density relative to the norm for providing infrastructure services totalled at least between 6.5 and 9.5 per cent of the Länder and community governments' overall expenditure. For a similar view questioning the justification of redistribution premia for more densely-populated areas see Söllner (2001).
27. See Seitz (2000*b*).
28. See Baretti, Fenge, Huber, Leibfritz and Steinherr (2000).
29. The opening clauses permit companies to pay wages below the collectively-agreed level if certain conditions are fulfilled (small number of employees, employment protection etc.).
30. Schäfer and Wahse (2000).
31. See Hunt (1999).
32. Information on coverage by working time accounts is obtained from the establishment panel of the Institute for Labour Market Research of the Federal Labour Office. See Schäfer and Wahse (2000) *op. cit.*
33. Fels *et al.* (2000).
34. Generally the Federal Labour Office subsidises part-time retirement only if additional workers have been hired. Since most workers opt for the so-called lump model (they work in the first half of the span to retirement full time and spend the second half in early retirement), not many additional workers have been hired so far.
35. This applies to persons without additional taxable income. The income of spouses is not taken into account. See Rudolph (1999).
36. ISG Sozialforschung und Gesellschaftspolitik and Kienbaum Management Consultants GmbH (1999). Most of the workers who quitted secondary jobs received low income from their primary job. They probably switched to the shadow economy.
37. ISG Sozialforschung und Gesellschaftspolitik and Kienbaum Management Consultants GmbH (1999), *op. cit.* which based their study on data up to August 1999, estimate that overall 90 000 regular half-time or full-time jobs haven been created as a result of the new regulation. According to figures published by the Federal Statistical Office, the number of part-time jobs (regular plus small hours jobs) in wholesale trade declined by about 10 per cent during the 12 months following the new regulation. In retail trade, the drop amounted to about 6 per cent, in hotel and restaurant trade part-time employment declined by 12 per cent. The number of full-time jobs in wholesale trade increased by about ½ per cent, in retailing it did not change, and in hotel and restaurant trade it declined by ½ per cent. In previous years, full-time employment in these branches had on average declined by about 1½ per cent, 2½ per cent and 3 per cent, respectively. In absolute numbers the net gain in full-time jobs in these sectors might add up to almost 45 000.
38. See OECD *Germany Survey* 1996 and Kolb (2000).
39. ISG Sozialforschung und Gesellschaftspolitik and Kienbaum Management Consultants GmbH (1999) *op. cit.*

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40. Temporary help agencies in Germany are allowed to place employees in a company for a maximum span of only 12 months, can grant fixed-term contracts under special conditions only, are not allowed to contract workers just for one specific assignment ("prohibition of congruency"), must not rehire workers they dismissed up to three months ago, and may not hire-out workers to the construction industry.

41. See OECD (2000*b*).

42. See OECD (1999) for a discussion of the issues involved.

43. The higher incidence of fixed-term employment in eastern Germany is the consequence of the higher share of persons in employment promotion schemes.

44. For details, see Rudolph (2000).

45. A recent empirical study by J. Heckmann and C. Pagés (2000) covering Latin America and OECD countries concludes that employment protection reduces employment. The composition of unemployment, moreover, appears to be heavily influenced by the regulation of fixed-term contracts and by dismissal protection. Empirical evidence reveals that a relaxation of employment protection tends to favour outsiders, because employers usually are less choosy when they have the chance to dismiss employees at low cost.

46. Workers aged 58 or older are exempted from all restrictions on the renewability of fixed-term contracts. Previously this was true for those older than 60 years.

47. Elmeskov *et al.* (1999) and Nicoletti *et al.* (2000).

48. These persons were not registered as unemployed in the Labour Market Statistics of the Federal Labour Office. Thus for a fixed pattern of unemployment, figures for registered unemployment and the rate of registered unemployment as published by the Federal Labour Office will be higher under the new regulation. In December 2000, 190 000 unemployed persons aged 58 and more received benefits under this provision. Recording them as unemployed would have increased the headline unemployment count by 5 per cent and the unemployment rate by 0.5 percentage point. There will be no sudden increase in unemployment numbers, however, as older unemployed already subject to this clause will not loose their privileged status.

49. On these issues, see Empter and Frick (1999).

50. According to a survey by the Deutsche Städtetag, in 1998 local authorities employed 300 000 persons entitled to social assistance, 50 per cent above the level of 1996.

51. German law stipulates that beneficiaries of social assistance are obliged to accept a job when available.

52. Older unemployed now may receive employment promotion and reintegration grants even if they had not yet been long-term unemployed, and the maximum period older long-term unemployed are allowed to stay in some employment creation schemes has been extended to five years.

53. On this topic, see Brinkmann (1999 and 2000).

54. Schmidt (2000).

55. Preliminary results have been published, showing that only 31 per cent of the participants in special programmes got a job immediately after in the primary labour market or started vocational training, 6 per cent returned to school and 21 per cent participated in other labour office programmes, while 36 per cent registered as unemployed. However, after six months just 25 per cent were still unemployed. See Dietrich (2001).

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56. At an hourly wage rate of DM 10, monthly wage cost subsidies for a full-time job amount to DM 345. With rising hourly wage rates the subsidy declines, reaching DM 0 at an hourly wage rate of DM 18.
57. Jacobebbinghaus and Steiner (2000).
58. Up to 10 000 residents from non-EU countries may be given a residence plus a work permit for a period up to five years, given that they have completed university or polytechnical college education with the main emphasis on information and communications technology, or have signed a work contract for an IT job implying an annual remuneration of at least DM 100 000. If there is further need, the number of permits may be increased to 20 000. Up to the beginning of December nearly 3 500 "green cards" had been issued.
59. However, according to another ifo-survey, less than 5 per cent of all companies in manufacturing are hindered by lack of skilled labour. For details, see E. Magvas and E. Spitznagel (2000).
60. See OECD (2000*b*).
61. While it takes a long time in Germany until a young scientist may teach and do research independently, tenure is granted relatively early once this position has been reached.
62. Monopolkommission (2000*a*).
63. Boylaud (2000).
64. Halk and Täger (1998).
65. On this issue see the legal expertise forwarded by the Banking Federation of the European Union (Gleiss/Lutz/Hootz/Hirsch, 2000) and the arguments in defence of the status quo by the association of German savings banks (Deutscher Sparkassen- und Giroverband, 2000).
66. Explicit and implicit guaranties for public banks imply that there are no default risks. This tends to improve ratings and lessens the cost of refunding. For instance, in 1999/2000 a major rating agency awarded a German state bank with the long-term credit-rating Aa1, whereas the leading private banks typically were rated Aa3. The same agency awarded a D-rating for the bank's financial strength net of public-sector guaranties, which is substantially below the typical rating for leading private banks in Germany (B). See Gleiss/Lutz/Hootz/Hirsch (2000) *op. cit.*
67. See Gonenc *et al.* (2000) for an extensive discussion on vertical separation issues.
68. Bundesregierung (1999).
69. See press release of the Federal Cartel Office on 08/09/2000. On this issue see also Link (2000).
70. It is true that there is strong intermodal competition which may serve as a substitute for infrastructure based competition in railways. However, since it does not seem to be feasible to privatise the railway network, a vertically-integrated railway company will always be in a position to obstruct the business of competitors.
71. Kommission Verkehrsinfrastrukturfinanzierung (2000). See also Link (2000) *op. cit.*
72. Regulatory Authority for Telecommunications and Post (2000).
73. Providers of postal services need a licence for the conveyance of letters weighing less than 1 000 grams. When the market for letters was opened for competition in 1998, the Deutsche Post AG was granted an exclusive licence (with some minor exceptions) for

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services relating to letters weighing not more than 200 grams. This exclusive licence expires on 31 December 2002.

74. The issue of universal service requirement is often raised in connection with liberalisation of network industries. However, Swedish Post (the former monopolist) regards this requirement as a competitive advantage due to network effects and is not compensated for providing universal services. See OECD (1999) and R. Gonenc *et al.* (2000) *op. cit.*

75. International remailing agreements are stipulated in the World Postal Agreement under the Universal Postal Union. However, within the EU the member states have granted each other additional remailing obligations. In addition, in 1999 most European postal operators agreed upon cost-based terminal dues.

76. The Deutsche Post allegedly granted rebates to big mail-order retailers under the condition that all parcels were sent by the Post. In summer 2000, the regulatory authority ruled that rebates for large-scale mail-order parcel services hinder the evolution of competition and thus are in conflict with legal provisions.

77. WIK (2000).

78. Monopolkommission (1999).

79. Bundesregierung (2000).

80. A recent study sponsored by the European Commission concludes: "Of the countries that liberalised in January 1998, Germany has seen the fastest development of competition over the period", Teligen (2000).

81. The liberalisation of voice telephony became effective on 1 January 1998.

82. In 1992, a licence for a second network was granted to a new market entrant. Two further licences were issued in 1994 and 1998, respectively. In 1995, price recording for mobile phone services commenced.

83. The market for telecommunication terminal equipment was almost completely liberalised in 1989. In 2000, the remaining *ex ante* conformity assessment and approval procedure for terminal equipment was abolished.

84. The German price index for telecommunication services is based on the consumption pattern in 1996, two years before the liberalisation of markets took place. Thus it is rather likely that there is a substantial substitution bias (relative prices have changed dramatically because of the rebalancing of tariffs). However, the relative importance of mobile phone calls is derived from the consumption pattern in 1998. Furthermore, it seems that there is a kind of new outlet bias (prices charged by new market entrants typically are included only with a certain delay, with an inadequate weight and sometimes with an inadequate linking procedure). On this topic see Schatz (2000) and Bundesbank (2000). For details of the quite complex German consumer price index for telecommunication services and the pros and cons of different approaches, see Beuerlein (1999).

85. Notwithstanding the massive price reductions by DTAG, protracted price differences between the incumbent and new market entrants persist for national and international long-distance calls. In autumn 2000, minimum prices for call-by-call voice telephony typically were 50 per cent and in a few sub-markets up to 90 per cent lower than standard prices of DTAG. Some tariffs with preselection offer still more favourable terms. It is difficult to explain why substantial price differentials coexist with only slowly changing market shares as the accessibility of new providers has improved considerably and transaction costs are minimised. (The incumbent is obliged to do the billing and the

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collection for call-by-call services of new providers. Information costs are reasonably low since many newspapers publish surveys of telecommunication tariffs. A private consulting company even collects data on tariffs and puts them on the Internet free of charge.)

86. As DTAG has reduced prices for digital service subscribers substantially in 2000, many new competitors could no longer compete with the incumbent. A competitor has complained to the regulatory authority that DTAG offers digital service subscriptions which allow high-speed Internet access at prices below costs. The regulator has decided recently to accept the access fees for digital services (T-DSL) and requested access to high frequency bands for competitors.

87. Charges for the use of elements of DTAG's network other than the local loop.

88. More than 50 companies already offer customers the full range of local telecommunication services. In about half of the cities with more than 50 000 inhabitants (about 25 per cent of the population) customers now may choose between two or even more service providers.

89. These are facilities at which providers connect leased lines to their own network.

90. Internet access is typically facilitated by calls to a national provider. Either the national provider bills its customers both for the call and for the provision of Internet services or DTAG does the billing for both (Internet by call). As in most OECD countries, tariffs for Internet-access including fees for the use of the telecommunication network now typically are below the prices for local calls (Paltridge, 2000). To some extent this price differentiation can be explained by cost savings resulting from the bundling of calls to Internet providers. However, it is quite likely that DTAG also follows a strategy of separating markets with differing elasticities of demand, charging lower price in those market segments with a high elasticity.

91. Subsequently, the subsidiary announced in winter 2000 that it would withdraw the offer for standard lines.

92. Most of the providers of telecommunication services favoured this element-based tariff model because it fosters the build-up of competing infrastructure. It was, however, disputed whether charges should relate to the number of elements actually used in the traditional network of DTAG or the number of elements in an abstract network, as it would be technically feasible today. DTAG claimed an adequate return on investments. Therefore, interconnection charges should relate to its existing network. DTAG's competitors then would have to connect their networks at up to 1 000 points to DTAG's grid to qualify for the most favourable interconnection tariff. The competitors of DTAG complained that a policy like that would demand that they replicate the inefficient traditional network of DTAG. Furthermore, they demanded that the level of tariffs should be adjusted to best practices in countries that already had liberalised telecommunication, and not to an average of countries as stipulated previously. The German Telecommunications Act demands that the regulatory authority approve rates "on the basis of the costs of efficient service provision falling to the individual service ...". In autumn 2000, the regulatory authority ruled that the number of interconnection points which qualify for the lowest interconnection charge should be fixed at 475.

93. E*x ante* regulation implies that the incumbent has to ask the regulatory authority for permission if it wants to change prices. Price changes may come into effect only after authorisation by the regulatory authority. E*x post* regulation means that the regulatory authority may review the actions of the market-dominating enterprises.

94. Regulatory Authority for Telecommunications and Post (2000) *op. cit.*

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95. Monopolkommission (1999) *op. cit.*
96. Bundesministerium für Wirtschaft und Technologie (2000).
97. A piecemeal approach of market-by-market relaxation of the regulation as emphasised for example by Schedl (2000) presupposes that the behaviour of the incumbent on markets for essential inputs and on output marktes still dominated can be regulated in such a way that no extra profits will be generated which could be used for predatory pricing strategies in emerging competitive markets.
98. Welfens (2001) and Bunte (2001).
99. Without the tax-induced price increase in spring 1999. A further increase of the energy tax at the beginning of 2000 and the introduction of general contributions covering the costs of the price-fixing for renewable energies and combined heat and electricity generation did not show up in the figures until autumn 2000.
100. Price increases, however, have to be authorised.
101. Originally, the second agreement between electricity generators, distributors and industrial consumers divided the market for electricity in Germany into two regions Transactions crossing this fictional border would have entailed additional expenses. But when in spring 2000 two vertically-integrated energy companies asked for permission to merge, the Federal Cartel Office granted the permission only upon the condition that the new company should refrain from applying this potentially anti-competitive measure. Afterwards all transmission declared that they would forfeit this national transmission fee.
102. However, the competition authority may demand immediate implementation for its rulings without any pending to the courts.
103. For details, see ewi and BET (2000).
104. Furthermore, the revised Energy Industry authorises gas and electricity network operators to refuse transmission of energy originating from countries which do not allow third-party access in a similar fashion.
105. Main features of the West German economy since the 1950s have been a high income per capita, a high degree of competitiveness and a solidarity based approach to social policy and intra-state fiscal relations that is being reflected in the constitutional postulate to secure roughly equal living conditions across Germany ("Social Market Economy", *Soziale Marktwirtschaft*).
106. By the middle of the decade employment in manufacturing had almost halved in comparison to 1990.
107. The value-added share of construction in the new states also exceeds that of the European transition countries. The share of construction in real value added totalled 13.5 per cent in 1999 (down from 17.1 per cent in 1994. In contrast, the share was 4.6 per cent in the Czech Republic (1999), 4.4 per cent in Hungary (1998), and 7.9 per cent in Poland (1998).
108. In the third quarter of 1990, immediately after the introduction of the Deutschemark in eastern Germany, industrial production dropped by some 50 per cent (index of industrial production of the German Democratic Republic). Following this decline, production was roughly stabilised for a few months by massive government aid supporting the trade with eastern European countries. When this aid was phased out, the recession accelerated again, leading to a further drop in manufacturing output of some 25 per cent in the first half of 1991 (index of manufacturing production of the *Statistisches Bundesamt*). See Deutsches Institut für Wirtschaftsforshung *et al.* (1999). National

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accounts statistics do no longer separate between the western and the eastern part of Berlin. To facilitate comparisons, the GDP and employment figures stated above refer to eastern Germany, western Germany and total Germany (GDP) without Berlin.

109. Sachverständigenrat zur Begutachtung der gesamtwirtschaftlichen Entwicklung, Wirtschaftspolitik unter Reformdruck, – Jahresgutachten 1999/2000, Stuttgart, 1999.

110. Between 1990 and 1995 1.3 million new companies were formally registered, and 0.6 million were dropped. However, the actual balance has been estimated considerably lower by some 0.3 million. In the second half of the decade, the rate of foundations declined significantly. See Deutsches Institut für Wirtschaftsforschung *et al.* (1999).

111. Measured in terms of real gross value added per employed; Berlin excluded.

112. Labour productivity in the overall economy totalled around 61 per cent of the western level in 2000 if measured in terms of real gross value added per hours worked rather than per employed persons.

113. Unemployment stocks are proportionately related to inflow rates into unemployment and unemployment duration. For example, one and the same unemployment rate can be generated by high inflow rates into unemployment and short unemployment duration or by the opposite combination. For an exposition about the relation between stocks and flows on the labour market see Wurzel (1993).

114. Deutsches Institut für Wirtschaftsforschung *et al.* (1999).

115. Unemployment duration sampled as complete spells from the flow.

116. See Schulz (1998).

117. See Hagen, Steiner (2000).

118. See Franz and Steiner (2000).

119. There was also income support paid on a large scale for workers in regular jobs working for shorter periods to maintain the companies' labour force. This type of support – which is being applied in the west to ease redundancies over the business cycle – has been scaled back to the extent other active labour market measures have been installed.

120. This includes persons who previously have been in pre-retirement measures.

121. High labour force participation at the time of unification reflected the inefficient operation of the old planned economy. While female labour force participation in the east declined substantially over the 1990s, it still exceeds participation in the west.

122. According to census information of the Institute of Labour Market Research of the Federal Labour Office. According to survey information from the German Socio-Economic Panel the difference between eastern and western employees amounts to 8.5 per cent, measured in terms of the number of hours worked per week (1998). See Deutsches Institut für Wirtschaftsforschung *et al.* (2000).

123. See Paquè (2000) and Leibfritz (1999). The data are subject to a selection bias in that higher income households are likely to be under-sampled in relation their distribution in the population. This, in turn, would imply that the true income gap between the new and the old states is somewhat larger than the one sampled.

124. Household "net equivalent incomes" are computed to account for the fact that the financial needs of households do not rise proportionately with household size. For this purpose, each family member is attributed a weight, which is a certain fraction of the weight attributed to the head of household. Total household net income is then divided by the sum of the weights rather than the sum of household members. In the

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present case, a weight of one is attached to the head of household, weights of 0.5 for each other household member aged 15 or older, and weights of 0.3 for each further household member aged 14 or younger. See Münnich and Ilgen (2000) and Münnich (2000).

125. Household surveys on income and consumption patterns by the Central Statistical Office are available for 1993 and 1998. See Münnich and Illgen (2000) *op. cit.*

126. As all flows, public pensions in eastern Germany were converted to Deutschemarks at a conversion ratio of 1:1. The adaptation of the eastern pension system to the western standard proceeded in two steps: in the first stage, the ratio of pensions to net earnings was raised proportionately such that the replacement ratio equalled 70 per cent for those with 45 years or more of employment, as in western Germany. In mid-1990, after the currency conversion the average pension level increased by 30 per cent. The second major step occurred in January 1992 when the western German pension system was legally extended to the east in accordance with the Pension Transition Law (R*ent-enüberleitungsgesetz*). In cases where the new pension fell short of the pension previously paid, supplementary pensions (A*uffüllbeträge*) were introduced. Since 1996 the Auffüllbeträge are decreasing as pensions rise. Up to 1996 the level of eastern German pensions was adjusted twice a year in accordance with the development in net wages in the new states, and this now takes place once a year.

127. Sinn, Hans-Werner (2000).

128. The highest net imports observed in the OECD in 1999 amount to 10.4 per cent of GDP (Greece) followed by 8.4 per cent (Portugal). While the imports of an entire country are not fully comparable with those of an area such as eastern Germany, this illustrates the high level of the new Länder's imports in relation to their economic activity.

129. Ragnitz *et al.* (1998).

130. See Seidel and Vesper (2000). Assessment of the infrastructure capital stock is subject to a number of methodological problems. Mainly, these concern the valuation of the pre-unification capital stock still in use, and definitional issues arising from investment activity outside the general government sector. To facilitate comparison between the east and the west, a narrow definition of the capital stock was computed as well. The comparison with the financially weaker western Länder considers only the old Länder which are receivers in the Fiscal Equalisation system (L*änderfinanzausgleich*).

131. See Müller, A. (2001) for estimates of the capital intensity in the business sector overall. Estimates of the capital intensity have been affected by the adoption of new national accounting conventions (ESA 95). For older studies, which were based on national account data prior to the revision, see the following references: for 1998 the capital intensity in the business sector has been estimated to total three-quarters of the western level if related to the number of employed (Ragnitz, 2000*b*) and if related to the size of the working-age population (Müller A., 2000). For 1999 six leading German research institutes jointly estimated that the capital intensity in the business sector totalled three-quarters of the western level in terms of the population. See Ragnitz (2000); Müller A. (2000) and Deutsches Institut für Wirtschaftsforschung, HWWA *et al.* (2000).

132. In the Enterprise Establishment Panel of the Institute for Employment Research establishments in all sectors of the economy are asked to rate their equipment on a five brackets scale extending from "newest state of technology" to "very obsolete". There are hardly any response differences between firms in the new and the old states. See Schäfer and Wahse (2000).

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133. For summaries of the research see Rothfels, Wölfl (1998); Müller, Rothfels, Wölfl (1998); Institut für Wirtschaftsforschung Halle (2000).

134. This is based on the subsidy accounting of the Federal Government as laid out in the Subsidy Report. See Bundesministerium der Finanzen (1999).

135. This is based on the Subsidy Report of the Federal Government. See Ragnitz (2000*a*).

136. The impact of the various types of investment support schemes on the user cost of capital has been studied by several authors. See Sinn, G. and H. Sinn (1991).

137. For a similar argument see Sinn (2000) and Klodt (2000). Sinn (2000) argues that high rates of subsidisation may even have induced firms to employ capital up to a point were labour productivity has been decreased. The argument is based on estimates indicating negative user cost of capital in the first half of the 1990s.

138. For an account of investment by sector see Albert Müller (2001). Investment data have been subject to significant change due to the adoption of the new European national accounting conventions, ESA 95. The information presented in this Chapter is based on the new, revised figures.

139. Most of the tiles' buildings are owned by housing societies, which in turn belong to the communities. Given the weak revenues base of the communities the Länder will bear the costs of dismantling the P*lattenbauten*. In addition, the Federal Government will remit the debt of the housing societies in proportion to their holdings of vacant flats.

140. For the group of manufacturing industries with capital intensity below the western level (70 per cent of the total) the correlation coefficient across sectors between productivity and capital intensity (both relative to the west) amounts to +0.2. For the group of industries with capital intensity above the western level (30 per cent of the total) the correlation coefficient amounts to –0.6. Overall the disaggregation of the manufacturing sector considered comprises 30 industries. The inverse relationship between relative capital intensity and productivity is forcefully demonstrated by mineral oil processing whose capital intensity exceeds the western level by 90 per cent while the productivity level reaches only 35 per cent of that in the west.

141. The branch data, on which this analysis is based, have been made available by the German Institute for Economic Research (DIW). Information on productivity and capital intensity is also available from the Establishment Panel of the German Institute for Employment Research covering smaller establishments as well. See Bellmann and Brussig (1999).

142. For empirical work on capital for labour substitution in eastern Germany see Gerling (1988) and Gerling (2000).

143. See R. Barrell and D. W. te Velde (2000).

144. Distortions in eastern Germany's market structure might also contribute to relatively high input prices in comparison to the west. G. Müller (1998) estimates that the productivity gap to west Germany would drop significantly if west German input and output prices were assumed.

145. For western Germany average labour productivity in rural administrative units has been estimated to fall short by eight per cent or more of productivity in administrative units with a medium population density. See Seitz (2000*b*).

146. However, with respect to input prices it is worth mentioning that eastern German firms are subject to higher energy prices than their western counter-parts. This is due to the fact that electricity generation from brown coal – which dominated energy generation in the GDR – is sheltered against competition from other energy sources.

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147. See Deutsches Institut für Wirtschaftsforschung *et al.* (2000).
148. See Lindlbauer (1999) and (2001).
149. See Schäfer and Wahse (2000).
150. See Bellmann and Brussig (1999). The 1995 Labour Force Survey leads to the same conclusion. See Franz and Steiner (2000).
151. This finding is consistent with the fact that net east-to-west migration increased since 1998, after eastern German growth rates had declined and even fell below west German levels.
152. For empirical work on east-west migration and commuting in Germany see Burda (1993); Burda *et al.* (1998), Kempe (1999) and Hunt (2000).
153. See Kempe (1999).
154. Unit labour costs are measured here in terms of compensation per dependent employee divided by value added in 1995 prices per employee.
155. Downsizing of employment in the manufacturing sector was much stronger in Germany than in the other European transition countries.
156. At the time of the GDR the official exchange rate between DM and East Marks was 1:1. This represented a vast over-valuation of the East Mark. However in the absence of on exchange market for the East Mark, no reliable data on the market evaluation of the currency are available. The black exchange market was too narrow to yield such information. Estimates based on purchasing power parity, such as Sinn (1991) suffer from the fact that the conditions for purchasing power parity to hold were not given.
157. The "tax wedge" between consumption and production wages increased rapidly in the new states, with potentially adverse effects on labour demand. For an analysis of the tax wedge in eastern Germany see: Franz (1999). For an account of the concept see also OECD (1994).
158. Sinn and Westermann (2000) compare eastern Germany with the Italian Mezzogiorno, arguing that in both cases high and persistent transfers have changed the incentive structure in the economy away from productive activities.
159. Investment subsidies lower the price of a factor of production, which in isolation increases output and employment (output effect). On the other hand, they lower the price of capital relative to labour, which favours the substitution of capital for labour (substitution effect). The net effect of capital subsidisation on employment can be either positive or negative, depending on which of the two effects dominates. Indeed, in an economy exhibiting very high fixed capital investment, as observed in the new Länder, the technical feasibility of substituting one factor of production for the other – capital for labour or *vice versa* – is higher than if the capital stock is largely given. Empirical evidence suggests that the output effect of investment subsidies has been larger than the substitution effect. However, there is evidence for the manufacturing sector that investment subsidies have implied significant substitution effects in favour of capital and at the expense of labour. This has reduced the employment intensity of economic growth in particular for low skilled labour. By contrast, in research and development the subsidisation of labour input has lead to over-staffing in the production process if assessed against western Germany. See Gerling (1998). Also, Gerling (2000).
160. See Franz and Steiner (2000).
161. Deutsches Institut für Wirtschaftsforschung *et al.* (1999).
162. See Barjak (1999) and Laschke (1999).

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163. The wage drift is even larger than Figure 31 indicates since social security contributions by enterprises – which are included in compensation – have risen over most of the 1990s.

164. The Federal Government is aiming at more comprehensive evaluation studies of ALMPs and has made a commitment to facilitate access of third parties to the data of the Federal Labour Office. See the Annual Economic Report of the Federal Government 2001.

165. See Hagen and Steiner (2000).

166. See OECD (1991).

167. In the second part of the 1990s revenues of the states and communities combined levelled off due to tax shortfalls that were largely unforeseen. The investment spending restraint in the east has been attributed to the shortfall in tax revenues (see Seidel, Vesper, 2000). However, since these shortfalls were largely driven by tax expenditures in favour of the new states, industrial subsidies for the east have effectively crowded out eastern public sector infrastructure investment. At its peak, annual revenues foregone from special depreciation allowances for eastern Germany alone totalled ¼ per cent of pan-German GDP (1996) declining thereafter when tax concessions were reshaped (*Investitionszulagen*).

168. In real terms investment now matches the level that prevailed in the old Länder at the beginning of the 1980s, at a time where the infrastructure capital stock in the west was more advanced as the one in the east is now. In west Germany growth in infrastructure investment after the war exceeded GDP growth for two decades. During the 1950s and 1960s nominal GDP grew at an annual rate between 11 and 8 per cent, while government fixed investment expanded by between 17 and 13 per cent annually. This changed in the seventies, with the average annual growth in infrastructure investment (5 per cent) falling short of average nominal GDP growth (8 per cent) by 3 percentage points. In the 1980s west German infrastructure investment stagnated on average (falling at the beginning of the decade but rising in the second half), while nominal GDP grew by 5 per cent on average.

169. Budgetary spending patterns by the new and the old states exhibit broad differences that one would associate with the differential stages of structural adjustment: The new Länder (states and community governments combined) devote a higher share of their budget to operating expenses (*Sachaufwand*), infrastructure investment, and transfers to enterprises (Table 28, upper panel). Overall outlays for personnel per inhabitant (wages and outlays for retired civil servants) fall short of those in the west.

170. In the German system, full pensions are paid out of the governments' budgets – rather than by the pension insurance – only for tenured civil servants *Beamte*. Since this type of tenure could only be given after reunification the average seniority of active *Beamte* as well as the stock of those having already retired is much lower than in the west.

171. See Deutscher Bundestag (2000).

172. See Deutsches Institut für Wirtschaftsforschung *et al.* (1999).

173. There are fewer associations outside of the government sector providing social and cultural services. In part, such tasks are therefore provided within the government sector.

174. For a discussion of the public finances in the new states see Seitz (1999) and (2000*a*).

175. For simulations of the demand effects of reducing transfers to the new states see Dreger and Ragnitz (2000).

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176. This work is based on the 1997 wave of the Establishment Panel of the Institute for Employment Research. See Müller, G. (2000). See also Ford and Suyker (1990).

177. In the past, leading German economics research institutes have repeatedly claimed that high rates of subsidisation, notably for declining industries such as shipbuilding and steel, have hampered structural change in Germany. There appears to be some evidence to support this proposition although a recent study based on vector autoregressive analysis failed to establish a statistically significant link between subsidisation patterns and structural change. See Stammer (1998).

178. See Ragnitz (2000*a*).

179. See Sachverständigenrat (1999).

180. See Ragnitz (2000*a*), *op. cit.*

181. Long employment spells of the retirees during their economically active life – which constitute the main factor explaining the high level of pensions in the new states – cannot serve as an economic rationalisation for high pension levels. In the GDR long work spells, in particular for women, came about because of the inefficiency of the centrally-planned economy.

182. See OECD (1991) *op.cit.*

183. More generally, with substantial differences in productivity – such as those observable between east and west Germany – migration from low productivity, low wage areas to high productivity, high wage areas is welfare enhancing. Wages need to be made sufficiently flexible, however, to accommodate the shifts in labour supply.

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Annex

Chronology of main economic events

1999

November

The European Central Bank (ECB) raises the repurchase rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 50 basis points to 3.00 per cent, 4.00 and 2.00 per cent, respectively.

December

Parliament passes legislation on a health reform ("Gesundheit 2000") to be implemented on 1 January 2000 with measures to control health costs. *Inter alia*, spending by hospitals and doctors are subjected to budgets, and health funds obtain greater powers for negotiating directly with health care providers. Additional measures to contain costs and secure more comparable cost developments are the introduction of a positive list over reimbursable pharmaceuticals in 2002 and uniform case-fees for hospital treatments from 2003 onwards.

Following a previous ruling from the Constitutional Court requiring restructuring of family allowances, financial support for child minding is changed, raising such expenses by an estimated DM 5.5 billion.

2000

January

In line with the Tax Relief Law the statutory lowest and highest income tax rates are lowered to 22.9 and 51 per cent, respectively. The top entrepreneurial tax rate is lowered to 43 per cent. The Child allowance and the basic tax-free allowance were increased. As stipulated in the Ecological Tax Law, the pension contribution rate is lowered from 19.5 to 19.3 per cent and will be financed by higher energy taxes. Moreover, the programme to reduce youth unemployment was extended.

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February

The European Central Bank (ECB) raises the repurchase rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 25 basis points to 3.25, 4.25 and 2.25 per cent, respectively.

March

The ECB raises the repurchase rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 25 basis points.

April

The ECB raises the repurchase rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 25 basis points.

May

The government introduces temporary work permits for up to 20 000 IT specialists (The "Green Card" initiative) to meet concerns regarding shortages of high-skilled labour.

June

The Federal Government presents its 2001 budget, which foresees a reduction in the federal budget deficit of DM 6 billion relative to the 2000 budget excluding proceeds from the UMTS auction. Whereas nominal expenditures are projected to remain unchanged, the government projects tax revenues foregone of DM 45 billion with the implementation of the tax reform, which will be offset by increases in other revenues.

The ECB raises the minimum bid rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 50 basis points.

August

The tax reform law passes the second chamber of parliament to take effect in January 2001. The law foresees reductions over the period 2000-05 of the lowest and highest statutory rates for personal income taxes to 15 and 42 per cent, respectively. Retained and distributed profits of corporations are taxed at a uniform 25 per cent from 2001 onwards. The imputation system for taxation of distributed profits will be abolished. Capital gains taxation for corporations will be abolished by 2002.

The auction of UMTS licenses leads to higher-than-expected proceeds of DM 99.4 billion.

The ECB raises the minimum bid rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 25 basis points.

September

The government introduces relief measures to compensate for the oil price increases. Tax relief for commuter expenses is raised from DM 0.70 per kilometre to DM 0.80 per kilometre and is extended to commuters using other means of transportation than private cars. A one-off heating subsidy of DM 5 per m^2 is introduced.

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October

The government presents an update of its Stability Programme. It foresees balancing the general government budget by 2004 and that of the Federal Government by 2006. Moreover, it is envisaged that by 2004 the tax-to-GDP ratio declines by 1 percentage point to 23½ per cent and the expenditure-to-GDP ratio by 3 percentage points to 40½ per cent combined with a concurrent fall in the gross debt ratio from 60 to 54½ per cent (Maastricht definition).

The ECB raises the minimum bid rate and the interest rates on the marginal lending and deposit facilities of the Eurosystem by 25 basis points to 4.75, 5.75 and 7.75 per cent, respectively.

2001

January

The first chamber of parliament ("B*undestag*") passes the new pension reform, which gradually introduces a voluntary funded system (government subsidised through tax concessions and grants) to complement a gradual reduction of contribution rates of the public pay-as-you-go pension system. The reform is designed to contain increases in contribution rates. Replacement ratios for new retirees are planned to be reduced from almost 70 per cent to between 67-68 per cent in 2030.

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